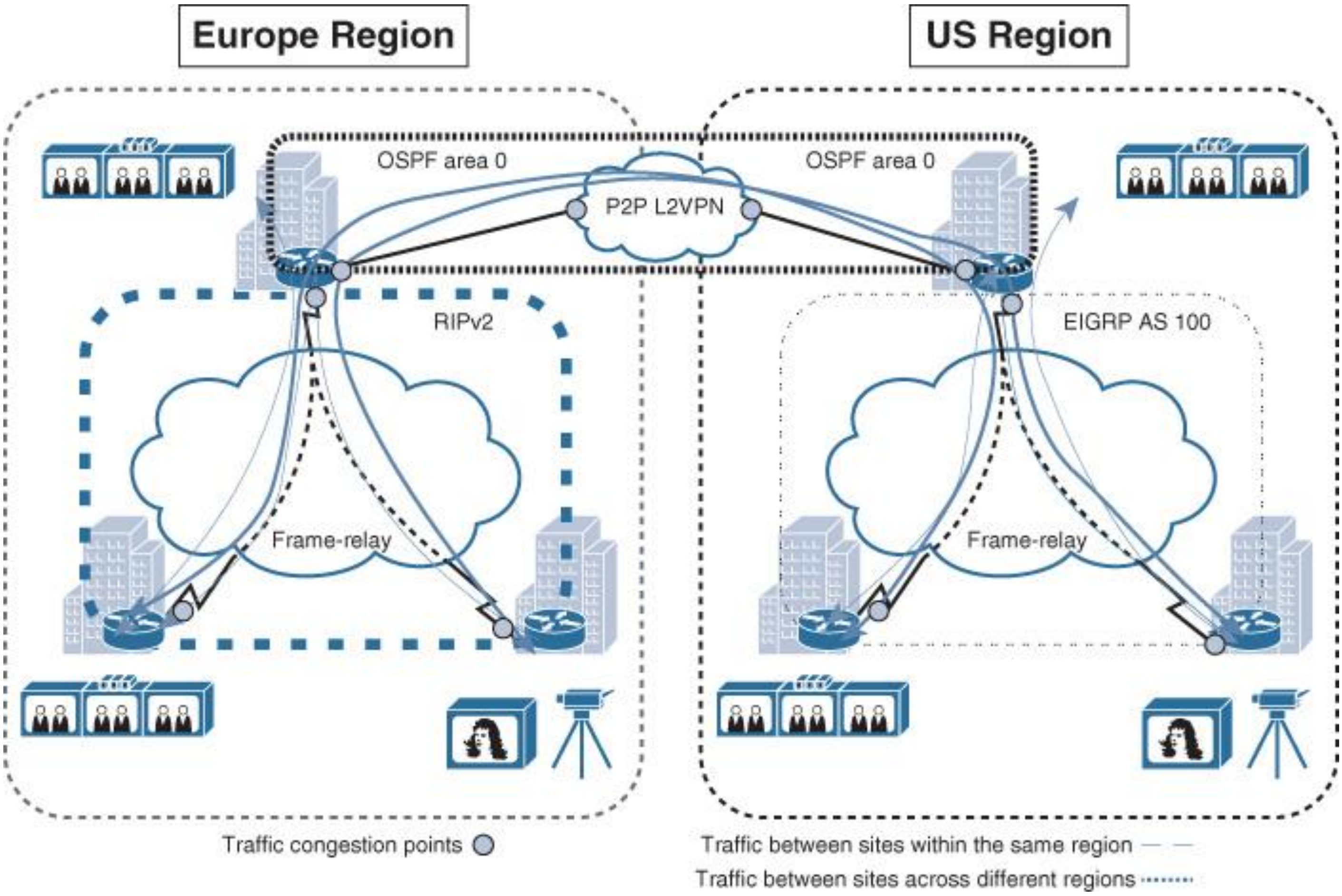

Figure 4-28 *Current Network Architecture and Traffic Flow*

Current state

- WAN is based on hub-and-spoke topology over Frame Relay.
- EIGRP is the routing protocol used over the U.S. region WAN.
- RIPv2 is the routing protocol used over the Europe region WAN.
- OSPF is the routing protocol used by each hub site and between the hub sites over the VLL.
- OSPF area 0 is used over the VLL and each hub LAN network.
- External BGP (eBGP) is the proposed protocol to be used over the new MPLS L3VPN WAN.

Assumption: WAN IP addressing and the desired amount of bandwidth per site is organized with the MPLS VPN provider before the migration starts.

Migration steps

Step 1. (Illustrated in Figure 4-29):

a. Select the hub site to act as the transit site during this migration process (per region).

b. Establish physical connectivity between the hub site (per region) and the new MPLS L3VPN SP.

c. Establish an eBGP session between the hub (CE) and the SP (PE).

d. Configure redistribution between OSPF and BGP at each hub site.

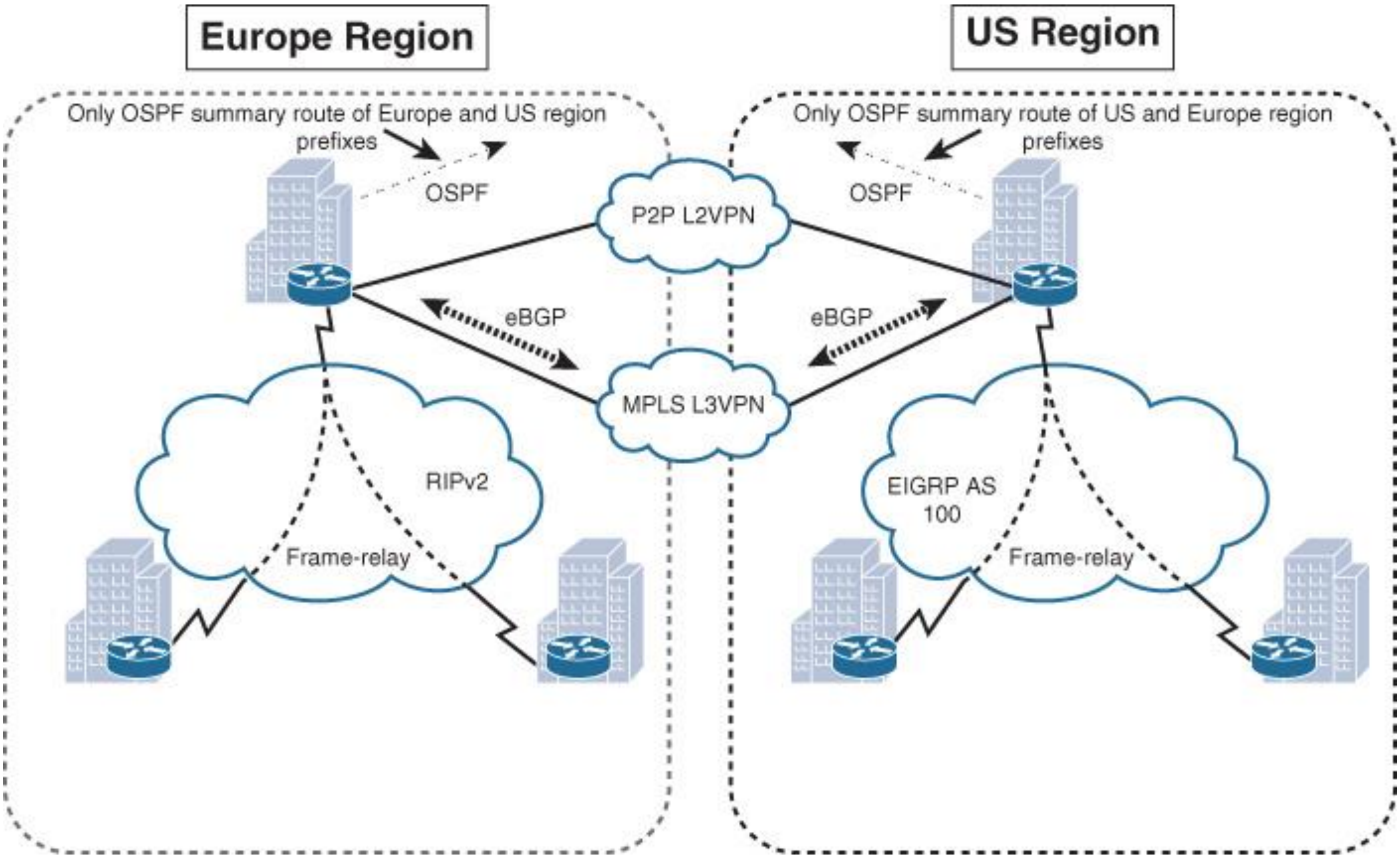

Figure 4-29 *WAN Migration Step 1*

Note

After the migration to the MPLS VPN, there will be two points of route redistribution with a backdoor link (over the L2VPN link between the hub sites). This scenario may introduce some route looping or suboptimal routing. Therefore, it is advised that over this L2VPN link each hub site (ASBR) advertises only the summary routes and not the specific to ensure that the MPLS WAN path is always preferred (more specific) without potential routing information looping. Alternatively, route filtering, as discussed in Chapter 2, “Enterprise Layer 2 and Layer 3 Design” (in the “Route Redistribution” section) must be considered.

Traffic between the LAN networks of each hub site will use the VLL (L2VPN) path as the primary path. Because the HQ LAN networks, along with the VLL link, are all part of the same area (area 0), the summarization at the hub (Autonomous System Border Router [ASBR]) routers will not be applicable here (same area). By reducing OSPF cost over the VLL link, you can ensure that traffic between the HQ LANs will always use this path as the primary path.

Step 2. (Illustrated in Figure 4-30):

a. Connect one of the spoke routers intended to be migrated to the MPLS VPN.

b. Establish an eBGP session with MPLS VPN SP and advertise the local subnet (LAN) using the BGP network statement (ideally without route retribution).

c. Once the traffic starts flowing via the MPLS VPN (eBGP has a lower AD 20), disconnect the Frame Relay link.

d. At this stage, traffic between migrated spokes and nonmigrated spokes will flow via the transit hub site.

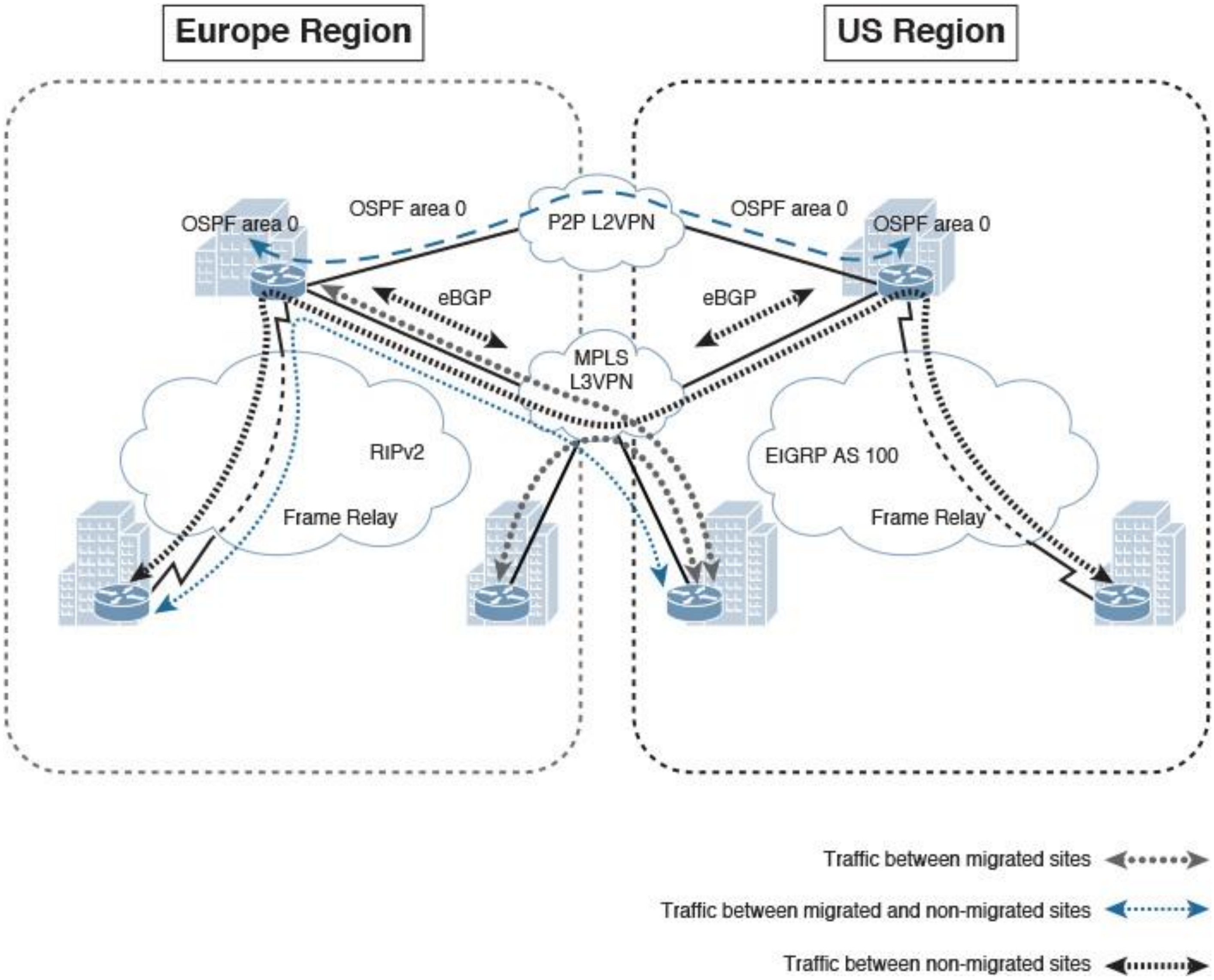

Figure 4-30 *WAN Migration Step 2*

Step 3. Migrate the remaining spokes using the same phased approach.

Note

With this approach, connectivity will be maintained between the migrated and nonmigrated sites without introducing any service interruption until the migration of the remaining remote sites is completed.

Enterprise Internet Edge Design Considerations

The Internet edge is another module or block of the modular enterprise architecture and part of the enterprise edge module, which provides external connectivity to the other places in the network (PINs) across the enterprise. The Internet block in particular acts as a gateway to the Internet for the enterprise network. From a design point of view, the Internet edge design may significantly vary between organizations, because it is typically driven by the security policy of the business. In addition, large enterprises typically put the Internet edge or module either collocated within the campus network or within its data center network. Nevertheless, in both cases, the location will not impact the module design itself; it will only change the overall enterprise architecture and traffic flow. Therefore, the design concepts discussed in this section apply to both scenarios (location neutral, whether the Internet block is located within the campus or within the data center).

Internet Edge Architecture Overview

To a large extent, the Internet edge design will vary between different networks based on the security policy of the organization and industry type. For instance, financial services organizations tend to have sophisticated multilayer Internet edge design. In contrast, retail businesses usually have a less-complicated Internet edge design. Consequently, this can lead to a significant difference in the design, which makes it impractical to provide specific design recommendations for this block. However, Figure 4-31 illustrates a typical (most common) Internet block foundational architecture. It highlights the main layers and network components that form this module as part of the overall modular enterprise architecture.

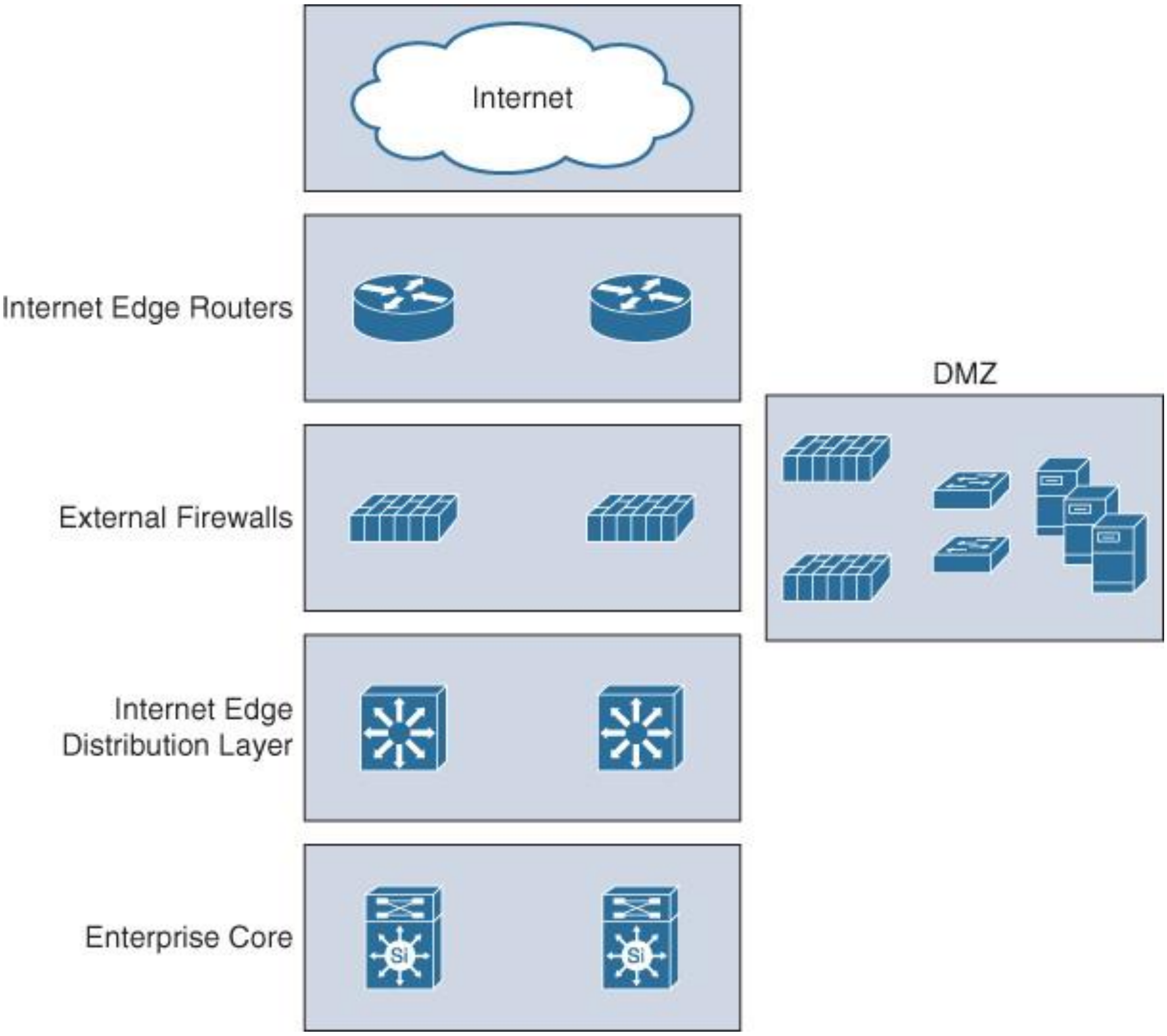

Figure 4-31 *Enterprise Internet Block Foundational Architecture*

This block should ideally follow the same principle of the hierarchical design model as part of the modular enterprise building block architecture. This hierarchical model offers a high level of flexibility to this block by having different layers, each focused on different functions. The distribution layer in this block aggregates all the communications within the block, and between the enterprise core and the Internet block, by providing a more structured design and deterministic traffic flow. This flexible design can be considered foundational architecture. It can then be changed or expanded as needed. For example, the demilitarized zone (DMZ) in Figure 4-29 can be replicated into multiple DMZs to host different services that require different security policies and physical separation. Some designs also place the VPN termination point in a separate DMZ for an additional layer of security and control.

Figure 4-32 highlights the typical primary functions and features at each layer of the Internet block architecture.

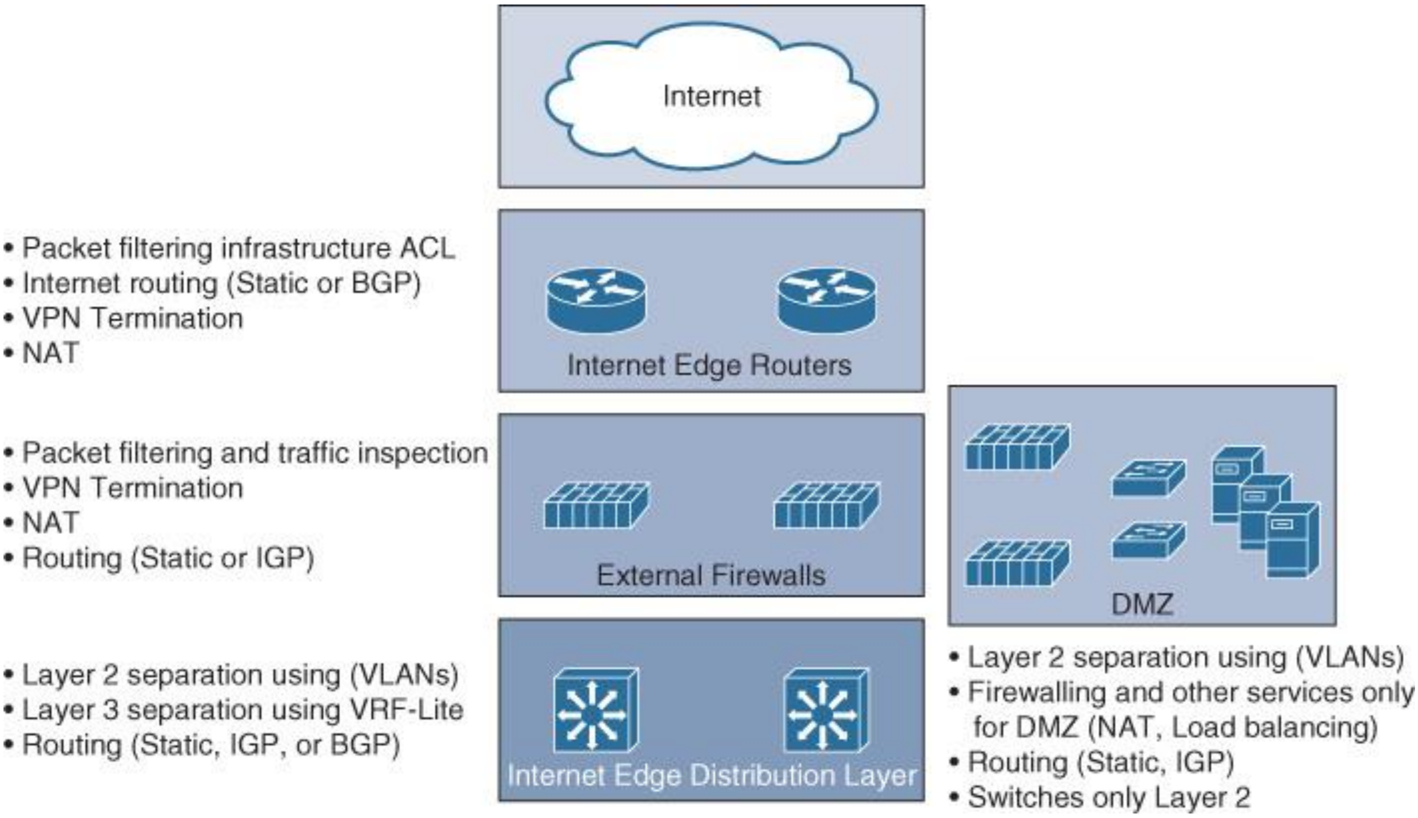

Figure 4-32 *Internet Block per Layer Functions and Features*

Note

These functions can vary from design to design to some extent. Moreover, all of them are not necessarily required to be applied. For example, NAT can be applied at only one layer only instead of doing double NATing.

Consider, for instance, a scenario where the security requirements specify that all traffic passing through the external Internet block firewalls must be inspected, not tunneled. This slightly influences the design. Technically, it can be achieved in different ways, depending on the available options and platform capabilities. (For example, some next-generation firewalls can perform deep packet inspection, even for tunneled traffic.) Alternatively, the VPN tunnels can be terminated at the Internet edge router or a separate DMZ can be created for the VPN, where VPN tunnels can terminate (using VPN concentrator or dedicated firewalls/routes). The decapsulated VPN traffic will then be sent back to the Internet edge firewalls for traffic inspection before reaching the internal network.

Enterprise Multihomed Internet Design Considerations

As discussed earlier in this section, the design of the Internet edge will vary to a large extent based on different variables and requirements. The major influencing factor is the organization's security policy. Similarly, the multihoming to the Internet follows the same concept, because it is part of this block. Because of this high degree of variability, this section covers the most common scenarios with multihoming to the Internet.

Multihoming Design Concept and Drivers

Multihoming refers to the concept of having two or more links to external networks, using either one or more edge nodes. This concept or connectivity model is common in large enterprises; however, the actual drivers toward adopting this connectivity model vary, and ideally the decision to do so should be driven by the business and functional requirements. In general, the most common drivers for enterprises include the following:

- Higher level of path and link redundancy
- Increased service reliability

- Offer the ability for the business to optimize the return on investment (ROI) of the external links through traffic load-balancing and load-sharing techniques
- Cost control, where expensive links can be dedicated for certain type of traffic only
- Flexibility and efficiency:
- This design approach increases the overall bandwidth capacity to and from the Internet (by using load-balancing or load-sharing techniques).
- Provides the ability to support end-to-end network and path separation with service differentiation by having different Internet links and diverse paths end to end from the ISP to the end users (for example, to serve different entities within the enterprise using different Internet links based on business demand or a security policy).

From a design point of view, network designers need to consider several questions to produce a more business-driven multihoming Internet design. These questions can be divided into two main categories:

- **Path requirements**
- Is the business goal high availability only?
- Is the business goal to optimize ROI of the existing external links?
- Should available bandwidth be increased?
- Should there be path and traffic isolation?
- **Traffic flow characteristics**
- Is the business goal to host services within the enterprise and to be accessible from the Internet?
- Is the business goal to host some of their services in the cloud or to access external services over the Internet?
- Or both (hybrid)?

The reason behind considering these questions is to generate a design (typically BGP policies) that aligns with the business and functional requirements. In other words, designing in isolation without a good understanding of the different requirements and drivers (for example, business goals and functional and application requirements) will make it impossible to produce an effective business-driven multihoming design.

Note

This information can be obtained in different ways, based on the gathered requirements. For example, it may be shown as functional requirements through utilization reports that show 75 percent of the traffic is outbound and 25 percent is inbound, in which it is clear that the traffic pattern is inclined toward accessing content over the Internet.

Note

BGP is the most flexible protocol that handles routing policies and the only protocol that has powerful capabilities that can reliably handle multiple peering with multiple autonomous systems (interdomain routing). Therefore, this section only considers BGP as the protocol of choice for Internet multihoming design; however, some designs may use IGP or static routing with multihoming. Typically, these designs eliminate all the flexibilities that you can gain from BGP multihoming scenarios.

BGP over Multihomed Internet Edge Planning Recommendations

Designing a reliable business-driven multihoming connectivity model is one of the most complex design projects because of the various variables that influence the design direction. Therefore, good planning and an understanding of the multiple angles of the design are prerequisites to generating a successful multihoming design.

The following are the primary considerations that network designers must take into account when planning a multihomed Internet edge design with BGP as the interdomain routing protocol:

- A public BGP autonomous system number (ASN) versus private ASN. Typically, public ASNs offer more flexibility to the design, especially with multihoming to different ISPs.

- Provider-independent (PI) public IP addresses offer more flexibility and availability options to the design compared to provider-assigned (PA) IP addresses.
- A PI address combined with a public ASN obtained from the Regional Internet Registry (RIR), for example, provides the most flexible choice with Internet multihoming, especially for enterprises that host services to be accessed from the Internet across different ISPs.
- Receiving a full Internet route can help to achieve more detailed traffic engineering policies. (For example, you can specify outbound traffic to use a certain link on a per geographical region basis, based on the IP prefixes along with its assigned BGP community values.)

Note

BGP community values can provide flexible traffic engineering control within the enterprise and across the ISP. Internet providers can match community values and predefined application policies per community value. For example, you can influence the path local_preference value of your advertised route within the ISP cloud by assigning an *x* BGP community value to the route. Refer to RFC 1998.

BGP Policy Control Attributes for Multihoming

As discussed earlier in this book, several BGP attributes influence BGP path selection. However, Table 4-13 lists the most common simple and powerful BGP attributes that you can use to control route advertisements and influence the path selection in BGP multihoming design.

Attribute	Usage Description
Local preference (LP)	Influence outbound traffic flows
AS-PATH prepend	Influence inbound and outbound traffic flows
Community values + (LP, AS-PATH, or weight)	Influence inbound and outbound traffic flows within the customer AS and across the ISPs autonyms systems
BGP weight	Influence local router decision for outbound traffic flows (Cisco proprietary attribute)

Table 4-13 *Common BGP Attributes for Internet Multihoming*

Note

BGP community values technically can be seen like a "route tag," which can be contained within the one AS or be propagated across multiple autonyms systems to be used as a "matching value" to influence BGP path selection. For instance, one of the common scenarios with global ISPs is that each ISP can share the standard BGP community values used with its customers to distinguish IP prefixes based on its geographic location (for example, by region or continent). This offers enterprises the flexibility to match the relevant community value that represents a certain geographic location and associate it with a BGP policy such as AS-PATH prepending to achieve a certain goal. For example, an enterprise may want all traffic going to IP prefixes within Europe (outbound) to use Internet link 1, while all other traffic should use the second link. As illustrated in Figure 4-33, BGP community values can simplify achieving this goal to a large extent in a more dynamic manner.

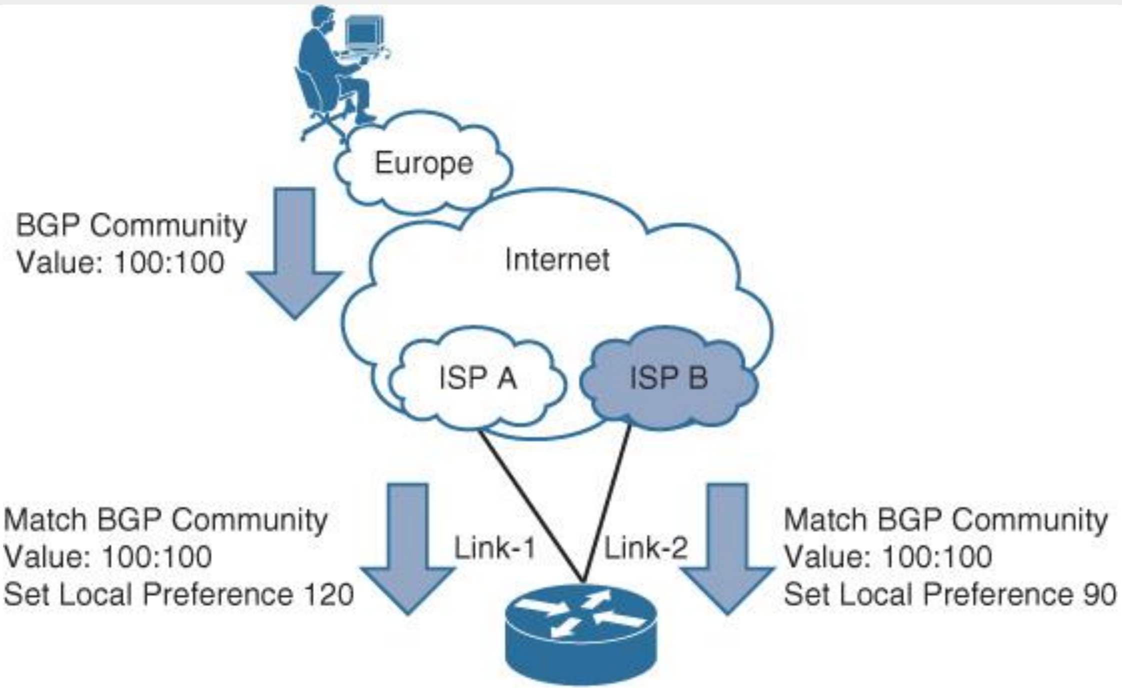

Figure 4-33 *BGP Community Value Usage Example*

Common Internet Multihoming Traffic Engineering Techniques over BGP

This section covers the primary traffic engineering models and techniques that you can use with multihomed Internet connectivity over BGP.

Note

Throughout this section, *ingress* always refers to the inbound direction toward the enterprise, and *egress* refers to the outbound direction toward the SP cloud.

Scenario 1: Active-Standby

This design scenario (any of the connectivity models depicted in Figure 4-34) is typically based on using one active link for both inbound and outbound traffic, with a second link used as a backup. This design scenario is the simplest design option and is commonly used in situations where the backup link is a low-speed link and is only required to survive during any outage of the primary link.

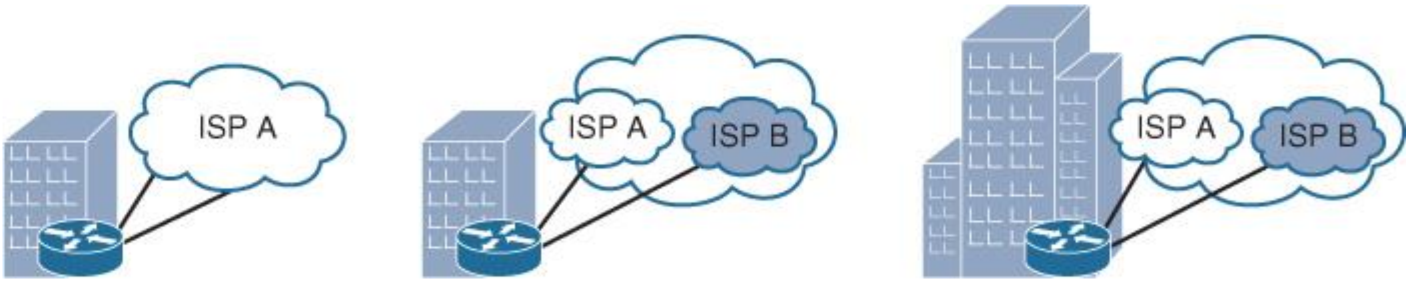

Figure 4-34 *Internet Multihoming Active-Standby Connectivity Models*

Figure 4-35 shows an active-standby scenario where ISP A must be used as the primary and active path for both ingress and egress traffic flows.

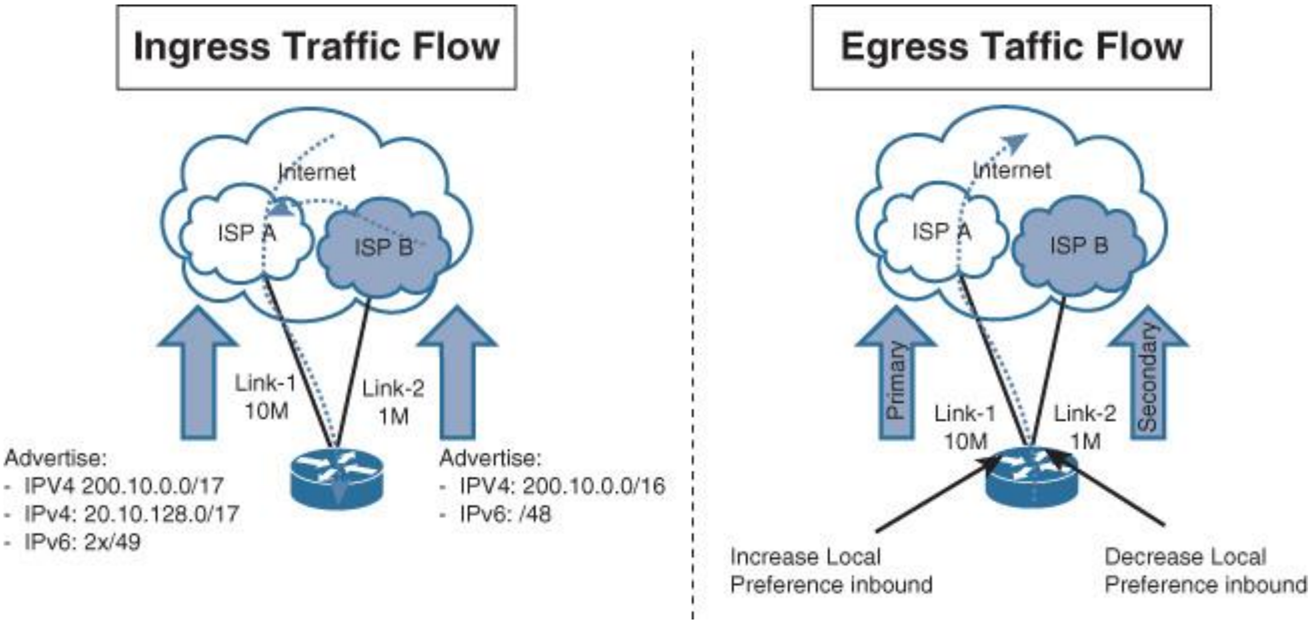

Figure 4-35 *Internet Multi-homing Active-Standby traffic flow*

Table 4-14 outlines a possible solution for this scenario.

Traffic Direction	BGP Policy
Ingress	Longest match over the preferred path, by dividing the prefix into 2 halves (For instance, advertise /16 as 2x /17 over the preferred ingress path toward ISP A in the scenario in Figure 4-35.)
Egress	Use local_preference (Set the preferred route/ path with higher value.)

Table 4-14 *Internet Multihoming Active-Standby*

Although AS-PATH prepending is one of the most common and obvious techniques that you can use here in this scenario to influence ingress traffic, ISPs usually allocate a higher local_preference value to prefixes learned by their customers than the ones learned through other peering ISPs. In other words, even though the prefix 200.10.0./16 is advertised to both ISP A and B with AS-PATH prepending applied toward ISP B, ISP B's customers will always use the path over ISP B because it will be assigned higher local_preference (within ISP B cloud) than the one learned via ISP A, when traffic is passing through ISP B. This means that the targeted goal will not be achieve optimally in this case because any traffic flow coming from any of ISP B's customers will use the low-bandwidth link (link 2 in this case) to reach 200.10.0.0/16. Therefore, by dividing the /16 network into two /17s and advertising it over ISP A, if there is a customer of ISP B that wants to reach your prefix (one of the 2 x /17s advertised by ISP A), ISP B will never get to the /16 sent over ISP B link because the longer match over ISP A will always win in this case, as illustrated in Figure 4-35.

Note

This behavior (where ISPs allocate a higher local_preference value to prefixes advertised within their cloud) is altered sometimes when the ISP enables their customers to control what BGP attribute and value they need to assign their prefixes when injected into the ISP cloud. This is normally achieved by using standard BGP community values to influence traffic routing within the ISP cloud (most commonly to change AS-PATH or local_preference values within the ISP cloud). For example, in the scenario illustrated in Figure 4-36, if you assign a BGP community value 300:110 to prefix 200.10.0.0/16 toward ISP A (AS 300), its BGP local_preference value within the SP cloud will be set to 110, which will be given preference across this SP cloud and its directly connected customers. Similarly, if you assign a BGP community value 500:70 to prefix 200.10.0.0/16 toward ISP B (AS 500), along with BGP AS-PATH prepending 3 x ASN, (100 100 100), this will make this prefix carry a low local_preference value within AS 500 (assuming, 70 is less than the common default BGP local_preference "100"). In addition, it will be seen (by AS 500) as a longer AS path (100 100 100) compared to the prefix learned via ISP A (AS 300). Technically, the prefix 200.10.0.0/16 within AS 500 will be seen as follows:

Via AS 100: Local_preference = 70, AS-PATH = (100 100 100)

Via AS 300: Local_preference = 100, AS-PATH= (300 100) (preferred path)

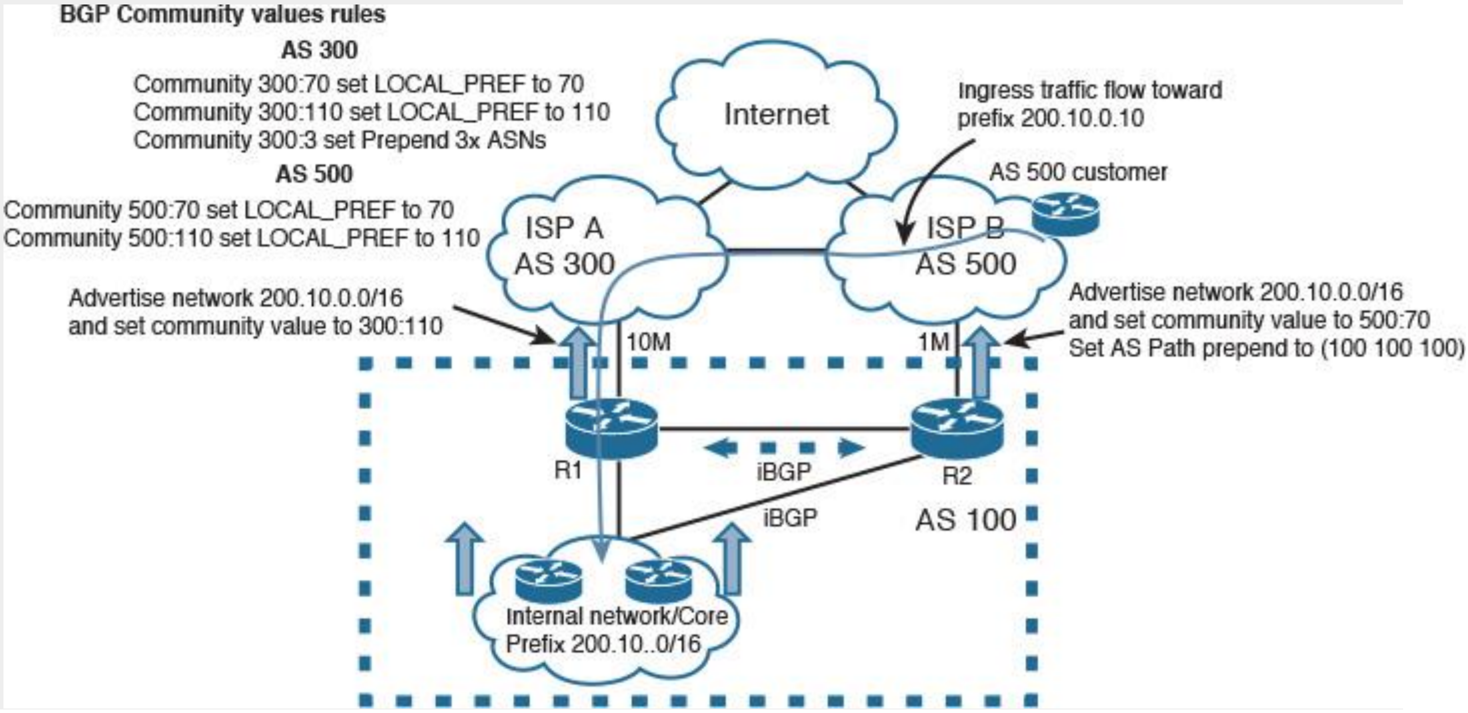

Figure 4-36 *Altering BGP Attributes Within the ISP Cloud*

With this approach, network designers can achieve a true and optimal active-standby Internet connectivity model. However, two limitations apply to this approach. First, not every ISP provides this flexibility to their clients. Second, when customers are multihoming to different ISPs, the advertised prefixes must be PI to avoid having the upstream ISP aggregate the advertised prefix and break the entire design. For instance, if ISP A advertises the prefix 200.10.0/16 toward ISP B as 200.0.0.0/8, the path via R2 over ISP B will be always preferred by ISP B customers (longest match) regardless of which BGP attribute has been used.

Scenario 2: Equal and Unequal Load Sharing

These types of design scenarios are commonly used either when there are two Internet links with the same bandwidth value and traffic needs to be distributed evenly across the two links to increase the overall bandwidth and reduce traffic congestion (equal load sharing) or when the Internet links have a different amount of bandwidth where traffic can be distributed in a weighted manner. Ideally, the link with higher bandwidth should handle more traffic than the link with lower bandwidth, as summarized in Figure 4-37.

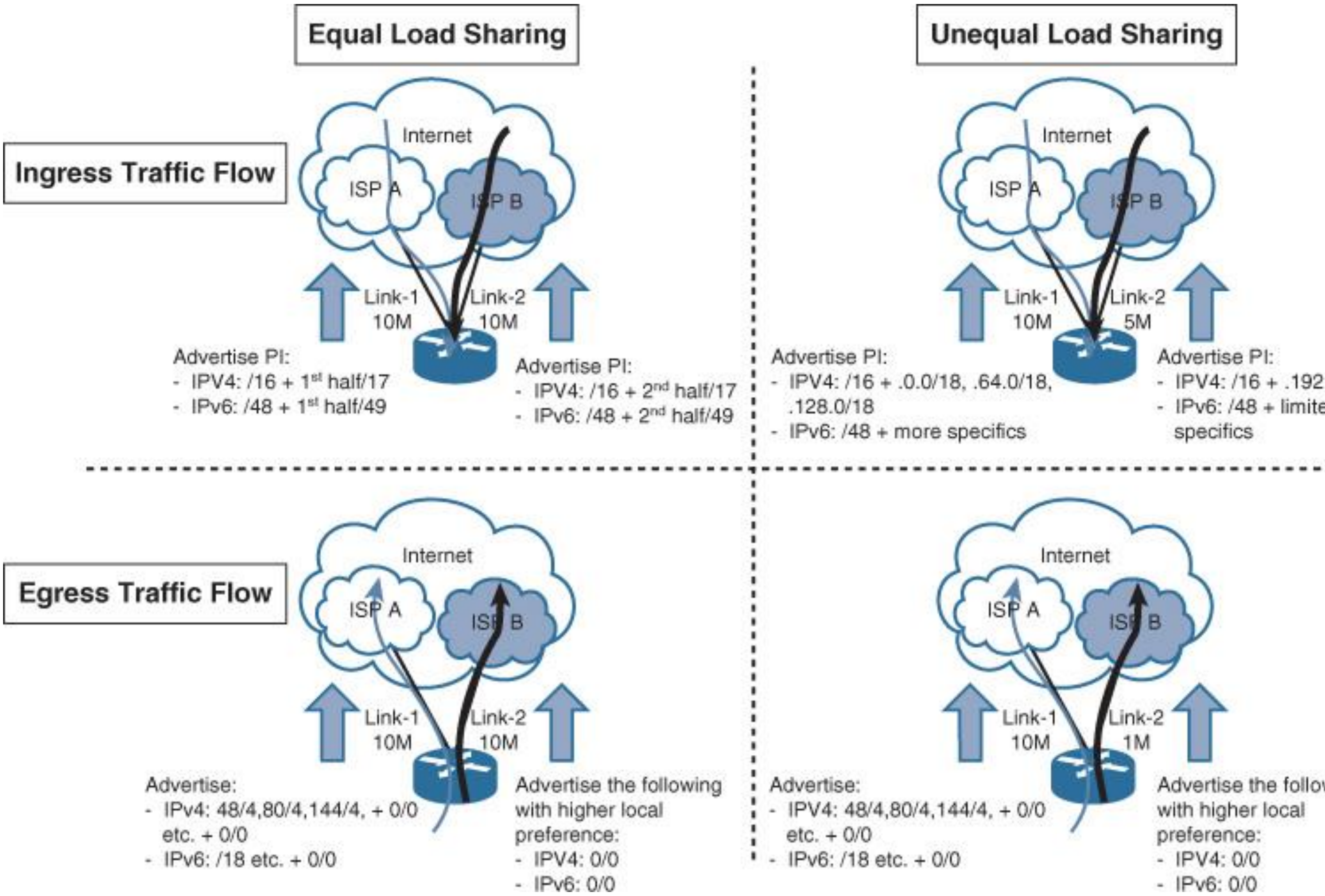

Figure 4-37 *Internet Multihoming Equal and Unequal Load Sharing*

Table 4-15 shows a typical solution.

Traffic direction	BGP Policy
Ingress	The typical mechanism to be used here is to divide the PI address into 2 halves. For example, an IPv4 subnet /16 can be divided into 2 subnets /17, similarly for IPv6 /48 to 2 /49. Then advertise each half over a different link along with the aggregate (IPv4 /16 or IPv6 /48 in this example) over both links to be used in case of link failure. For unequal load sharing, you can use the same concept with more small subnets to be advertised over the path with higher capacity.
Egress	For the outbound traffic direction, you need to receive the full Internet route from one of the ISPs along with the default route from both. Accept with filtering only every other /4 for IPv4 (for example, 0/4, 32/4). IPv6 can use same concept (IPv6 either selectively or the same concept). From the other link, increase the local_preference for the default route. In this case, the more specific route (permitted in the filtering) will be used over one link. Every other route that was filtered out will go over the second link using the default route. For unequal load sharing, more subnets can be accepted/allowed from the link with higher capacity.

Table 4-15 *Internet Multihoming Equal/Unequal Load Sharing*

Note

ISPs usually deploy route filtering policies with their customers and with other peering ISPs, where only certain subnet lengths are accepted, to avoid propagating a large number of small network such as v4/24 or v6/ 64. However, the subnets presented in this section are hypothetical to simplify the explanation of the discussed points.[4]

4. http://www.space.net/~gert/RIPE/ipv6-filters.html

Although the typical mechanisms used to influence path selection equally (to certain extent) are the ones described inTable 4-15, this cannot guarantee fair or equal load distribution across both links (because traffic load cannot be measured based only on its network or subnet size). For instance, if a /24 subnet divided into two halves (.0/25 and .128/25), advertising each half (/25) over two separate Internet links will (in theory) provide equal load distribution across both Internet links in the inbound direction. In fact, what is hypothetically correct is not always practically achievable; the reason why is because if there are a few hosts such as an FTP or web server as part of the subnet (0/25), with high traffic volume destined to these servers, the link advertising (0/25) will be used more than the second Internet link. This same concept applies to the outbound traffic direction. In other words, achieving true equal load distribution must not be derived only by IP subletting (dividing and advertising the available IP range equally across the available links). Instead, it must be based on actual service utilization reports. Alternatively, you can use a specialized load-balancing appliance that that can distribute traffic flow equally based on real-time link utilization.

Similarly, if unequal load sharing is required (for example, 70/30 load distribution, which means 70 percent of traffic should use the link with higher bandwidth and 30 percent should use the other link with lower bandwidth), the same principle and concerns discussed earlier with regard to equal load sharing apply here. In other words, proper planning and a study of utilization reports must be performed before designing and applying BGP policies to influence which network or host should be reachable over which link. Also, the same applies to outbound traffic. (What are the most targeted services in terms of traffic volume and number of traffic flows, such as cloud-hosted applications like Cisco Webex?) After these facts are identified, network designers can provide a more practical multihoming with the desired proportional load distribution model (whether it is equal or unequal load distribution).

Note

Cisco Performance Routing (PfRv3) technology can achieve an advanced level of traffic load distribution based on different criteria and real-time link utilization.[5]

5. http://docwiki.cisco.com/wiki/PfR:Technology_Overview

Scenario 3: Equal and Unequal Load Sharing Using Two Edge Routers

The design techniques described in the preceding scenarios all apply to this scenario for the ingress traffic flow direction. However, the egress traffic flow direction depends on the LAN side design behind the routers:

- **Using a firewall with a Layer 2 switch between the firewall and routers:** Use multiple groups of Hot Standby Routing Protocol (HSRP) or Virtual Router Redundancy Protocol (VRRP) for load sharing on the routers' side.
- **Using Layer 3 network node, such as a switch connecting directly to the edge routers:** You can use IGP with ECMP in this case.

Note

In both cases, you need to make sure that there is a link between the two Internet edge routers (physical or tunnel) with internal BGP (iBGP) peering over this link, to avoid traffic black-holing in some failure scenarios.

Asymmetrical Routing with Multihoming (Issue and Solution)

The scenario depicted in Figure 4-38 demonstrates a typical design scenario with a potential for asymmetrical routing. This scenario is applicable to two sites or data centers with a direct (backdoor) link, along with a layer of firewalling behind the Internet edge routers. In addition, these firewalls are site specific, where no state information is exchanged between the firewalls of each site. In addition, both sites are advertising the same address range toward the Internet (PI or PA) range. Therefore, a possibility exists that return traffic (of outbound traffic) originated from site 1 going to the Internet using the local Internet link will come back over the site 2 Internet link. The major issue here is that the firewall of site 2 has no “state information” about this session. Therefore, the firewall will simply block this traffic. This can be a serious issue if the design did not consider how the network design will handle situations like this, especially during failure scenarios.

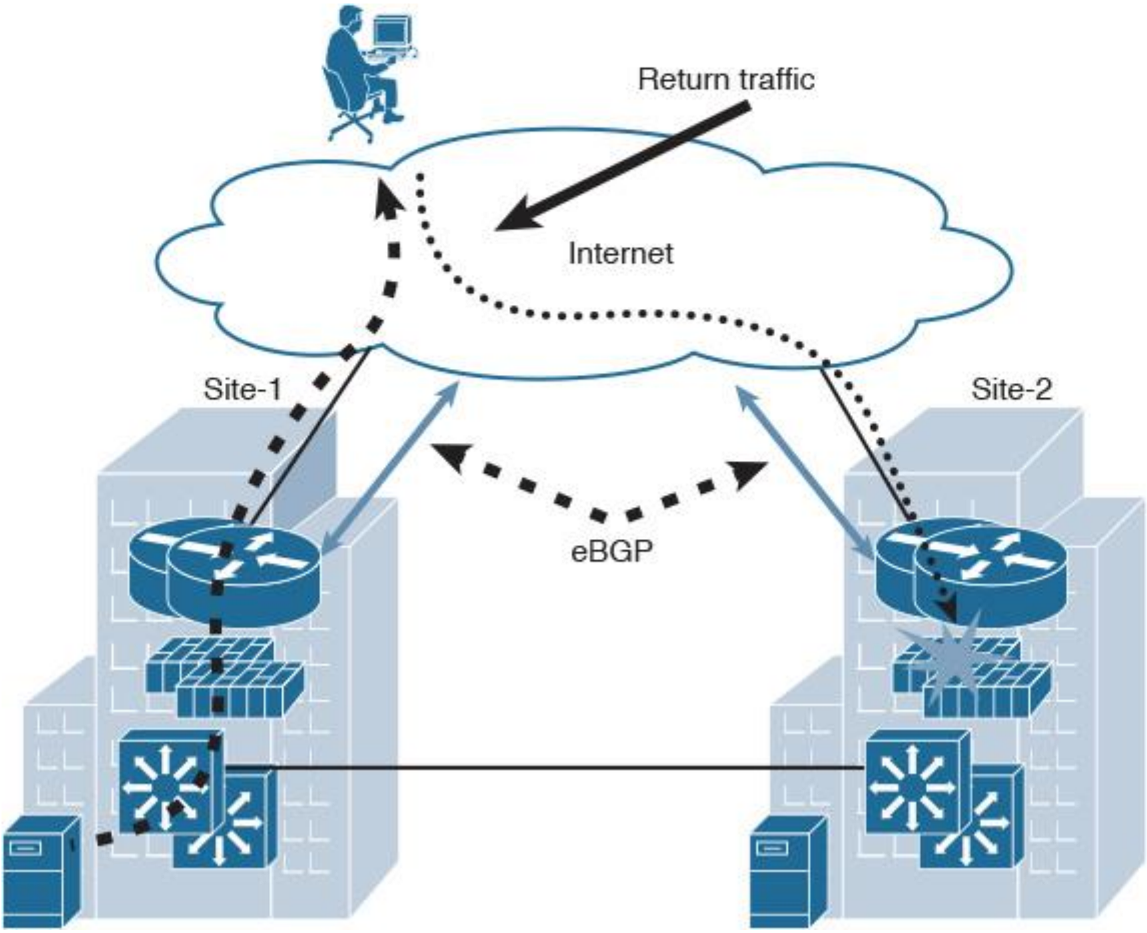

Figure 4-38 *Asymmetrical Routing*

To optimize this design to overcome this undesirable behavior, network designers need to consider the following:

- **Control plane peering between edge routers in each site:** The first important point here is to make sure that both Internet edge routers are connected directly and use iBGP peering between them. This link can be physical or over a tunnel interface, such as a GRE tunnel. A network designer can optimize this design and mitigate its impact by adding this link along with associated BGP policies (make site 1 internal prefixes more preferred over the site 1 Internet link), which will help to avoid the blocking by the site 2 firewall.
- **Organized IP addressing advertisement:** To make the preceding point work smoothly, we need to make sure that each site is advertising its own route prefixes as more specific (longest match principle), in addition to the aggregate of the entire PI subnet, as discussed earlier. For example, /16 might be divided to two /17s per site. (You can use the same concept with IPv6.).
- **NAT consideration:** The other point to be noted here is that if these prefixes are NATed by the edge firewalls, one of the common and proven ways to deal with this type of scenario is by forming direct route peering between the distribution/core layer nodes and the Internet edge routers. Consequently, the edge firewalls can perform NAT for traffic passing through them (as long as the two earlier considerations mentioned in this list are in place).

For instance, in Figure 4-39 the Internet edge distribution is peering with the Internet edge router using multihop eBGP with private AS and advertising the PI prefixes over BGP (using static routes with a BGP network statement to advertise the routes). The firewall is using a default route toward the Internet edge router, along with NAT.

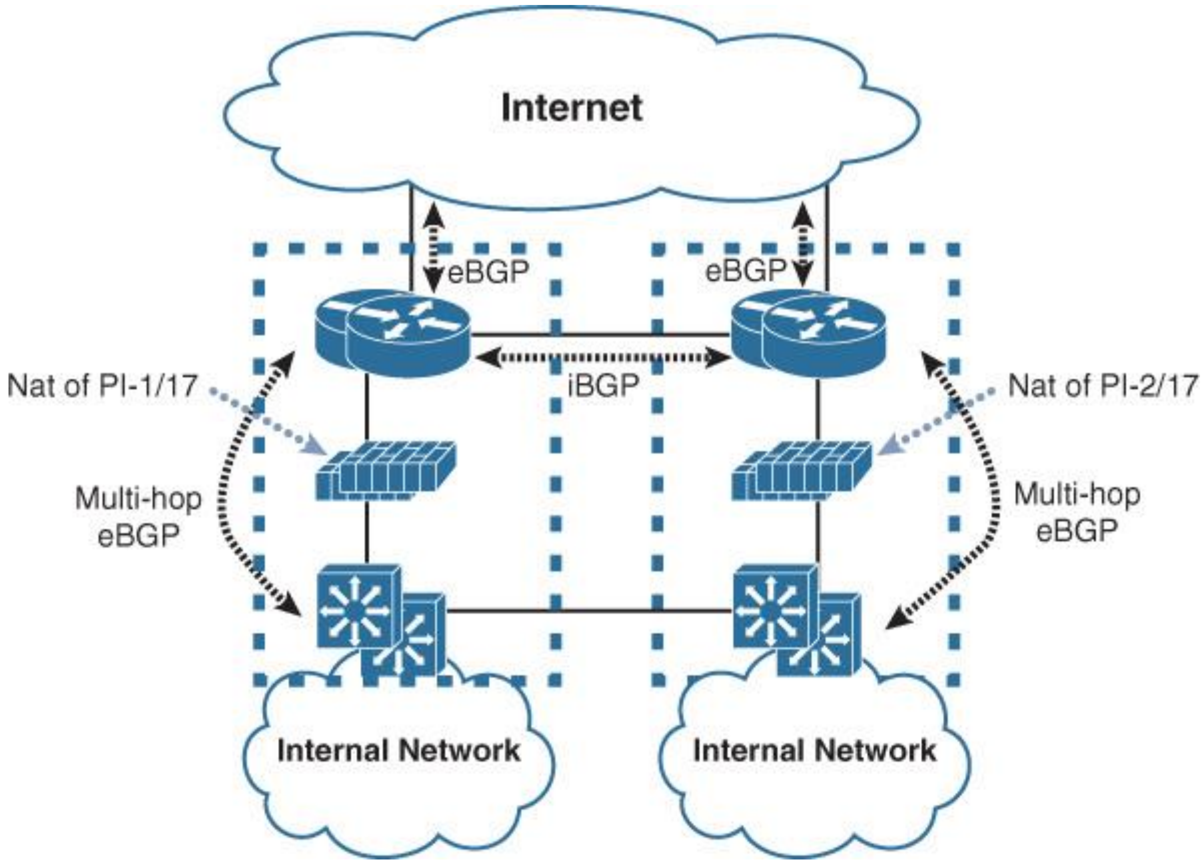

Figure 4-39 *Optimized Multihoming Design with Firewalls*

By incorporating the design recommendations to optimize the preceding scenario, a network designer can make the design more agile in its response to failures. For example, in the topology in Figure 4-39, if the link between site 1 and the Internet were to go down for some reason, traffic destined for site 1 prefixes (part of 1st half of the /17) would typically go to the site 2 Internet link (because we advertise the full /16 from both sites). Then traffic would traverse the intersites link to reach site 1. As a result, this design will eliminate the firewall blocking issue, even after an Internet link failure event, as illustrated in Figure 4-40.

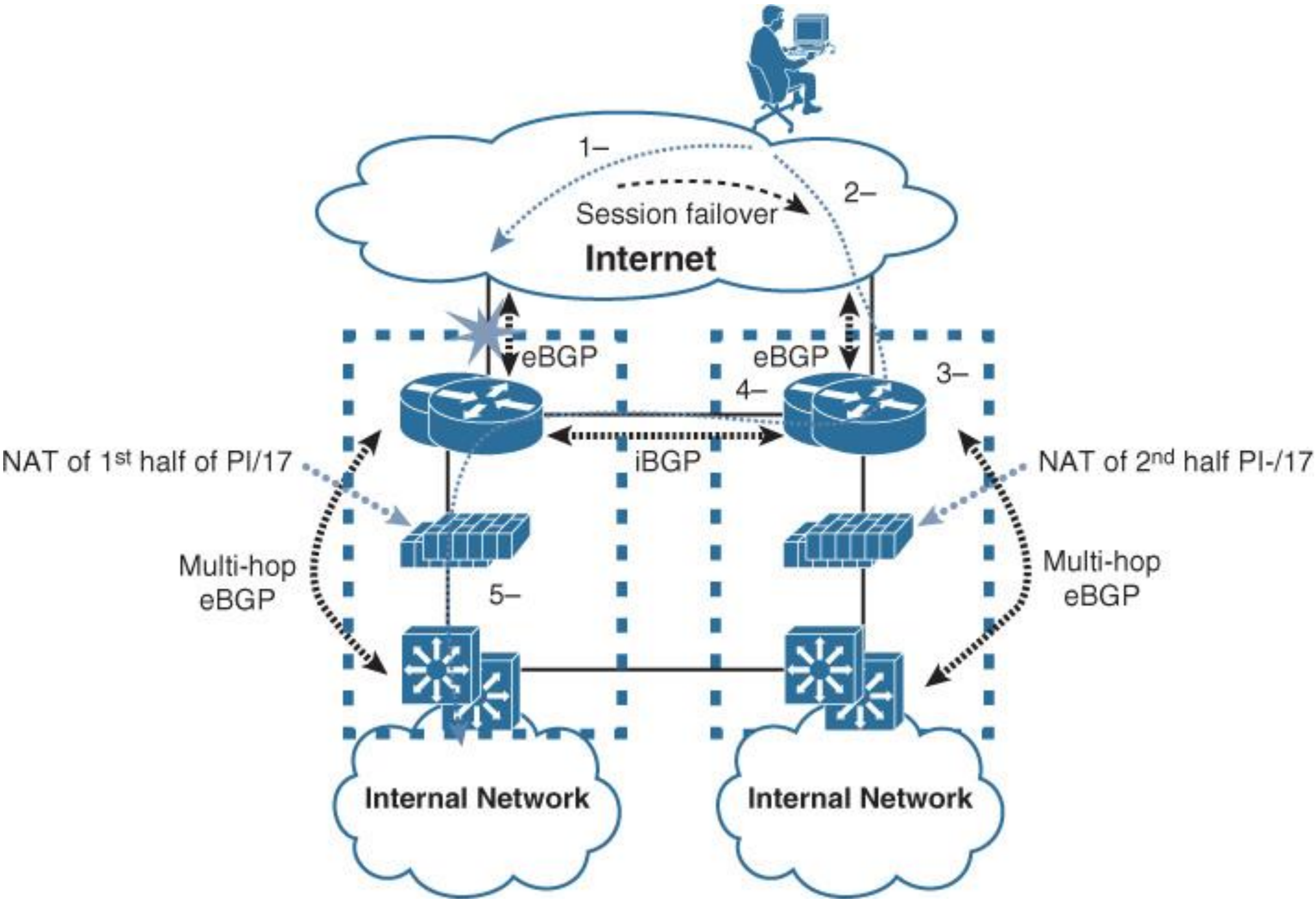

Figure 4-40 *Optimized Multihoming Design: Failure Scenario*

Summary

Today's enterprise businesses, in particular multinational and global organizations, primarily rely on technology services and applications to achieve their business goals. Therefore, the enterprise edge module (both WAN and Internet) is one of the most vital and critical modules within the modern modular enterprise architecture. It represents the gateway of the enterprise network to the other remote sites, locations, business partners, and the Internet. Therefore, network designers must consider designs that can provide a common resource access experience to the remote sites and users, whether over the WAN or the Internet, without compromising any of the enterprise security requirements, such as end-to-end path separation between certain user groups. In addition, optimizing Internet edge design with business-driven multihoming designs can play a vital role in enhancing the overall Internet edge performance and design flexibility and can maximize the total ROI of the available links. Last but not least, overlay integration at the enterprise edge in today's networks can offer enterprises flexible and cost-effective WAN and remote-access connectivity, even considering the additional layer of control plane and design complexity that may be introduced into the overall enterprise architecture.

Part III: Service Provider Networks Design and Architectures

Chapter 5. Service Provider Network Architecture Design

One of the common design questions is this: What is the difference between designing an enterprise network versus service provide network? Technically, both types of networks are entitled to use the same Layer 2 and Layer 3 technologies and protocols. The design principles with regard to each type of these networks can vary to some extent; however, due the fact that the goal of service provider (SP) business almost always is to provide transit connectivity services such as interconnecting the dispersed sites of their customers across the SP core infrastructure (providing WAN services) or providing transit path to reach the Internet and to reach services in the cloud, either hosted by the sameprovider or by other providers. Moreover, with enterprise networks, the boundary points with external networks areusually very limited due to the typical nature of enterprise communication that is either between internal employees or "to and from" services and applications that either hosted internally within the enterprise or externally in the cloud. In contrast, an SP is a transit network by nature for external customers, consisting of multiple (normally large number) boundary points "ingress and egress" (with each customer or peering SP interconnect an ingress and egress point). Therefore, the primary principle with regard to SP network design is to provide a scalable and reliable infrastructure that can transmit customer traffic fast and reliably enough from its ingress point to the intended egress point across the SP network "transit network."

Furthermore, SP networks (as transit transport networks) are commonly based on a two-tier hierarchy, as depicted inFigure 5-1. Typical SP networks are constructed of multiple points of presence (POPs), with the provider edge (PE) nodes normally linked to each other over the SP backbone. The core is often structured of a meshed network, partially meshed network, or a set of rings, and each POP is usually made up of a ring or a multilayer hierarchy network. In addition, in an SP environment, routing information is normally carried via internal Border Gateway Protocol (iBGP) or multiprotocol iBGP (MP-iBGP). The interior gateway protocol (IGP) within the SP network is used only to carry next-hop information for BGP peers. As a result, the optimal path to the BGP next hop is decided by the SP's IGP by default. Nonetheless, several techniques enable you to alter this default behavior within the SP network, such as Multiprotocol Label Switching Traffic Engineering (MPLS-TE).

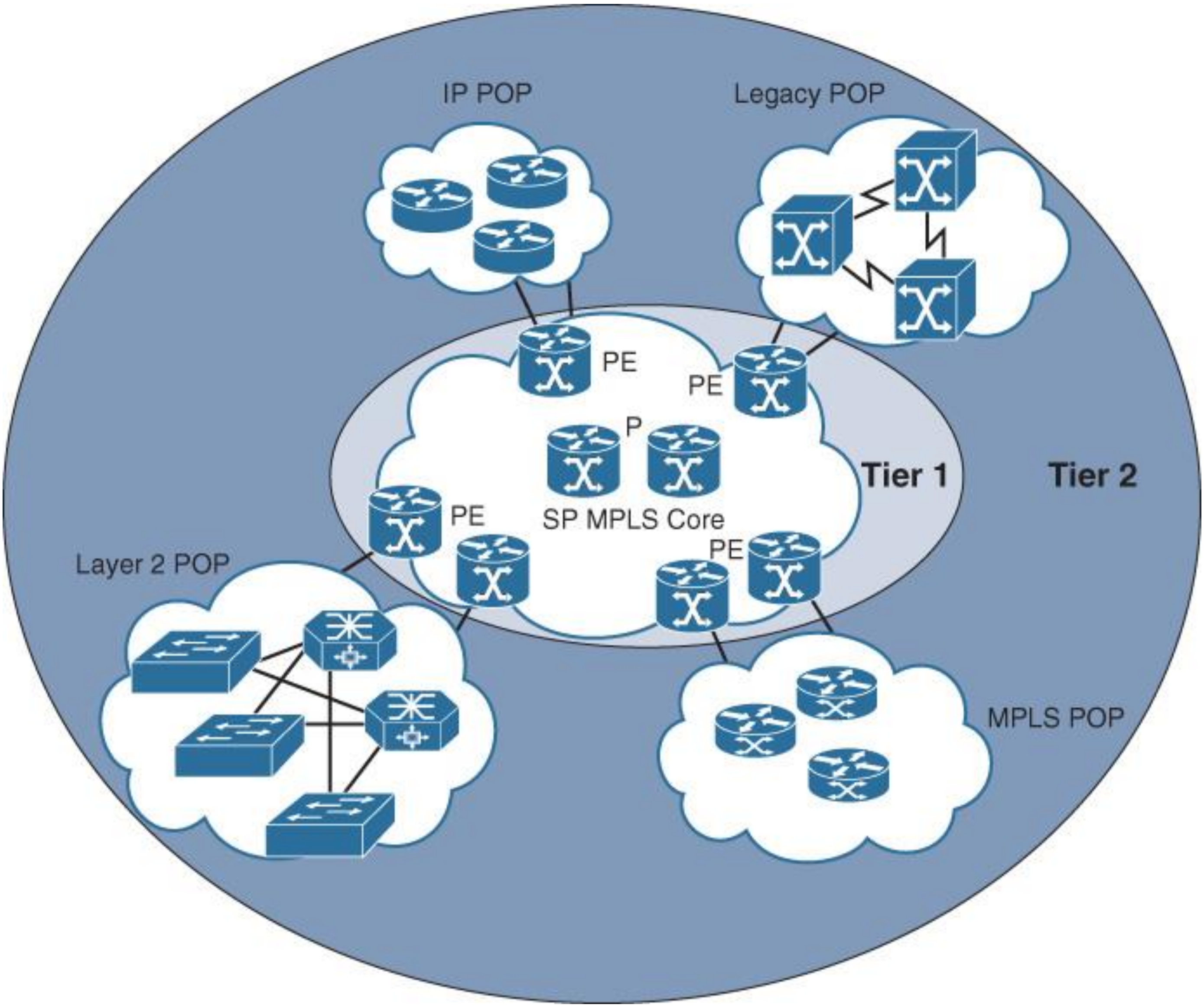

Figure 5-1 *Classic Service Provider Network Architecture*

Note

There are some enterprises that provide transit services to their customers; therefore, what should define the design principles is the nature of the business functions and offered services, even if its categorized as an enterprise business.

However, it is still debatable whether, by using overlays and virtualization techniques like MP-BGP VPN with MPLS, a "layer in layer" hierarchy can be achieved, in which these technologies can provide a sort of abstraction layer of the underlying physical infrastructure where customer routes are carried over the MP-BGP in its own virtual private network (VPN) (which can be considered another logical layer). This is a completely valid point. Other examples where more layers can be added to the SP architecture either physically or virtually (overlaid) include the following:

- Hierarchal MPLS architecture
- Hierarchal VPLS
- MPLS VPNv6 over an IPv4 SP core
- SP network with multilevel POPs (to cover wide geographic areas)

Note

The simplicity of SP network architecture compared to large enterprise networks architecture is relative and not absolute. For instance, a large SP network with thousands of nodes deployed under a single BGP autonomous system (AS) with one IGP flooding domain, such as single Intermediate System-to-Intermediate System (IS-IS) level, is simpler than a global enterprise network with BGP in the core and multiple IGP islands across different geographic regions combined with IPsec tunnels over the Internet as a backup path.

In contrast, an SP network with a few hundred routers only, distributed across different BGP autonomous systems with MP-BGP inter-AS, along with MPLS VPN, MPLS-TE, and MPLS-TE fast reroute (FRR) across both autonomous systems that each has its own IGP domain and protocols, will be to a large extent a complex type of SP architecture. In other words, as highlighted in Chapter 1, "Network Design Requirements: Analysis and Design Principles," network complexity is not only measured by size or number of nodes. However, multiple additional factors can drive the level of complexity, such as control plane design and structure, number of network protocols used, and how the routing domain design may affect other protocols. For example, the design and deployment of MPLS-TE in conjunction with MPLS-TE FRR over different routing domains is more complicated than over a single routing domain from both design and operational point of views.

Service Provider Network Architecture Building Blocks

The next-generation networks (NGNs) of SPs offers a converged infrastructure for video, mobile, and cloud services interconnected over different technologies and services over one converged MPLS-enabled core that provides various Layer 3 and Layer 2 interworking functions (IWFs), commonly referred to as *IP NGN SP networks*.

In a typical SP network (classic and NGN), each architecture has two main components: the core and POP. This section covers these two components from a network architecture point of view. Generally, as mentioned earlier, an SP network architecture consists of two tiers of hierarchies: SP core and the distributed POPs connected to it. POPs, however, can be divided into multiple tiers depending on the size, coverage of the geographic area, and type of service access provided by the POP, such as residential services, mobile backhaul aggregation, or MPLS L2/L3 VPN. Figure 5-2 illustrates a typical SP network with a POP that consists of multiple tiers.

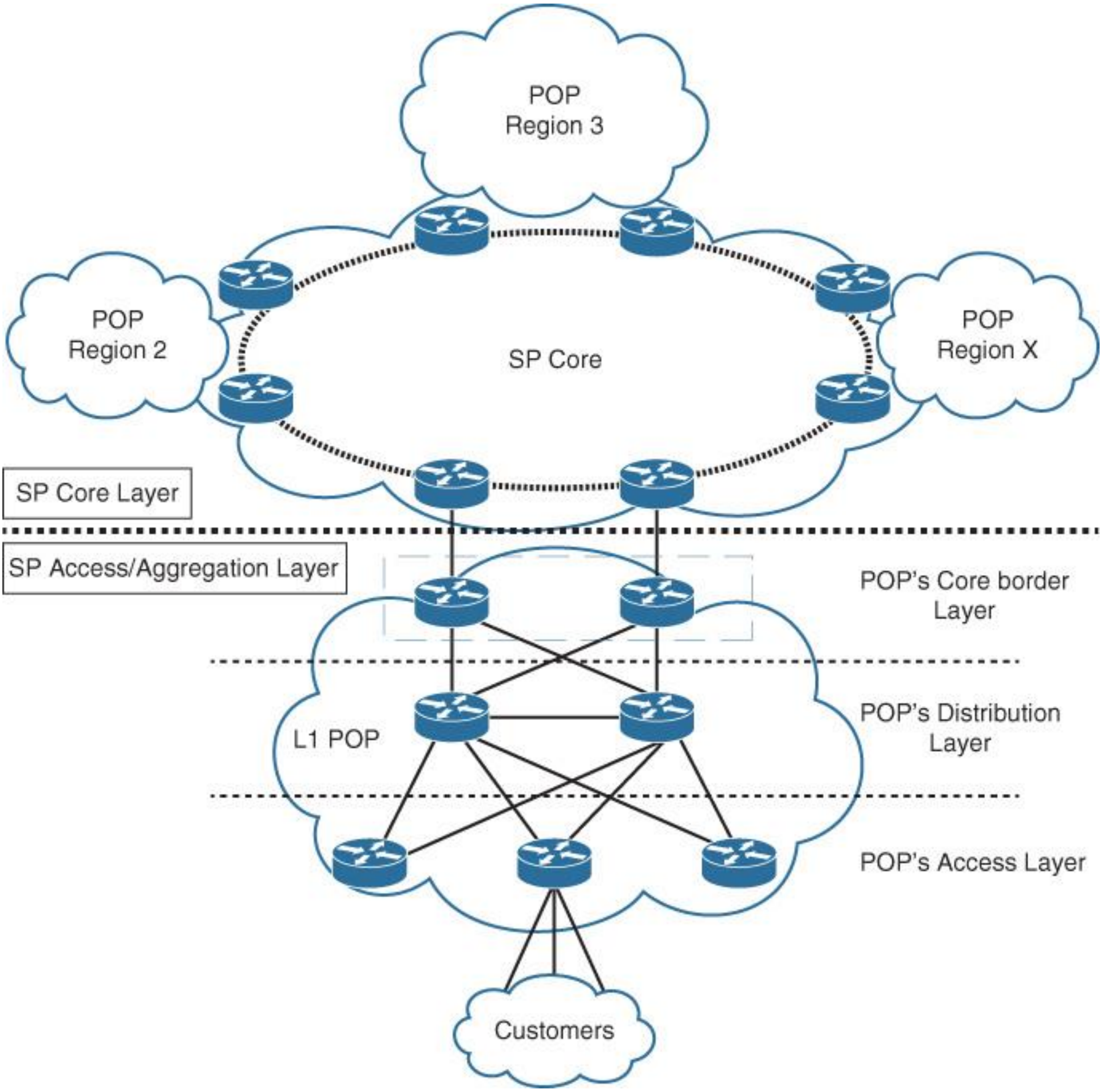

Figure 5-2 *Multitier POP Architecture*

Point of Presence

In SP networks, a POP represents the network edge module in certain geographic areas where customer connectivity terminates. In general, the network design of a POP, which can sometimes be seen from a high-level architectural point of view, is like a small to medium-size LAN design. In this LAN-like design, there can be multiple interconnected Layer 3 nodes over point-to-point links or aggregated through Layer 2 or 3 switches in a hierarchal manner. However, a POP network might span multiple areas in a city over an optical network to cover a metropolitan-area network (MAN). In fact, there is no standard POP design because several factors drive its design and components, such as the following:

- **The services a POP is providing:** For instance, Internet access only, and Layer 3 or Layer 2 service terminations. Layer 2 termination itself can vary, as well. For example, it can be legacy connectivity such as Frame Relay or modern such as Metro Ethernet.
- **The geographic area a given POP is covering:** For example, if it is covering a large city with a large number of customers, the POP might need to be designed in multiple levels to optimally provide the required area coverage and access capacity.

Consequently, the topology of POPs vary and take different forms, such as the following:

- Ring topology (fiber/xWDM [*x* wavelength-division multiplexing] rings)
- Hierarchal topology (two tier or three tier)
- Mix of hierarchal and ring topologies

In general, a typical POP consists of some or all of the following architectural components/layers:

- **Core border:** High-speed and -capacity devices interconnect the POP with SP core. In addition, there are some POPs designed in multiple levels to cover larger geographic areas, with the core border layer aggregating the connections and traffic of smaller POPs within the same geographic area (level 1 and level 2 POPs), as illustrated in Figure 5-3.

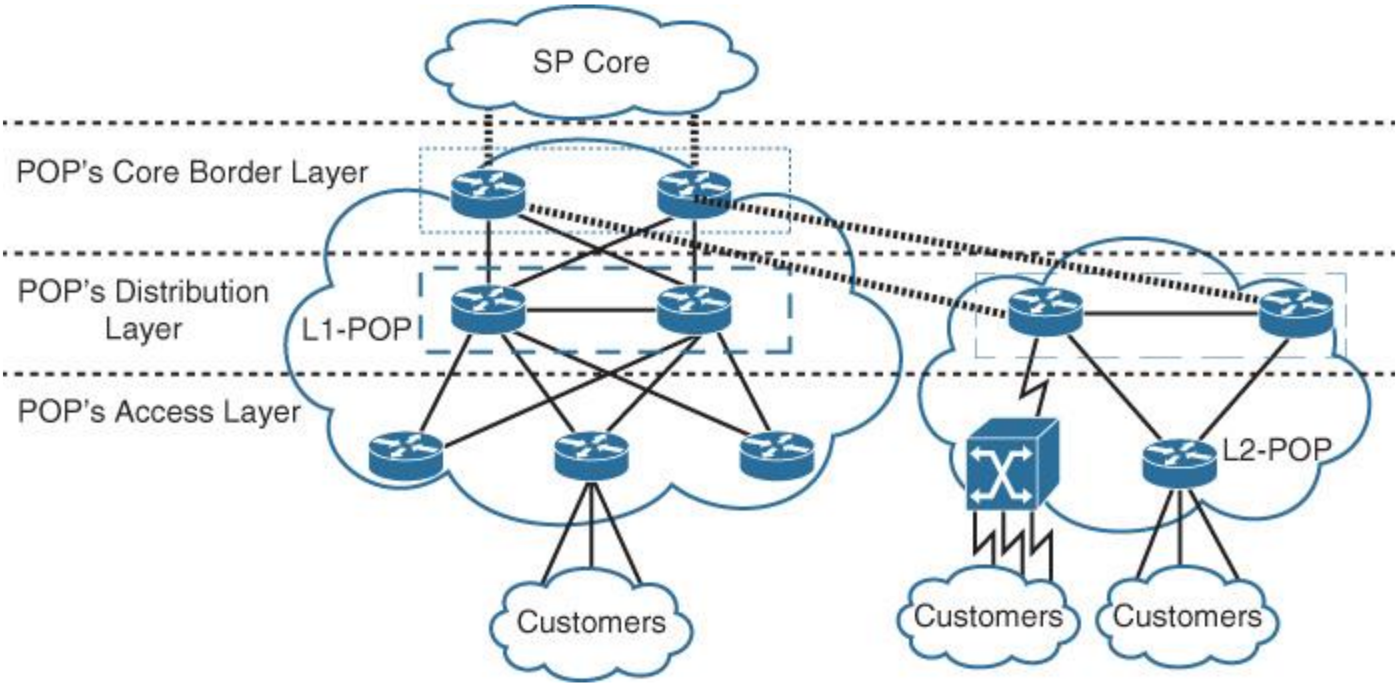

Figure 5-3 *Multilevel POP*

- **Distribution:** The POP distribution layer is required in multitier POP architectures when the number of access layer nodes is large. For example, in the Metro Ethernet POPs in large cities, there might be a high density of access nodes within the POP, and the distribution nodes can optimize the design to aggregate those links, such as with a Hierarchical Virtual Private LAN Services (H-VPLS) architecture.
- **Access:** The access layer represents the termination point and aggregation of the customer's access links and connections. Normally, the access node can take different forms; it could be a PE router in an MPLS VPN environment or a Layer 2 switch in an H-VPLS architecture or a digital subscriber line access multiplexers / broadband remote-access server (DSLAM/BRAS) for DSL services.

Note

As highlighted earlier, there is no one standard connectivity model for the POP network. For instance, sometimes only two tiers per POP is required, where the distribution and core border functions are combined in one tier, This is common in MPLS L3VPN POPs with several PEs in which the distribution/core border layer performs links aggregation, MP-BGP session aggregation (distribution nodes deployed as BGP route reflectors [RRs]), and traffic aggregation, as illustrated in Figure 5-4.

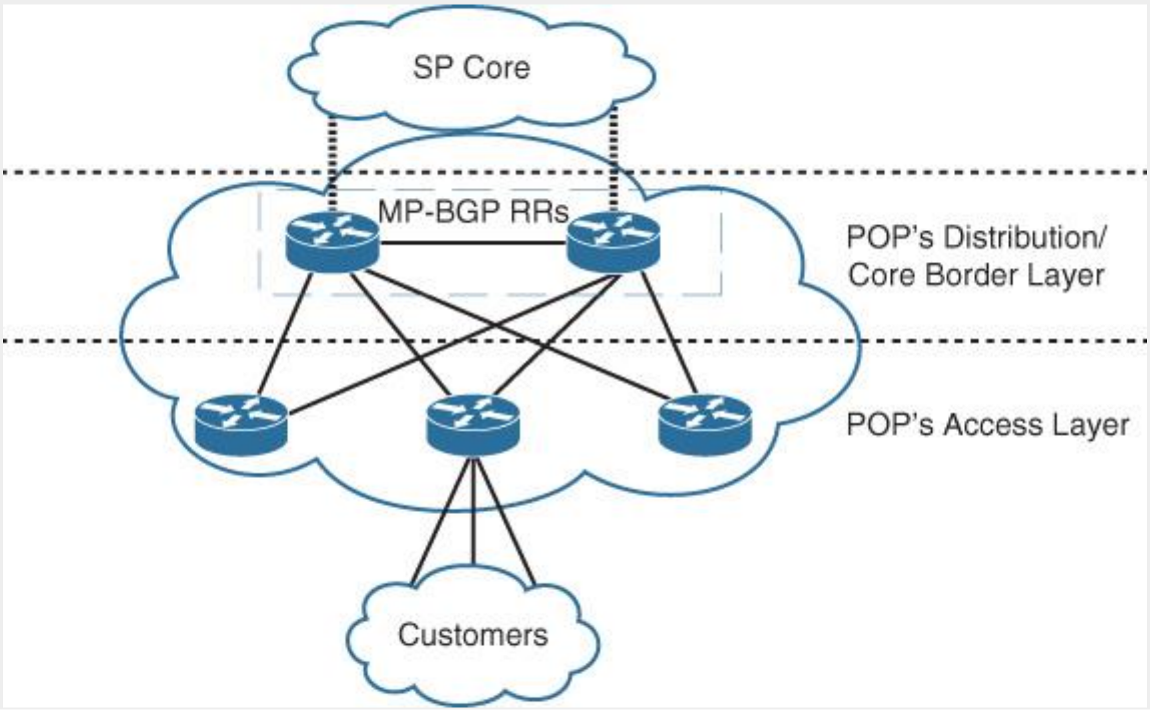

Figure 5-4 *MPLS L3VPN Multitier POP*

In addition, besides the multitier POP connectivity model illustrated in Figure 5-3, Figure 5-5 illustrates another two connectivity models of multiservice POPs that provide different services access such as enterprise Layer 2 Metro Ethernet, L3VPN, as well as residential services access such as IPTV. (Hypothetically, the POP architecture shown in Figure 5-3 should be more able to support large-scale POPs in terms of number of nodes, traffic volume, and geographic coverage.)

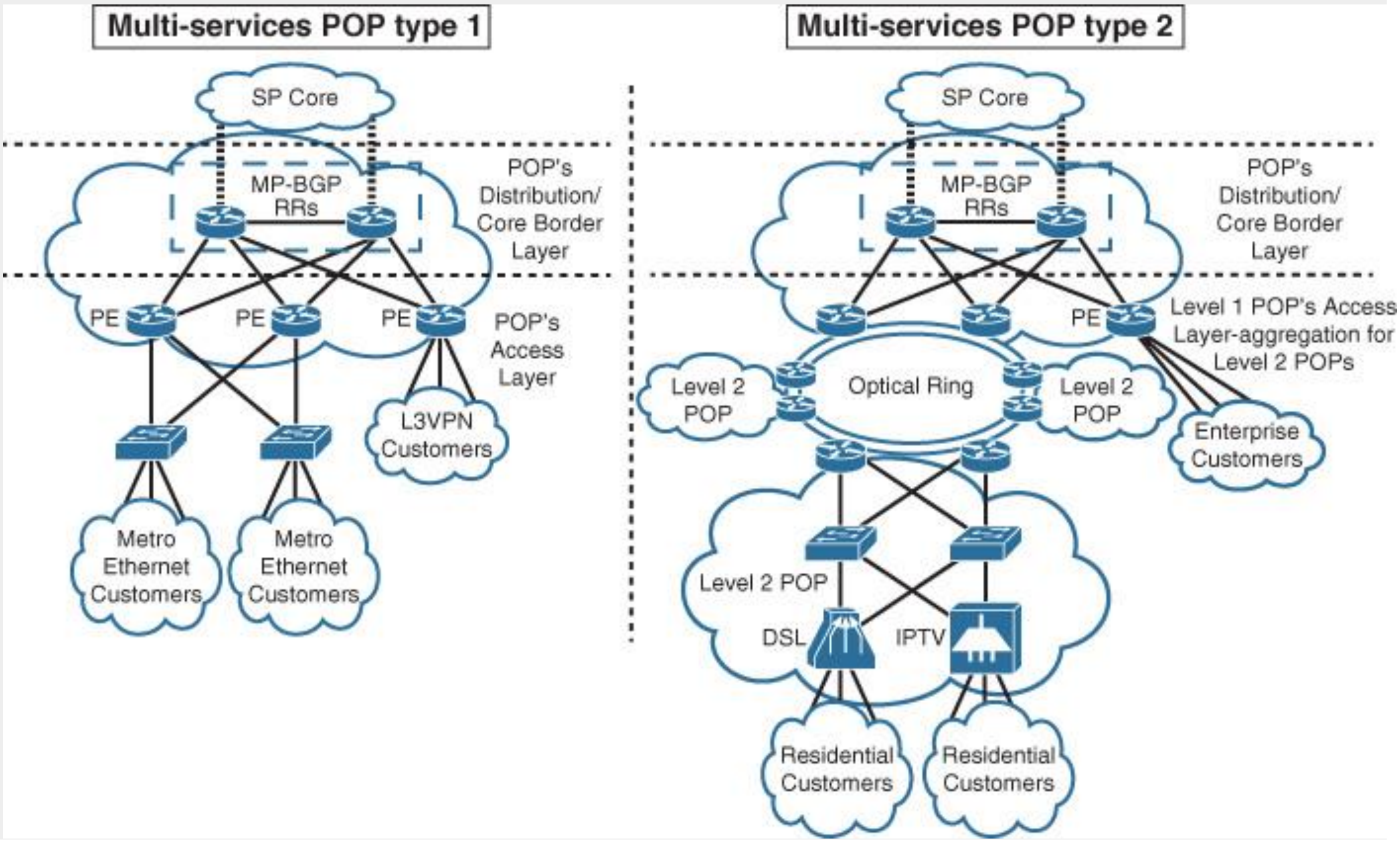

Figure 5-5 *Multiservice, Multitier POP Architectures*

Service Provider Network Core

The concept of the core in SP networks is not different from enterprise networks discussed earlier in this book; however, the core in SP networks from a topological point of view provides the interconnection between the different POPs and regions. Therefore, the underlying physical topology of an SP core may vary depending on the size of the SP, number of customers, geographic locations (national or international), and the services offered (Internet service provider [ISP], MPLS VPN, data center [DC] hosting SP). Even so, the most typical core topologies used in SP-grade networks are either ring or partial-mesh topologies, and commonly the core topology selection is driven by the following factors.

- Geographic area coverage.
- Available physical connectivity in certain areas.
- Cost.
- Primary business services and application requirements. For instance, content providers may invest in long-distance high-capacity optical links, whereas ISPs may use other intermediate providers to offload the cost of extending the core across certain areas.

Service Provider Control Plane Logical Architectures

In today's SP network architectures, several technologies and control protocols independently or collectively transform the network into a "carrier grade" type of network. This section covers the main control plane and forwarding protocols used by modern SPs from a high-level network architecture point of view. The subsequent sections cover the possible design options and considerations in greater detail. This section covers the following SP architecture elements that can all collectively construct the logical overlay and the core transport (forwarding and control plane) of a typical SP network:

- IGP

- BGP
- MPLS
- MP-BGP

Chapter 6, "Service Provider MPLS VPN Services Design," focuses on the following two primary services enabled by the protocols in the preceding list and offered by today's SPs:

- MPLS L3VPN
- MPLS L2VPN

IGP in Service Provider Networks

The principles of IGP routing designs covered earlier in this book, such as topology and reachability information hiding techniques, are all applicable with regard to the physical and logical layouts. However, there are some differences between designing IGP for enterprise networks compared to SP networks, as summarized in the list that follows. Therefore, network designers must identify the nature of the targeted environment when evaluating, planning, and designing the IGP:

- The goal of IGP in SP networks is not to carry customer routes but to carry infrastructure addresses and provide next-hop reachability to BGP. (BGP is the de facto protocol of SP networks, regardless of whether MPLS enabled or non-MPLS.)
- The IGP protocol for the SP core networks is typically either Open Shortest Path First (OSFP) or IS-IS for several known reasons, such as supporting MPLS-TE.
- In SP networks, IGP routes must be contained within the SP network only and are not to be exchanged or leaked with any customer network.
- In addition to external BGP (eBGP) and static routing, IGP is also used by some SPs as a PE-CE (customer edge) routing protocol. This IGP is intended to extend customer Layer 3 domain toward to the SP network (virtually separated per customer at the directly connected PE), and it is not an extension of the SP core IGP. (Chapter 6 discusses MPLS VPN designs in more detail.)

Despite the simplified IGP design of SP networks compared to enterprise networks, several factors may influence the logical design of the IGP in an SP network. For instance, breaking IGP flooding domains may optimize the IGP design and facilities, enhancing the scalability of the network. However, at the same it may add complexity to the design of other protocols, such as MPLS-TE. More details are covered later in Chapter 6. (See the section "Service Provider Control Plane Scalability Design Options and Considerations" in that chapter.)

BGP in Service Provider Networks

In SP networks—such as ISPs that have either no MPLS enabled across the network and traffic forwarding is based mainly on native IP or MPLS enabled across the network—BGP (iBGP) is the de facto protocol that carries all the routes that do not contribute to the internal SP IGP routing, such as the Internet routing table and customer-assigned addresses. This concept applies to both IPv4 and IPv6.

The main difference between BGP design in ISP networks and large enterprises is the number of the prefixes and the scale of the network size that BGP needs to handle. Accordingly, scalability within a manageable size is one of the primary goals of SP networks. However, BGP scalability design considerations, discussed earlier in "Chapter 2", using BGP RR and confederations are still all applicable to the BGP design in ISP-grade networks. Furthermore, later in Chapter 6, in the section "Service Provider Control Plane Scalability Design Options and Considerations," you will learn about additional scaling techniques that are more SP related, which can be considered to scale BGP design, such as multitier RR architecture.

In fact, the SP business by nature is a revenue-driven business. Therefore, SPs always aim to satisfy service-level agreement (SLA) requirements with their customers. In addition, network designers must keep in mind that any design decision that may lead to negatively impacting customer satisfaction must always be avoided, even if the proposed design option might (technically) offer significant benefits. BGP, as the primary control plane protocol in SP networks, can be a significant influencer. For instance, some BGP design decisions, such as route aggregation, hot-potato routing, and cold-potato routing, can have a direct impact on the SP business in terms of losing or generating revenue. The subsequent sections highlight how these technical design decisions may impact the SP's business.

BGP Route Aggregation (ISP Perspective)

One of the primary revenue-generating services of today's SPs is Internet access provision. Normally, Internet SPs peer with other peering or transit providers to build their Internet reachability. In ISP networks, it might seem easy to handle routes within and between peering ISPs. However, sometimes simple design decisions, such as aggregation, can make a significant difference in the overall network performance and may impact the ISP business. For instance, from the ISP perspective, it is sometimes advised to advertise the aggregate of the prefixes that are owned by the ISP and allocated to its customers. However, the question to be asked here is this: Will this make a tangible difference? And how will this decision impact the ISP business? Consider the scenario depicted in Figure 5-6.

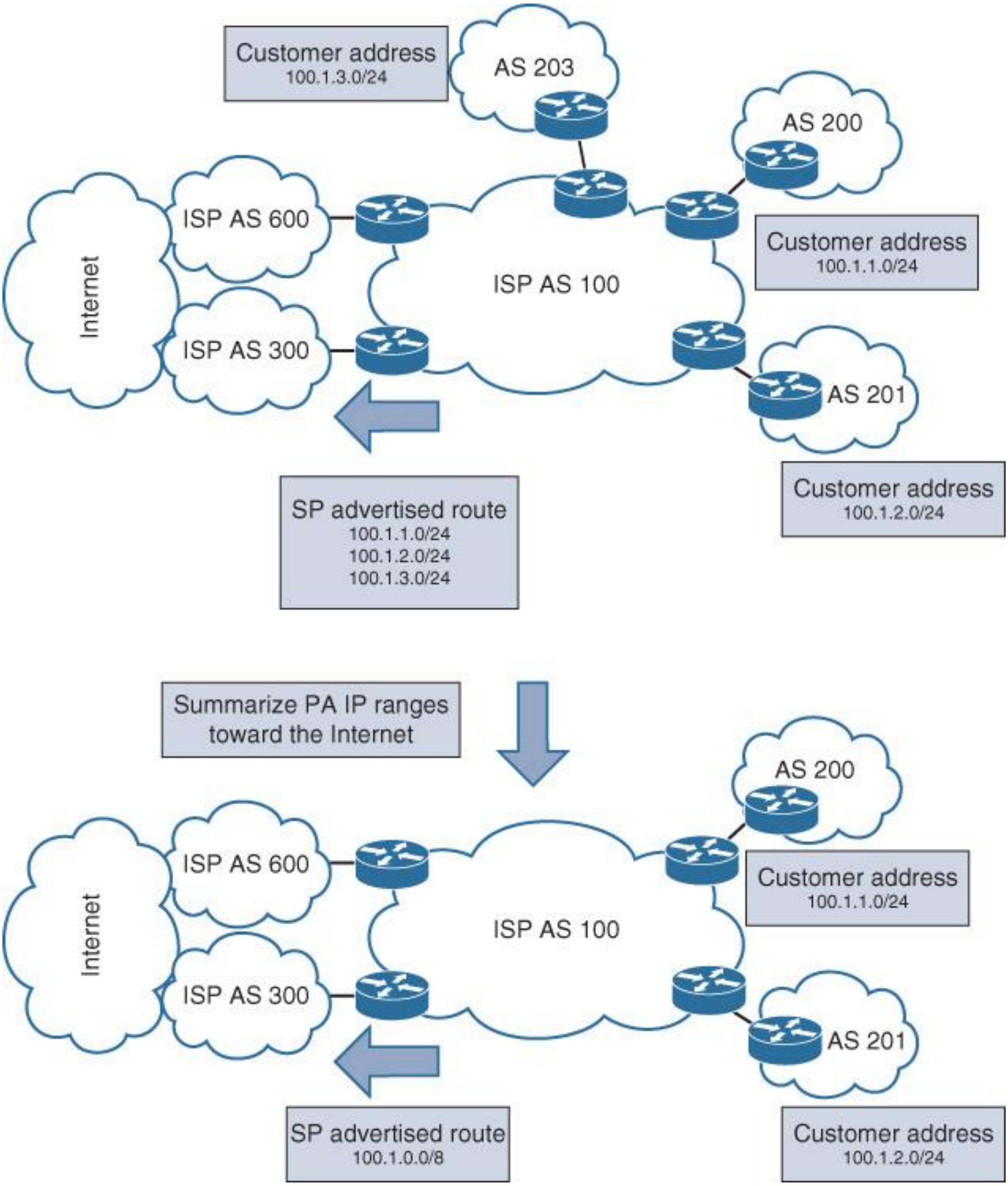

Figure 5-6 *The Impact of BGP Aggregation Between Peering ISPs*

Note

The subnet length advertised between the peering ISPs presented in this section (such as /24) is only to simplify the illustration of the advantages or implications of the decision to enable BGP aggregation or not, at the peering boundary points. However, some ISPs may not accept a subnet like /24 due to "net policing" they use with their peering ISPs. This filtering is commonly based on the Regional Internet Registry (RIR) minimum IP subnets allocation (/20). Consequently, several ISPs that enable net police filtering at their peering boundaries with other ISPs will drop or filter any advertised provider-assigned IP prefix larger than /20.

The scenarios illustrated in Figure 5-6 are common in ISP peering types of networks. The bottom scenario in this figure represents an optimized route's advertisement between AS 100 and its peering ISPs (AS 300 and 600), where the ISP (AS 100) summarizes the provider-assigned/aggregatable (PA) IPs allocated to its customers toward the Internet (typically toward the other peering ISPs). With this approach, any link failure or flapping with any of its customer links will not propagate this failure event information to other peering ISPs (fault isolation based on reachability information hiding principle). As a result, BGP aggregation in this scenario can optimize the overall performance of the network and over the Internet, and can provide a better quality of experience for their customers. It can also meet a tight SLA in terms of Internet reachability following a recovery from a failure event. In other words, the BGP design decision in this scenario (summarization) can be seen as a "business enabler."

Note

BGP PA IP range summarization by this ISP can specifically optimize customer experience following a PE-CE link failure event, because this ISP advertises only the summary. When the customer link recovers from the failure, there is no need for BGP to converge across the Internet (across the multiple peering ISPs) for this customer to be reachable over the Internet. This will also help to avoid BGP "path hunting" behavior in which a prefix withdrawal may create a major flap event over multiple AS hops away in the Internet.

However, in this particular scenario, if the customer is connected to a second ISP (multihomed) using its same PA IP range allocated by the first ISP (AS 100) and the customer aims to use the second ISP as a backup path only (for example by using AS-PATH prepending over the backup path), this will result in an undesirable outcome, regardless of what BGP attribute they use to influence BGP path selection. This is simply because the second ISP does not own this PA IP range; therefore, it cannot aggregate this range. This will usually lead to a situation where the Internet (other peering ISPs) will see a more specific route (longest match) via the second ISP, which will be the preferred path regardless of what BGP attribute has been used, as illustrated in Figure 5-7.

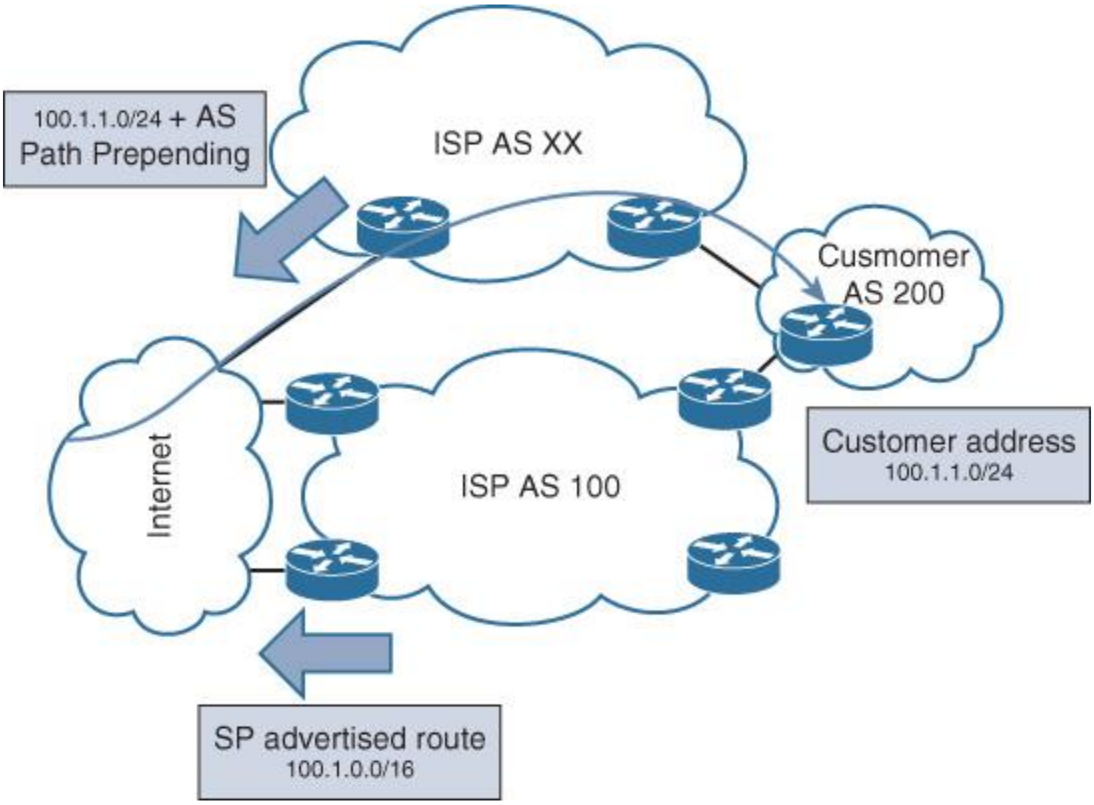

Figure 5-7 *ISP Summarization and Suboptimal Routing*

To avoid situations like this, customers commonly use a provider-independent (PI) IP range, which the ISP does not summarize; alternatively, address summarization can be negotiated with the ISP to send more specific prefixes, as illustrated in Figure 5-8.

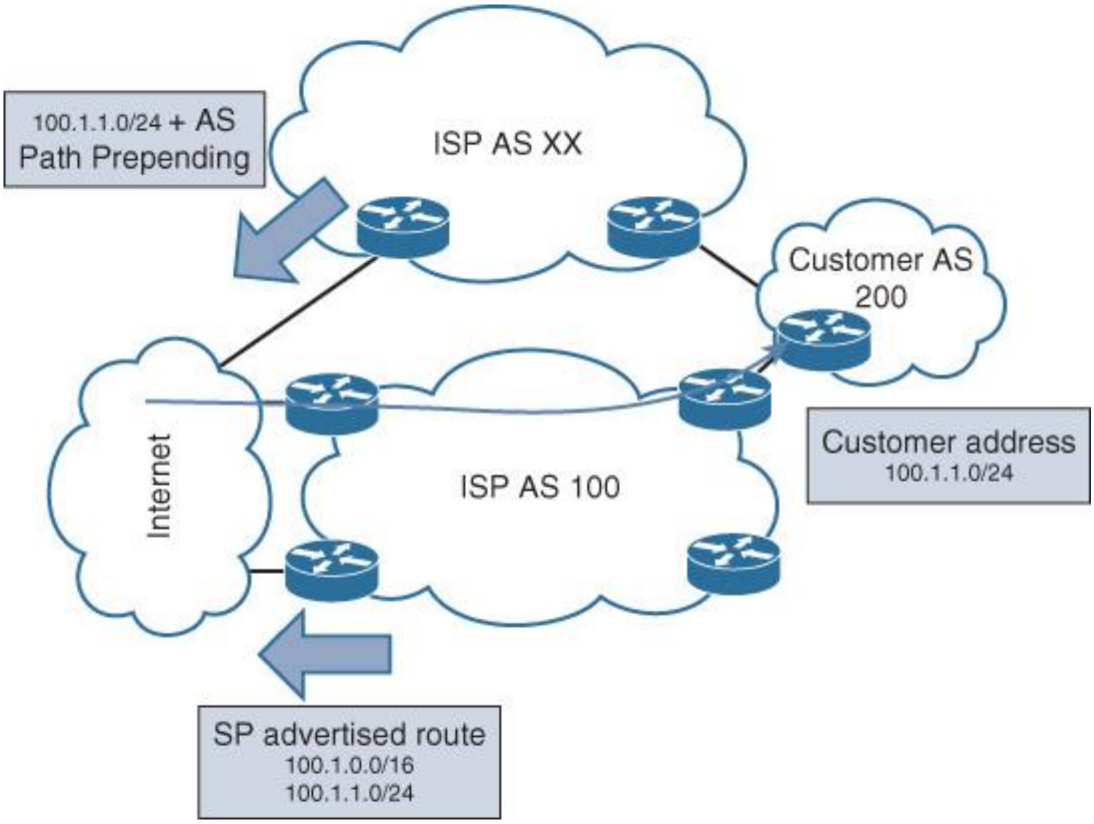

Figure 5-8 *Optimal Routing over Peering ISPs*

Note

Sending more-specific customer routes (PA IPs) to peering ISPs is not always achievable, because of the net police route filtering used between ISPs (discussed earlier).

Note

Traffic originating from customers of the second AS in this example may still prefer the path via the local AS if this AS assigned higher BGP local preferences for routes learned from its customers, as discussed in the section "Enterprise Multihomed Internet Design Considerations" in Chapter 4.

However, sending more-specific prefixes may lead to a longer recovery time (BGP convergence across the Internet also will be impacted by BGP path hunting behavior) following a failure event. This will be a concern from the ISP if there is a strict SLA they have to meet with regard to service recovery time, considering that what is a priority for the business has to always be given preference and the technology must serve as a facilitator.

This was an example of how BGP design decision can directly impact the ISP business in terms of performance and customer satisfaction.

Hot- and Cold-Potato Routing (SP Perspective)

One of the common methods to forward transit traffic in SP networks (commonly ISPs) is by sending the traffic to the closest exit point considering IGP metrics/cost to achieve traffic fast forwarding. This approach is commonly known as*hot-potato routing*. The name of this approach exemplifies the situation where a hot potato is held by someone in the hands, and he/she quickly passes it to the next person. Anyone holding the hot potato will attempt to pass it to the next person as quickly as possible. Hot-potato routing is the typical behavior on the Internet.

In Figure 5-9, the direct path via the western region represents typical hot-potato routing. One of the main benefits of this traffic routing approach is the fast forwarding of transit traffic from the point of entry to the closest exit point. This can be cost-effective to those SPs that have an agreement to use other providers' physical networks between cities to forward transit traffic in scenarios where adding more bandwidth involves paying extra cost. However, when there is a BGP RR in the path, this behavior may be broken.

Typically, an RR in a BGP environment will advertise the best path from its point of view of the topology, which might not be the closest exit point from a given PE/border router point of view. This BGP RR path hiding behavior can prevent efficient use of BGP multipath, lead to multi-exit discriminator (MED) oscillations and suboptimal hot-potato routing, as illustrated in Figure 5-9 (the path via the northern region) [34].

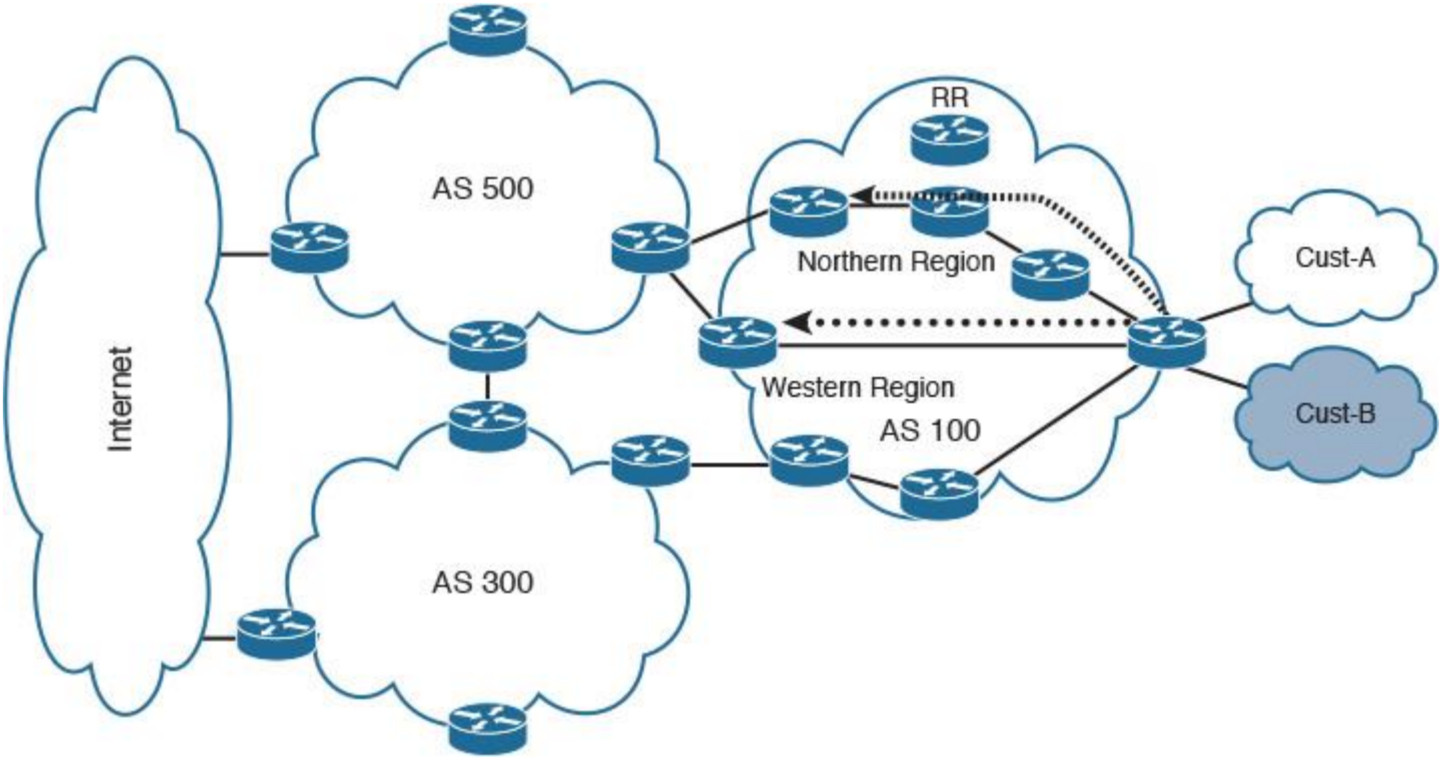

Figure 5-9 *BGP RR and Hot-Potato Routing*

Multiple options can be considered to overcome this behavior (this behavior might be undesirable for some requirements and not always), such as the following:

- BGP ADD-PATH capability
- BGP diverse paths (RFC 6774)
- BGP partial mesh of iBGP sessions in addition to existing sessions to the RR, to add the optimal path for those edge routers that need to be used for hot-potato routing

BGP ADD-PATH

ADD-PATH is a BGP feature that allows BGP speakers to advertise more than one path over a single neighbor session. This capability can prove useful when there is an RR in the BGP architecture to send multiple paths to the RR clients (overcomes RR issues when hot-potato routing in a transit network is required), as illustrated in Figure 5-10. However, some considerations must be taken into account when considering this feature as a design option:

- All routers may require software upgrade to support this feature.
- Network designers need to identify how many paths and what paths to announce.
- There will be additional memory overhead on the receiving PE because of the additional paths and the increases in CPU utilization (updates processing and other internal processing such as next-hop trigger).

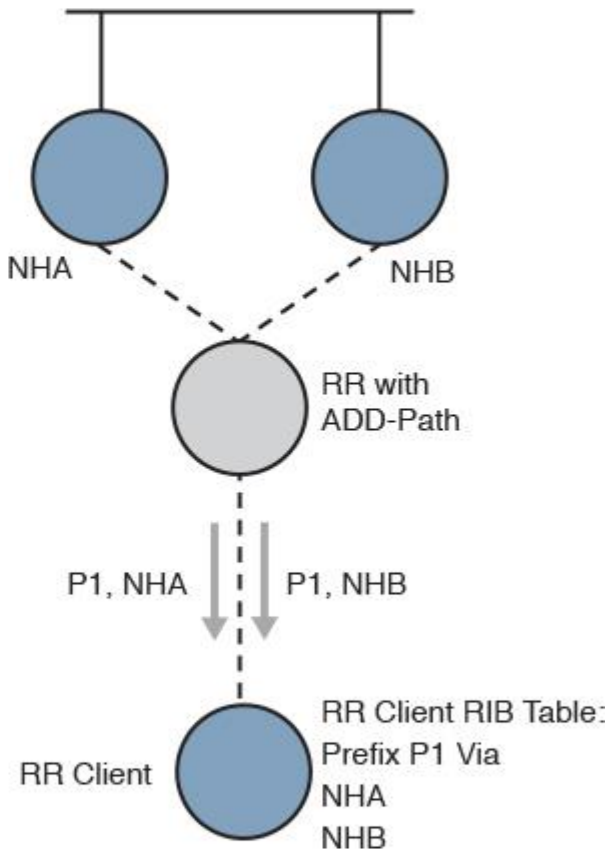

Figure 5-10 *BGP ADD-PATH*

BGP Diverse Paths

With this approach, BGP can achieve the same goal of what ADD-PATH does; however, it does not require an upgrade of the entire network or any protocol changes. Diverse paths can be announced using two deployment models, as illustrated in Figure 5-11:

- Single RR with a second iBGP session to the PE's "shadow session" that announce only diverse paths (advertises next best path)
- Adding a second RR to the cluster, where the second RR "shadow RR" calculates and advertises the diverse paths (next best path to the same prefix with a different next hop)

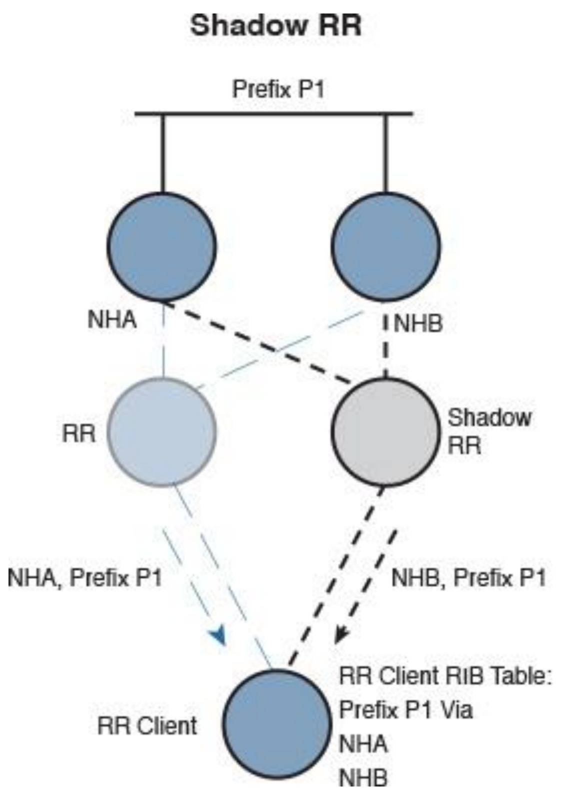

Figure 5-11 *BGP Diverse Path*

In addition, this model can easily integrate in a large-scale “multitier BGP RR” network, as illustrated in Figure 5-12.

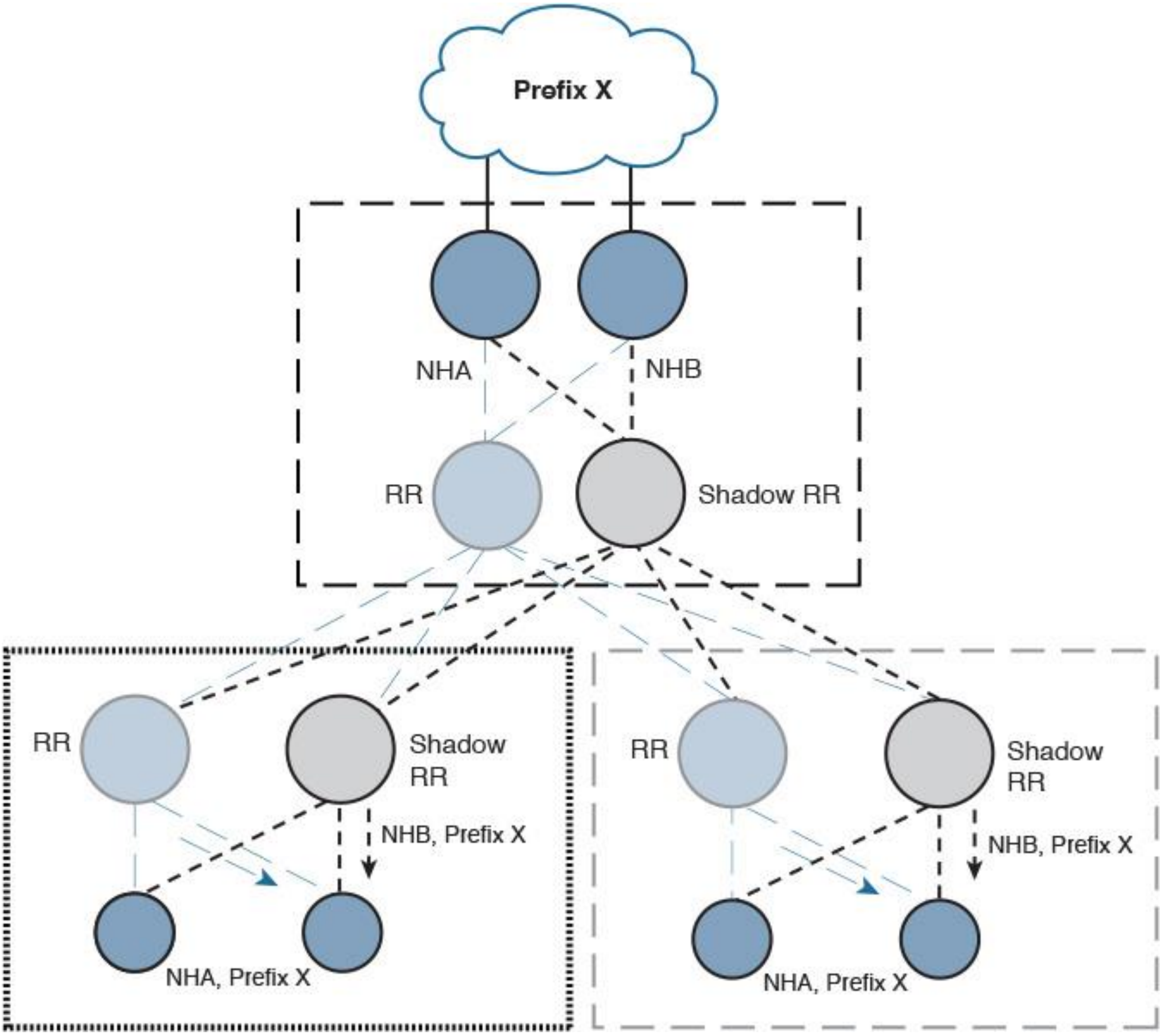

Figure 5-12 *BGP Diverse Path with in Multitier BGP Design*

However, some design aspects need to be considered with regard to this design approach:

- The required additional sessions to enable diverse paths
- The required additional memory to store diverse paths

Note

In general, the goal of the preceding two options is to send more than one best path known by the RR to the iBGP speaker. The iBGP speaker would then have these paths and can install them in the Routing Information Base (RIB) table if the multipath option is enabled. Routes could also be programmed into the Forwarding Information Base (FIB) as an alternate backup path if BGP prefix-independent convergence (PIC) is enabled. Nevertheless, these two options are not analogous. For instance, with the "diverse path," the RR iBGP client receives only two paths per prefix (best and the next best path). In cases of hot-potato routing, where an ISP's network has multiple exit points for the same prefix, let's say five equal-cost multi-path routing (ECMP) routes, and you want to use all of them, the ADD-PATH option with *N* paths is more feasible in this case compared to the diverse path option.

BGP Partial Mesh

As illustrated in Figure 5-13, in some scenarios a partial mesh can be considered an option as well to achieve optimal hot-potato routing when there is BGP RR in the BGP topology; however, this approach may lead to scalability limitations if many border routers need a direct iBGP session.

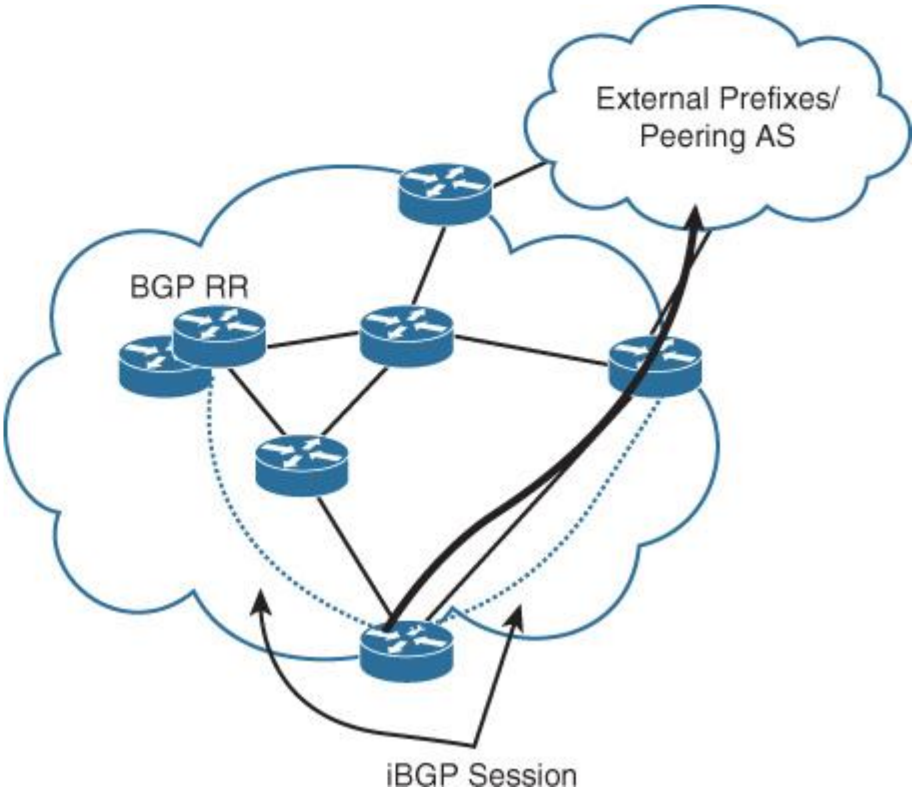

Figure 5-13 *BGP Partial-Mesh Sessions*

Although all these options are valid approaches, the decision of selecting the optimal option depends on the environment and some other design aspects such as the network scale and performance impact. Table 5-1 compares these options from different design perspectives.

Design Consideration	BGP ADD-PATH	BGP Diverse Paths	BGP Partial Mesh + RR
Software/code upgrade to enable the capability	Yes (network-wide)	Yes, partially (BGP RR only)	No
Required BGP sessions to receive multiple paths	1	Two	Depends on the network
Number of paths	2 or more (configurable)	2 paths *	Depends on the iBGP sessions + the RR BGP session
Scalability	Moderate	High	Moderate
Manageability	Simple	Simple	The more sessions, the more complex the network
Hardware resources utilization (memory, CPU)	Moderate to high (depends on the number of paths and prefixes)	Moderate	Moderate to high (depending on the number of iBGP session besides the BGP session to the BGP RR)

* Assuming two upstream BGP RRs used per BGP RR client to provide the diverse path functionality.

Table 5-1 *Hot-Potato with BGP RR Optimization Options*

Note

You can use the unique RD approach for similar situations in MPLS VPN environments (see Chapter 6 for more details). Furthermore, placing the RR physically close to eBGP speakers (at the AS boundary) along the edge of the network would allow traffic to more accurately follow the IGP metrics when destined toward external prefixes. Although technically this can be a valid option, practically it will not scale in large SP environments, especially when there are multiple boundary points (such as peering with multiple external networks/SPs).

Hot-potato routing is seen as a good approach for transit providers to route transit traffic as quickly as possible over the closest exit point; however, this behavior may lead to undesirable customer experiences in some situations, which can affect the SLA between the provider and its customers. For example, the customer with AS 500 in Figure 5-14 has an SLA with ISP AS 100 to offer a certain level of Internet bandwidth; however, forwarding the traffic over to AS 300 to reach the Internet using the hot-potato approach will lead to sending AS 500 customer traffic over the low-bandwidth link of AS 300. This will result in an undesirable experience for this customer. Therefore, in this situation, cold-potato routing can be a more efficient solution.

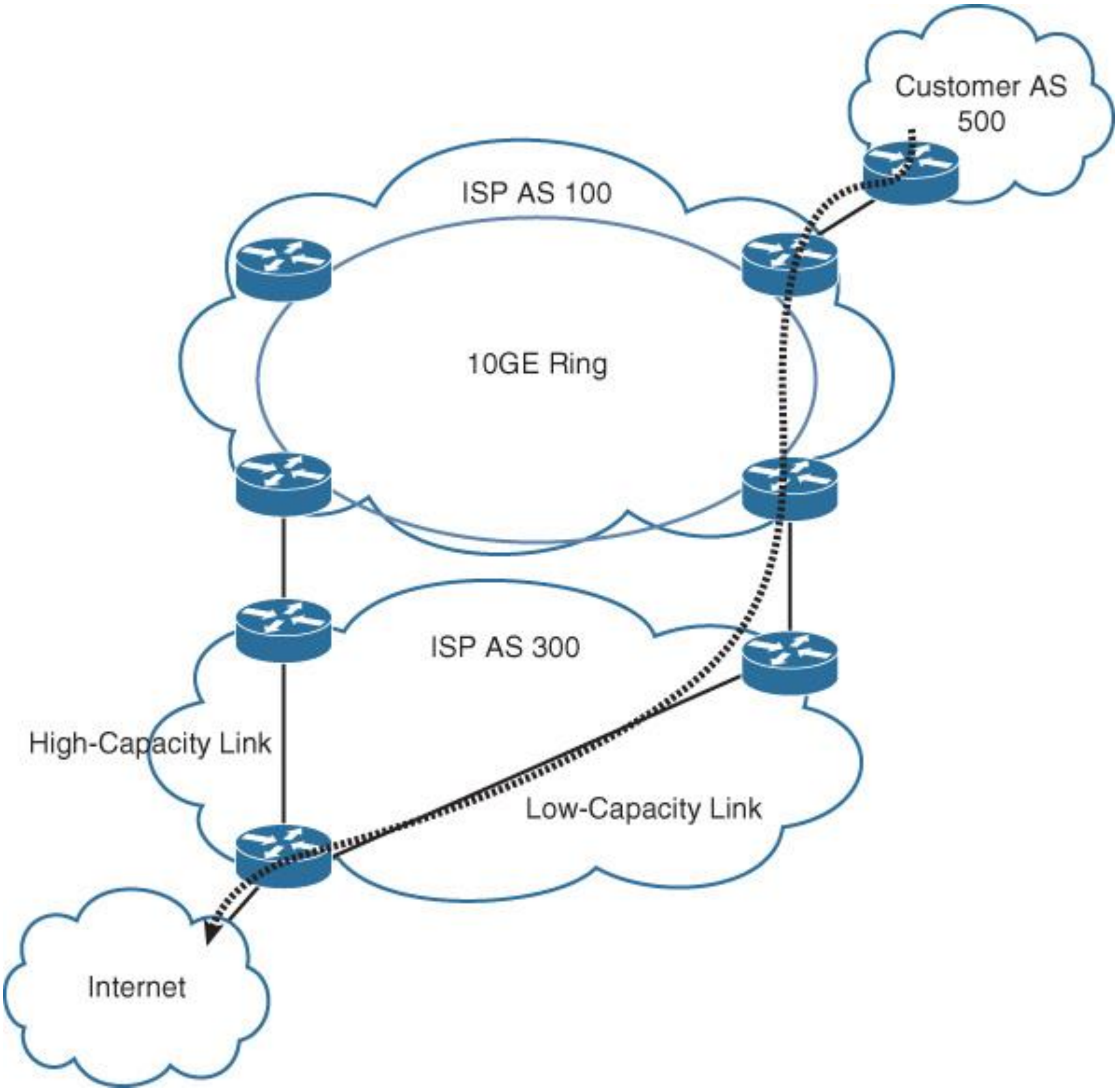

Figure 5-14 *Hot-Potato Routing Limitation Scenario*

Cold-potato routing, however, tends to keep traffic within the same AS until reaching the closest exit point to the destination. In other words, cold-potato routing aims to hold a packet within the AS from its entry point to the AS. This approach offers more granule "end-to-end" control of packet forwarding because they will have visibility and control over the end-to-end path. In addition, this approach can be beneficial to data center hosting and content SPs because they can control and offer better quality, which will lead to a better SLA and more satisfied customers. SPs normally use BGP attributes such as local_preference to keep the path within the same AS as the preferred one, in addition to BGP MED and the accumulated IGP metric attribute for BGP AIGP. However, cold-potato routing can be an expensive approach because SPs might be required to use their own intercity or international links more often for transit traffic. Sending transit traffic over other providers' infrastructures (interprovider) might offer a cheaper rate (cost versus quality).

Therefore, the decision to adopt a hot-potato or cold-potato routing approach with the SP network has to be aligned with the business priorities, constraints, and requirements. This alignment will achieve a successful business-driven design decision, which is not solely based on technical best practices recommendations.

Multiprotocol Label Switching

MPLS is a standard IETF technology, defined in a multiple RFCs, and can be described literally as a Layer 2.5 networking protocol. Typically in the OSI model, Layer 2 refers to protocols like Ethernet, which normally carries IP packets over a LAN or WAN network, while Layer 3 on the higher layer of the Open System

Interconnect (OSI) controls the routing of packets over Internet Protocol (IP routing control plane). In contrast, MPLS is provisioned between these two layers, offering additional capabilities for the transport of data across the network. MPLS offers a less-hardware-intensive mechanism to forward traffic based on label switching.

In today's SP networks, MPLS is driving the network overlay concepts that enable it to support multiple technologies and protocols over one common physical infrastructure. Consequently, MPLS-based SP networks can offer various services and are more responsive to market trends and changes by relying on the high degree of flexibility and simplicity provided by MPLS to the network to introduce new services.

The following are the primary drivers for SPs (also large-scale enterprises) to adopt and invest in MPLS-enabled core infrastructure:

- Accelerate time to the market. MPLS offers the ability to enable the delivery of different transport services across the same packet-switched network infrastructure (for example, L2VPN, L3VPN, and interworking between legacy and modern Layer 2 services [ATM to Ethernet]). This can enable SPs to introduce new services that meet market trends and customer needs in a timely manner using the same underlying core infrastructure.
- Offer differentiated services and meet tight SLAs. MPLS offers the ability to enable a source-based routing mechanism (MPLS-TE) that can control where and how traffic is routed across the network to manage capacity and congestion, and prioritize different services and path protections that can meet various strict SLAs requirements.
- Enhance network resiliency with MPLS FF (MPLS-TE, RLFA).
- Reduce operational complexity because the different control plane protocols that serve different services will be transported over one core infrastructure (multiservice infrastructure) such as L3VPN and L2VPN. This means that there will be fewer technologies and protocols to support, which translates into a lesser amount of expertise required.

Consequently, by adapting a service convergence strategy, SPs can decrease the time to the market to introduce new services and capabilities to their customers. Furthermore, it helps to significantly reduce operational expenditures (OPEX) by managing one multiservice-aware IP MPLS core rather than multiple separate networks that require different expertise and increase troubleshooting complexity (for instance, the combination of legacy services like ATM, Frame Relay, time-division multiplexing [TDM], and IP). However, the MPLS core can still support legacy services at the edge to maintain and protect current customer investments and to provide access diversity, which can offer a high degree of service access flexibility to enterprise customers with the least capital expenditure (CAPEX) by using same existing infrastructure in scenarios where minimizing cost is one of the highest business priorities.

MPLS Label-Switched Path

Label-switched path (LSP) is a fundamental requirement for any MPLS forwarding to occur. Typically, LSP is routed across an MPLS network via forming a unidirectional tunnel between any two routers,. The ingress node or router that first encapsulates a packet inside an MPLS LSP in MPLS terms is called the *label edge router* (LER), and the node that performs only MPLS switching in the middle of an LSP is called the *label-switching router* (LSR). Finally, the LSP termination node is the egress node where the MPLS label is removed (popped).

In an MPLS network, an MPLS signaling protocol performs the mapping of LSPs to specific label values. Two primary MPLS signaling protocols are in use today:

- **Label Distribution Protocol (LDP):** A simple nonconstrained protocol
- **Resource Reservation Protocol (RSVP) with Traffic Engineering (RSVP-TE):** A complex protocol compared to LDP, with more overhead, and most commonly used for MPL-TE setup

In today's SP networks, both protocols are actually used. The LDP is typically used for MPLS VPN and other services, and RSVP-TE is used for traffic engineering services.

MPLS Deployment Modes

The typical and most common deployment mode of MPLS is the frame mode. In the frame mode, the MPLS label is assigned between Layer 2 and Layer 3 headers (Layer 2.5), where Layer 2 can be a standard Ethernet or a legacy Frame Relay. As illustrated in Figure 5-15, this mode provides flexibility to SPs to maintain the legacy access media type if required. This connectivity model is most commonly used between the PE-CE links or between a level 2 POP/PE and level 1 POP (a regional POP), whether as a permanent or as a temporary solution (during the migration from FR to IP).

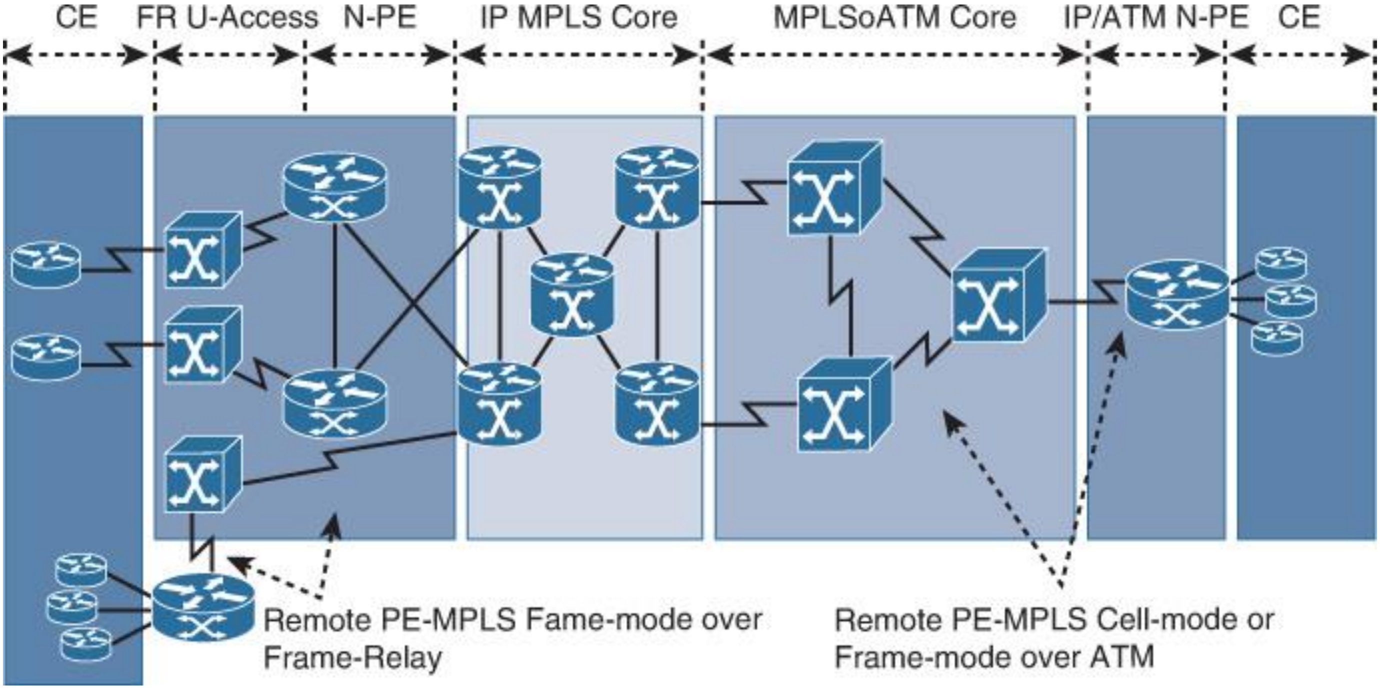

Figure 5-15 *MPLS Modes*

MPLS cell mode, however, is specifically developed and used in MPLS over ATM environments where ATM switches are used (typically to forward data based on virtual circuits [VCs]). In this mode, the MPLS label value is driven from the ATM virtual path or channel identifiers. This mode is commonly used when the core ATM network cannot be upgraded for some reason, such as limited budget. From the design point of view, this can be considered a design constraint for network designers that must be dealt with.

Accordingly, MPLS cell mode offers network designers the flexibility to deliver MPLS over ATM infrastructure. This can provide the business with the various benefits of MPLS-enabled infrastructure highlighted earlier in this chapter without any additional CAPEX to upgrade the underlying hardware infrastructure, taking into consideration its scalability limitations and slower overall convergence time, along with the increased operational complexity (this is commonly known as *multiple operations support systems*, which can lead to operational inefficiency associated with increased OPEX). As a result, some businesses, over the long term, may end up spending more on OPEX in an attempt to save on CAPEX. In other words, a network architect with good strategic planning can justify the suggested solution (for instance, to migrate to pure end-to-end MPLS and IP only) if the proposed or intended solution is for the long term, as illustrated in Figure 5-15.

Note

Scenarios where the core (transit) ATM switches either do not support MPLS (cell mode) or are in a transition period to complete IP-based infrastructure MPLS frame mode over ATM can be a viable choice.

Multiprotocol BGP

BGP is extended to provide multiprotocol support (MP-BGP) that enables BGP to carry routing information for different network layers and address families. In modern SPs, MP-BGP, along with MPLS, is mainly used to facilitate the control and forwarding of MPLS VPNs. This is commonly where private Layer 2 and Layer 3 for both unicast and multicast network services can be delivered over a common MPLS- and MP-BGP-enabled infrastructure, which can significantly optimize control plane design and operational complexity. Logically, MPLS in SP-grade networks represents the underlay control protocol across the SP transport network, and MP-BGP is the overlay control protocol that hosts and controls customer prefixes. Furthermore, in modern SP-grade networks, MP-BGP is used not only for Layer 3 IP routing information exchange; it is also used for Layer 2 VPN service as an autodiscovery mechanism and Layer 2 MAC routing information (as described in IEFTF draft "draft-ietf-l2vpn-evpn-11"), where the MAC learning between PEs happens at the control plane level. Therefore, MP-BGP plays a vital role as its the primary control protocol for the main services offered by today's SPs (such as Layer 3 and Layer 2 VPN services). Chapter 6 covers these MPLS VPN services in more detail. Figure 5-16 illustrates the building blocks of the various services and protocols that are enabled by MPLS and controlled on a higher level by MP-BGP.

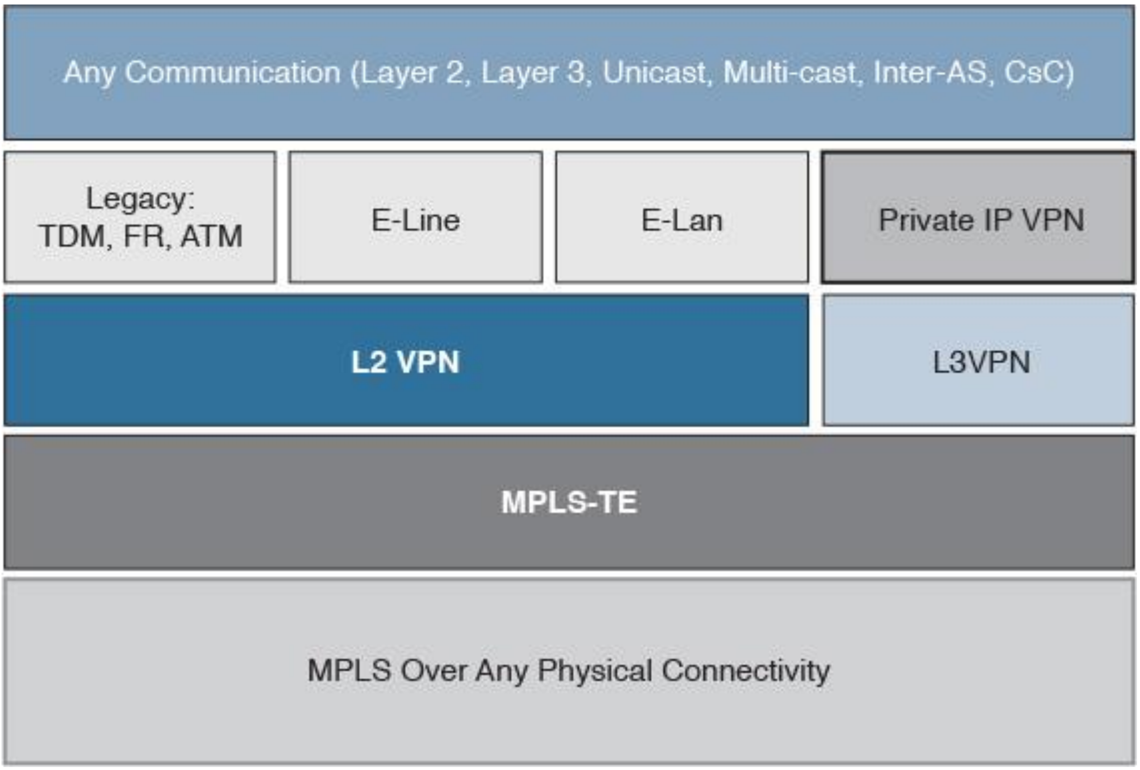

Figure 5-16 *MPLS-Enabled Service Building Blocks*

MPLS Traffic Engineering

Multiprotocol Label Switching (MPLS) Traffic Engineering is a constrained source-based routing mechanism that helps network operators to optimize traffic routing across MPLS-enabled networks based on multiple advanced path attributes that is impossible to achieve with the classic IP routing only. This section covers the drivers toward adopting MPLS-TE to act as a business enabler, along with the different MPLS-TE design considerations, advantages, and the associated implications.

Business and Technical Drivers

One of the common questions that network architects and designers face is this: Do we need MPLS Traffic Engineering and why? Typically, a good network design must accommodate traffic requirements in terms of traffic volume and traffic pattern, considering its characteristics as well, such as the delay of sensitive traffic. However, in reality, traffic requirements in many situations change over time (normally after years of the initial network deployment). This change results from several variables, such as business organic growth, merger and acquisition, or the business's adoption of new services and applications. This is typical in SP networks; after all, the number of customers may significantly grow within a few years. Consequently, a network designed for yesterday's traffic requirements may not optimally deliver today's traffic requirements.

Furthermore, many businesses cannot afford to upgrade their core infrastructure within a short period of time. (This might be cost related, or the business cannot afford upgrade outages and so on.) Consequently, network designers need to incorporate a solution that can offer the business the flexibility to overcome these limitations with minimal infrastructure changes or upgrades. For instance, in the scenario depicted in Figure 5-17, an SP based in Australia has five primary POPs (Sydney, Brisbane, Melbourne, Adelaide, and Perth) with different intercity optical link capacities.

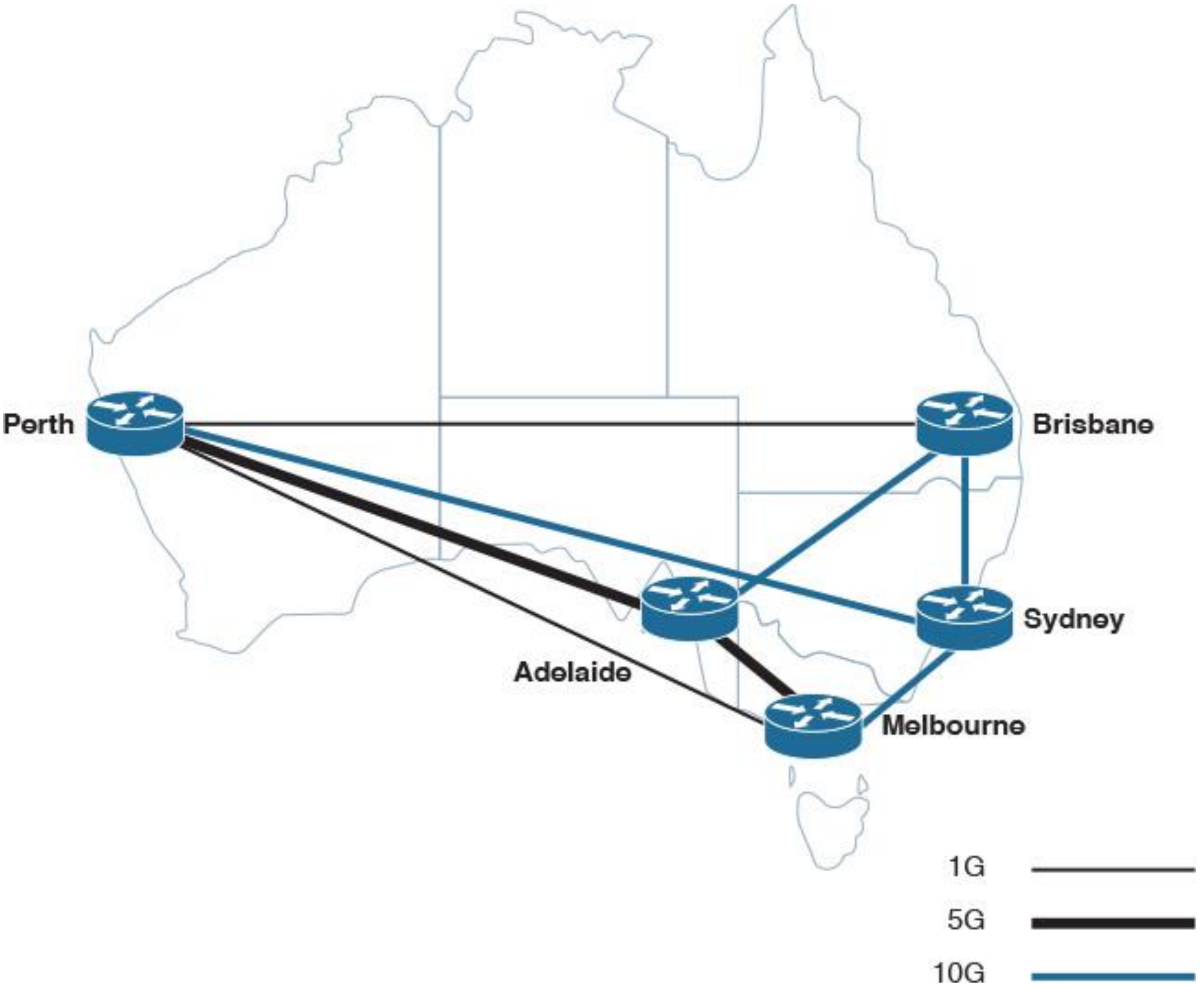

Figure 5-17 *MPLS-TE Scenario Network Layout*

This SP has a new demand to provide point-to-point virtual leased line (VLL) over Layer 2 MPLS VPN service for two of its customers (between Perth and Brisbane POPs) with bandwidth requirements of 1 Gbps each. In addition, path protection for this VLL service is part of the SLA in case of any link failure in the path.

However, with the current infrastructure using typical routing traffic engineering techniques (which is typically hop-by-hop destination-based routing), this SP will not be able to accomplish this requirement optimally. Moreover, although policy-based routing (PBR) may be used (technically) to achieve this goal, PBR often introduces several limitations to the design and the overall network performance. These limitations include operational complexities, instability, potential routing loops, and high hardware resource utilization.

With MPLS-TE, network designers can achieve a more scalable and constrained sourced-based routing solution. By applying MPLS-TE, this SP can selectively route the VLL L2 VPN traffic over the higher-capacity path (Brisbane to Sydney, then Perth), as illustrated in Figure 5-18. In addition, MPLS-TE fast reroute (FRR) can offer path protection toreroute traffic over an alternate path that meets minimum bandwidth capacity requirements to align with the offered SLA. In this example, the alternate path will be Brisbane to Adelaide, then Perth, as illustrated in Figure 5-18.

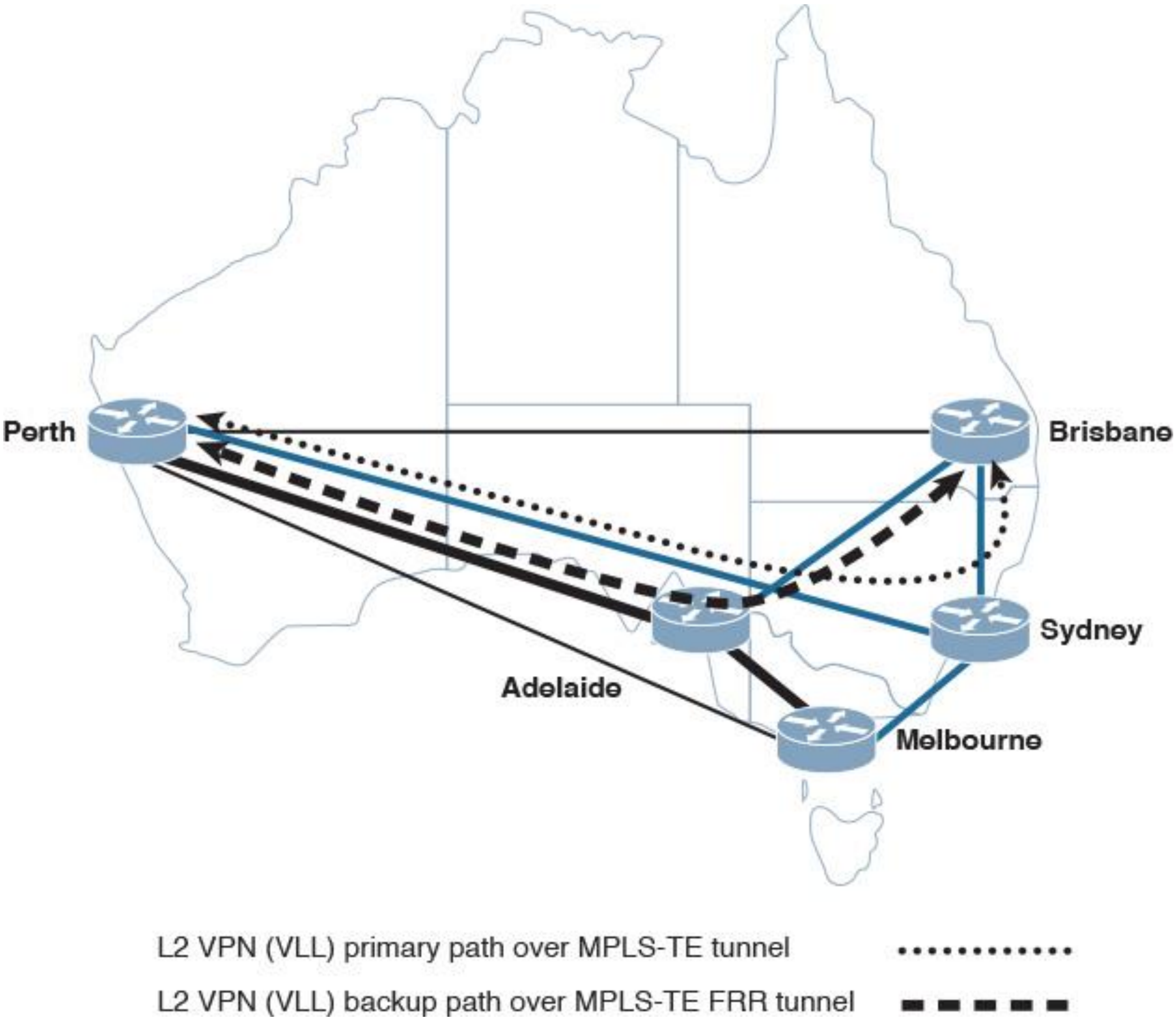

Figure 5-18 *MPLS-TE Solution 1*

Furthermore, this SP needs to accommodate new requirements for two of its key customers who have main offices in Melbourne and Perth. The first customer needs a VLL point-to-point service between Melbourne and Perth with 1 Gbps of bandwidth, and the second customer requires all their voice traffic between their main offices in Melbourne and Perth to be treated differently (routed over a lower-congested path with the least latency, along with path protection in case of a link failure).

MPLS-TE can help this SP to optimize traffic utilization over its intercity optical links with regard to the available bandwidth and customer requirements. For example, VLL traffic between Melbourne and Perth for the first customer can be redirected over the Sydney POP when its link between Sydney and Perth is underutilized. This offloads some of the traffic between Melbourne and Perth using MPLS-TE constrained routing.

At the same time, the voice traffic of the second customer can be sent over a TE tunnel that has better bandwidth and with minimal latency. In this example, the path will be Melbourne to Adelaide, then Perth, while the direct path between Melbourne and Perth can be used as a backup (alternate path) for voice traffic in case of any failure in the primary path, as illustrated in Figure 5-19. In addition, this SP is planning to enable unequal load sharing for data traffic with low priority (such as marked-down traffic with an EXP value of 0) between Melbourne and Brisbane POPs via both Adelaide and Sydney POPs, thereby taking advantage of the MPLS-TE capability to provide equal and unequal load balancing over multiple paths.

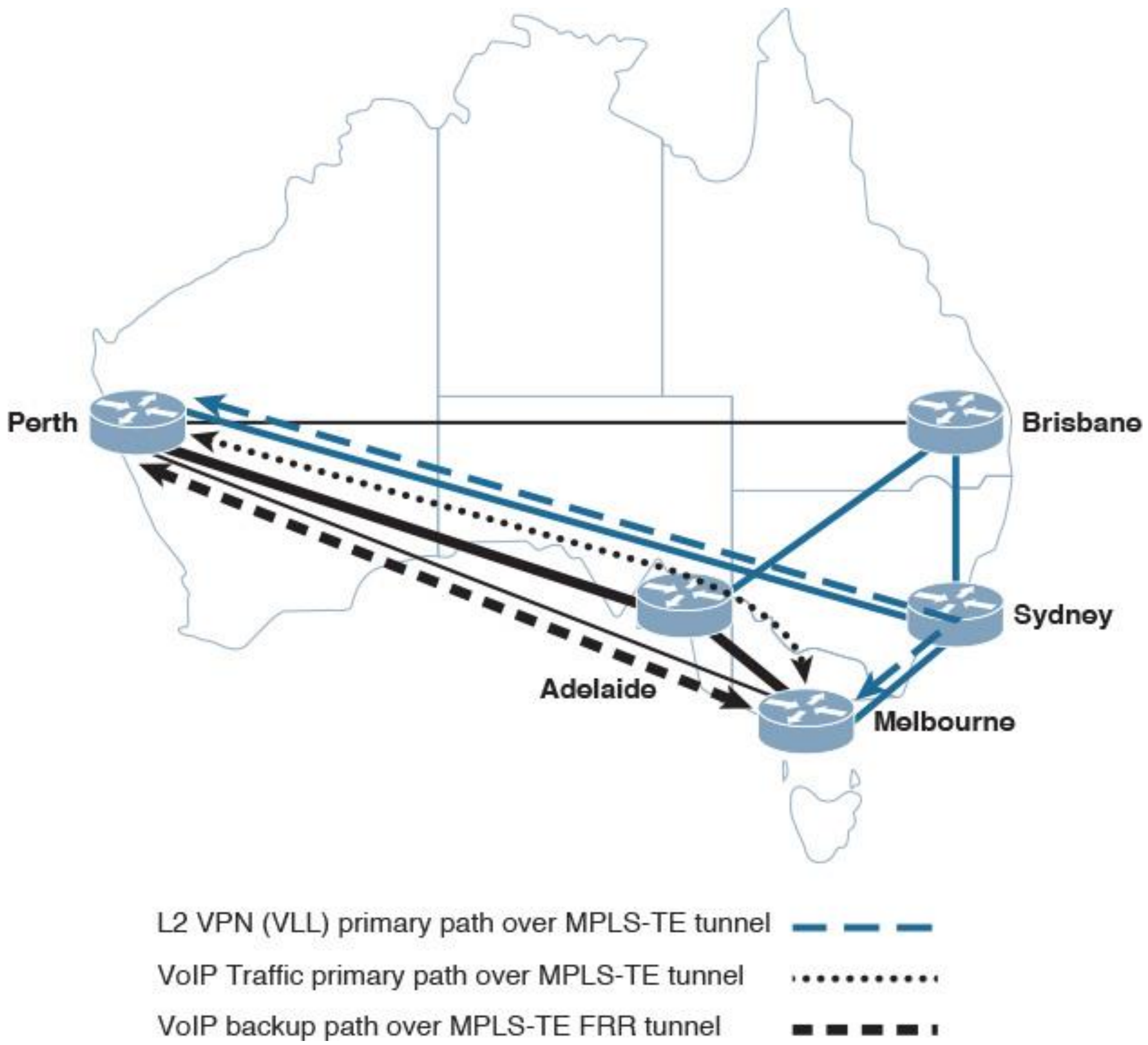

Figure 5-19 *MPLS-TE Solution 2*

From this example, we can conclude that several benefits may drive businesses and network designers to consider traffic engineering, including the following:

- Offer businesses a cost-effective solution to optimize the utilization of infrastructure links (optimized bandwidth management).
- Opens new markets for SPs by enabling them to offer more reliable services with very strict SLAs (for example, end-to-end high-bandwidth capacity, highly reliable service with MPLS-TE FRR, end-to-end low latency, and low packet loss services).
- Constrained source-based routing with MPLS-TE provides network designers a flexible and proven mechanism to have a more granular control over traffic passing through the transport network supported by various powerful MPLS-TE features, such as FRR and DS-MPLS-TE.
- Offer network designers multiple design options to enhance the overall network design in terms of path utilization and path protection by using MPLS-TE unequal-cost load balancing and MPLS-TE FRR, respectively.

However, MPLS-TE is not a feature or mechanism that must be used with MPLS-enabled networks. For example, networks overprovisioned with links and bandwidth normally do not need MPLS-TE to optimize and balance the utilization of links. Enabling MPLS-TE in networks such as this can sometimes be seen as added complexity without a real business value.

Therefore, network designers must understand the goals and requirements during the planning phase to suggest the suitable solution and the optimal MPLS-TE design approach. Otherwise, lack of good planning and a structured approach in designing MPLS-TE can significantly increase operational complexities.

This section covers the common planning approaches of MPLS-TE to generate a business-driven solution, along with the different design considerations that influence MPLS-TE design.

Note

In some situations, the network does not require MPLS-TE to optimize link utilization. However, MPLS-TE is mainly used to provide path protection with MPLS-TE (FRR). MPLS-TR FRR and MPLS with quality of service (QoS) are covered later in this book. Therefore, good planning with a good understanding of the targeted goal is crucial with MPLS-TE.

MPLS-TE Planning

In Chapter 1, planning was divided into two main approaches: strategic and tactical. These two approaches can be adopted to plan a business-driven MPLS-TE design to help network designers make the most suitable design choices based on the selected approach, which is typically driven by the business's needs and strategy.

MPLS-TE Strategic Planning Approach

In this approach, network designers must accommodate a long-term business and design objective to be reflected on the MPLS-TE design. For instance, an SP may decide to offer differentiated services to its clients to generate more revenue. One of these services may be to provide guaranteed end-to-end voice delivery with minimal to zero service disruption in the event of any network component failure within the SP network.

In the aforementioned scenario, network designers must make sure that customer Voice over IP (VoIP) traffic is routed over the least-congested paths without introducing any unnecessary latency to the traffic. In addition, a path-protection mechanism should be incorporated to meet the promised SLA by this SP after any failure event. Based on that, network designers can plan MPLS-TE combined with differentiated service (DS) MPLS-TE and FRR to meet this new long-term business strategy or direction. Similarly, Figure 5-20 represents a network where both R1 and R2 can send traffic peaking up to 5G, which can lead to potential congestion on the link between R3 and R5. Therefore, a long-term solution is required here.

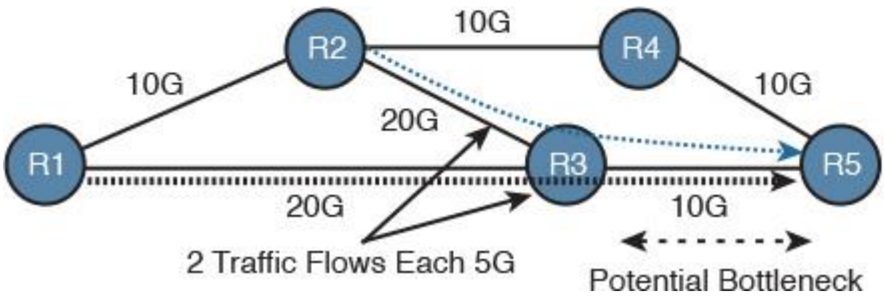

Figure 5-20 *Traffic Flow Before Applying MPLS-TE*

By introducing MPSL-TE, this network can permanently overcome this limitation (during normal, not failure, situations) by shifting the traffic flow of R2 to go over R4, then R5, assuming that the goal in this design is to provide optimization based on bandwidth only, as shown in Figure 5-21.

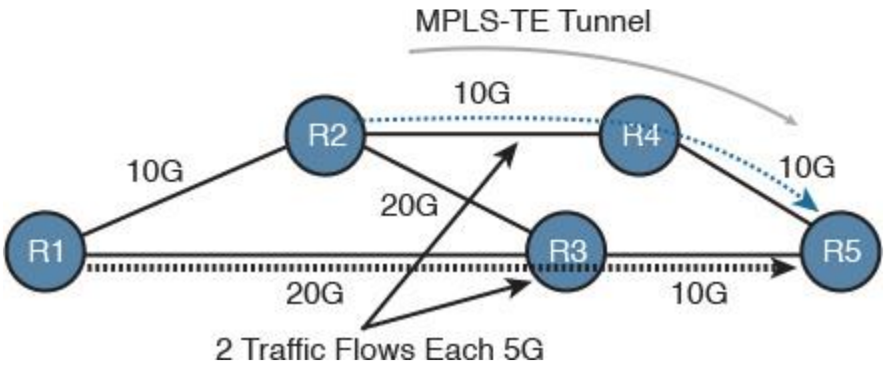

Figure 5-21 *Traffic Flow After Applying MPLS-TE*

MPLS-TE Tactical Planning Approach

In this approach, however, network designers typically are required to incorporate certain requirements that are normally temporary as a quick fix to an issue or a short-term solution that is to be changed later. For example, the scenario depicted in Figure 5-22 shows a global SP that currently has two interconnections between the North American and European regions. Currently, the first link (link 1) is facing high traffic utilization. The business decided to upgrade this link to accommodate the increased traffic volume. However, this upgrade may take several months to complete. Therefore, this SP needs to temporarily offload some of the traffic from link 1 and send it over link 2, which is currently underutilized.

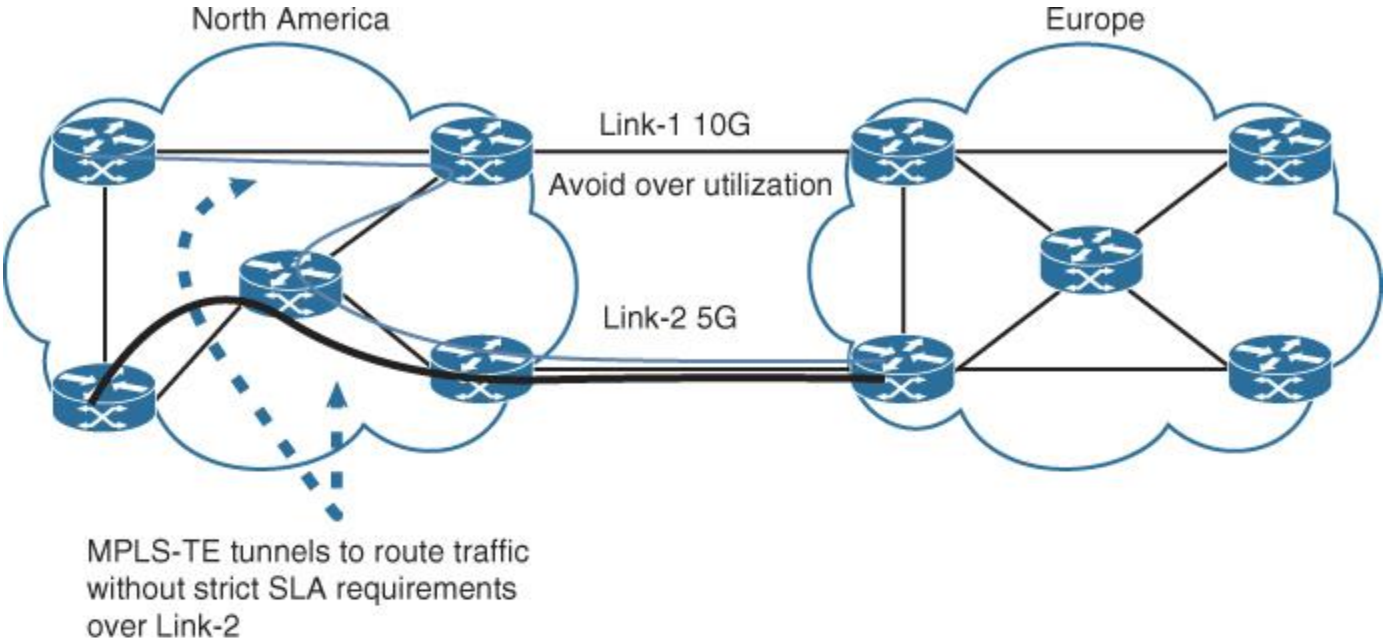

Figure 5-22 *MPLS-TE Technical Planning Scenario*

In this scenario, network designers can temporarily set up MPLS-TE to force certain traffic selectively (for example, traffic from customers without strict VoIP SLA) to traverse the second path. Even though the second path is longer from a classic routing point of view (more hops) and has a lower bandwidth, the reality is this path is underutilized, which can reduce packet drop possibility, because the first path is overutilized.

During the planning phase, it is important that network designers understand the goal to be achieved and whether MPLS-TE is the right solution. From a design point of view, the more protocols and features in the network, the more complex the design and operation will be. Answers to the following questions can help network designers, to a large extent, drive the MPLS-TE design in the right direction:

- Is the solution intended to be an interim solution or a long-term solution?
- What is the ultimate goal that needs to be achieved?
- Provide more balanced and efficient bandwidth utilization across the network
- Provide path protection (fast failover in milliseconds)
- Protect certain traffic from paths with a high degree of fate sharing (shared link risk group SLRG)
- Provide service differentiation to meet tight SLA requirements
- Combination of the above
- Is the targeted environment under single or multiple IGP or BGP domains?
- What is the scope of the solution (within a POP, between certain regional POPs, or across the entire network)?

MPLS-TE Design Considerations

Technically, there are two primary functions to set up MPLS-TE:

- Path computation that is driven by a set of predefined constraints to satisfy the desired design goals
- How to place and forward the desired traffic through the established MPLT-TE tunnel path

Therefore, it is important that network designers understand the mechanisms that MPLS-TE uses and supports to calculate the path in a constrained manner. Network designers must also be able to identify and understand the different options that can be used to forward the traffic along the path. In addition, the design architecture of the underlying Layer 3 routing (single versus multiple flooding domains) and the placement of the MPLS-TE tunnels (outer mesh versus inner mesh) are critical influencing factors to the MPLS-TE design and operation in large-scale networks. This section highlights and compares these aspects and discusses the advantages and the associated implications.

Constrained Path Calculation

The constrained shortest path calculation (CSPF) leverages the classic IGP shortest path first (SPF) algorithm to take into account different constraint variables. For instance, let's consider the same scenario of the previous section and assume that this SP has added a new POP in Tasmania with two interconnections, one to the Melbourne POP over an optical link and another to Perth's POP over a satellite link, as illustrated in Figure 5-23. With the typical SPF calculation, which is normally (directly or indirectly) calculated based the link bandwidth, the path via the satellite link will always be preferred over others. However, satellite links typically can offer high bandwidth with high latency. Based on that, if this SP wants to provide a tight SLA for real-time "delay-sensitive" traffic (VoIP) to its customers with sites connected in Tasmania and Perth, the satellite path must be avoided for this type of traffic, even though it has high bandwidth.

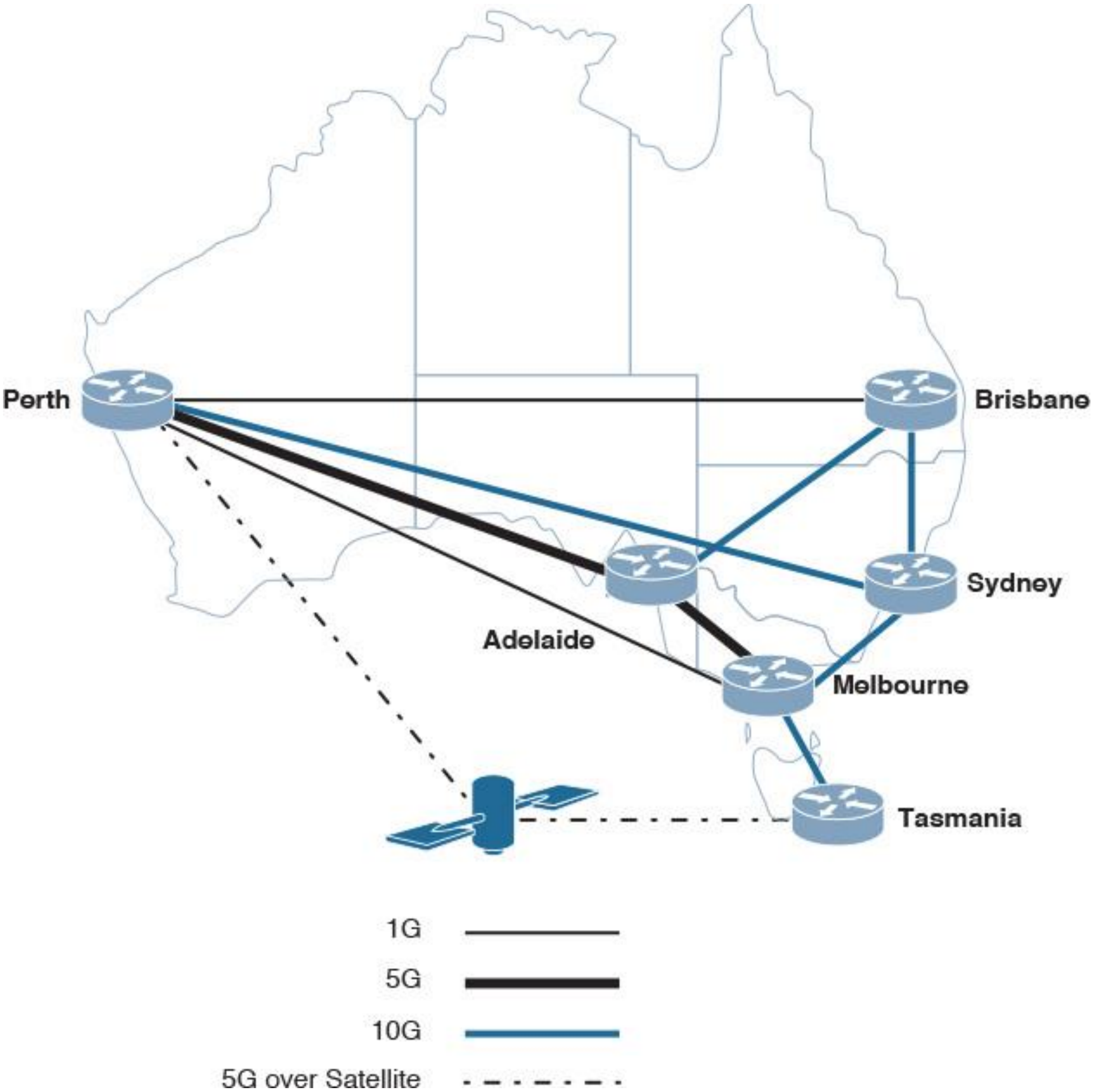

Figure 5-23 *Scenario 2 Pre-MPLS-TE*

With CSPF, network designers can influence the CSPF calculation to include additional attributes, also known as *link coloring* or *affinity attributes*, where each path can be assigned a value or attribute to its physical link. If this value does not match the one configured at the TE tunnel, this link or path will be excluded from the CSPF calculation. This attribute can reflect different design criteria such as fate-sharing or high-latency paths. These attributes are carried by the link-state IGP-specific TE extensions (RFC 3630, 2784). Based on that, each node within the MPLS-TE-enabled domain (typically a single IGP domain) will have all the relevant information (bandwidth required for LSP setup and link attributes) stored in the MPLS Traffic Engineering Database (TED).

Consequently, the SP in the example depicted in Figure 5-23 can take advantage of the CSPF and send voice traffic over the path with less bandwidth and low latency while data traffic can still travel over the high-bandwidth, high-latency satellite path, as long as the appropriate mechanism to place traffic into the TE tunnel is selected to send voice and data each into its respective TE tunnel, as illustrated in Figure 5-24.

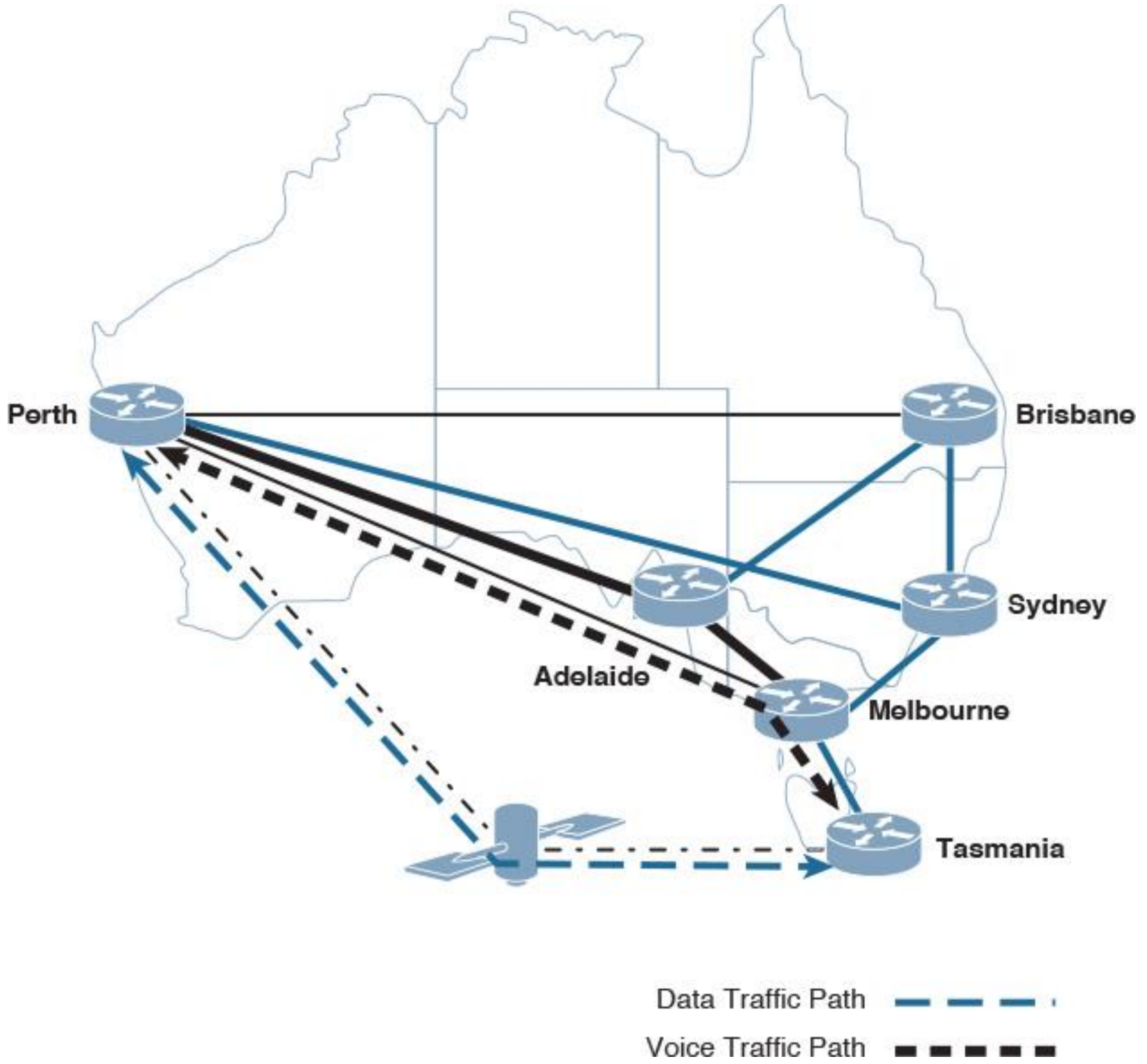

Figure 5-24 *Scenario 2 MPLS-TE Solution*

Furthermore, after the TE LSP setup is completed over the selected path that meets the CSPF criteria, the TED needs to be updated about the path state. For instance, the path selected for voice traffic in Figure 5-24 (Tasmania to Melbourne, then Perth) may become congested after a while because the link-state advertisement that is typically sent periodically, and also after any state change (for example, bandwidth or link attribute changes). In a process commonly known as *path reoptimization*, MPLS-TE will initiate CSPF to recalculate a new LSP that satisfies the minimum predefined requirements. In addition, MPLS-TE provides the ability to assign priorities to TE LSPs. This means that LSPs with a higher priority will have the ability to tear down lower-priority LSPs during the setup and during the recalculation of the LSP over a new path.

Path reoptimization with MPLS-TE can offer a more dynamic and updated traffic pattern from the TED point of view, which may provide better bandwidth optimization. However, in networks with a high degree of traffic pattern changes across them or in the scenario when high MPLS-TE physical paths have a flapping issue, it will lead to significant network instability. This instability is especially pronounced if there are multiple MPLS-TE LSP priorities defined (flapping). Therefore, if network designers need to consider this capability, proper preemption delay timers need to be considered, as well, to avoid network instability situations. In addition, MPLS-TE relies on the “make-before-break” mechanism when building a new LSP in a path optimization situation to avoid tearing down an active LSP before building a new one. However, this approach can lead to double booking of bandwidth during the setup of the new LSP and before tearing down the original LSP.

It is obvious from this example that the CSPF can offer more flexibility when it comes to path computation and selection criteria. However, if the considered CSPF approach does not align with the actual design goal, it will most probably be considered an added complexity to the design. For instance, if the design goal is only to provide better bandwidth utilization across the links, while the design considered an advanced CSPF criteria to select paths based on different attributes such as latency and shared risk link groups (SRLGs), this design typically falls under the principle discussed inChapter 1, called *gold plating*. Gold plating occurs when the design adds more than what is required. Typically, this will be seen as an added complexity from both the design and operational perspectives. Therefore, it is critical to identify the actual intended goal of the solution during the planning phase before selecting any approach, because each component of the MPLS-TE can achieve different design goals in different ways.

Note

Link coloring or affinity attributes with regard to SRLG is only used to identify and mark SRLG points/links to be used in scenarios where the calculated path has to be free of any SRLG fate sharing.

Most MPLS-TE deployments are based on one of the mechanisms to calculate available bandwidth for the MPLS-TE LSP setup:

- **Offline calculation:** As the name implies, the offline calculation is typically performed outside the network using third-party specialized tools that can build and create sophisticated LSP setup. In particular, with the offline tools, SPs can overcome LSP packing issues because the tool has awareness of the entire LSP that needs to be set up. However, LSPs with the offline tools are sometimes built based on a stable traffic situation, which may lead to congestion during traffic spikes due to the lack of online visibility and reaction to traffic changes.
- **Online calculation:** With the online calculation, the router (headend) is responsible for the LSP setup. This may be seen by some providers as a way to simplify operational complexity because it can dynamically react to network/traffic pattern changes without the need to rely on another tool, which typically requires additional expertise and offline work. However, with the online approach, the LSP packing issue must be taken into account.

MPS-TE Tunnel Placement

As discussed earlier in this chapter, typical SP networks are structured of two tiers (POPs and core). On top of these two primary tiers, there are multiple overlays and control protocols running, including MPLS-TE. The misconception of MPLS-TE deployment over this type of architecture is that it should be run between PEs (PE to PE MPLS; TE tunnels). However, as explained earlier in this section, the successful MPLS-TE design must be driven by the intended goal or facilitated by MPLS-TE. Therefore, the two primary potential places to create and terminate MPLS-TE (headend and tail end) are either PE-to-PE or P(core)-to-P(core). Table 5-2 compares the implications of using either approach with regard to different design aspects.

	PE-to-PE MPLS-TE (Outer Mesh)	P-to-P MPLS-TE (Inner Mesh)
Number of LSPs	Large	Moderate
End-to-end traffic differentiation	Supported with high control	Partially supported (not end to end) with limited control (over the core)
Path protection	End-to-end LSP	Across the core only
Design simplicity	The larger the number of PEs, the more complex the design	Almost always simpler than PE to PE
Operational complexity	The larger the number of PEs, the more complex to manage the network	Almost always simpler than PE to PE
Design scalability	Relatively scalable (the more PEs/LSPs, the more memory and CPU required)	Almost always more scalable than PE to PE because number core nodes usually fewer than PEs (ideally)
LSP stacking (LDP over TE tunnel with MPLS VPN)	Not required	Required

Table 5-2 *MPLS-TE PE Mesh Versus P Mesh Tunnels*

As highlighted in Chapter 1, for a design to be successful and reliable with regard to scalability, it has to be within a manageable scale. Otherwise, the added operational complexities may outweigh the advantage of scaling by size only. The following are primary factors that influence the scale of a MPLS-TE:

- LSP information state
- RSVP-TE information state
- Manageability level
- Processing overhead

In fact, in large environments with a mesh of TE tunnels, the placement of MPLS-TE headend and tail end (whether it is PE to PE or P to P) has significant impact on the factors in the preceding list, as shown in Figure 5-25.

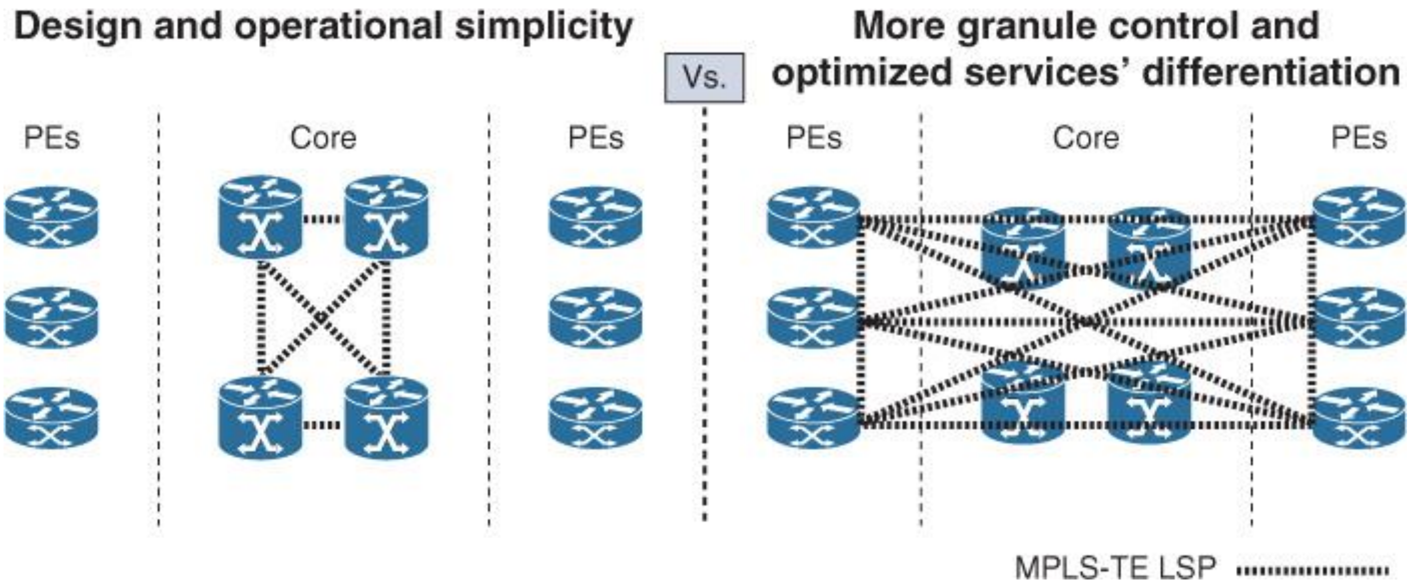

Figure 5-25 *MPLS-TE Tunnels: PE to PE Versus P to P*

Consequently, network designers and architects can decide suitable placement of the TE tunnels by aligning the design's goals, constraints (such as hardware limitations), and the design implications highlighted in Table 5-2. In addition, you can use a hybrid approach in some situations to satisfy different requirements. An example of a hybrid approach may be to provision a full mesh of TE tunnels across the core network to provide bandwidth optimization and path protection while using PE-to-PE TE tunnels to optimize end-to-end path selection and control between certain PEs only. Table 5-3provides some examples of design requirements and possible MPLS-TE solutions with regard to TE tunnel placement.

Design Requirements	Possible Solution
Path/bandwidth optimization for congestion points across the network	MPLS-TE on only the edge nodes (PEs) that face congestion such as regional Internet edge services PE (partial PE to PE)
Bandwidth optimization across the core network	MPLS-TE with a full logical mesh over a partial physical mesh or ring core, ideally with the help of an offline capacity planning tool service (ideally P to P)
Meet tight SLA requirements for providing L2VPN for a VLL service	MPLS-TE combined with tunnel priories using DS-TE and associated with MPLS-TE FRR (typically PE to PE)
Path protection to minimize convergence time to milliseconds across an overprovisioned bandwidth in the core	MPLS-TE FRR, ideally over one-hop primary TE tunnels and backup tunnel for each primary tunnel (ideally P to P) combined with automesh
Provide service differentiation for MPLS L3VPN customers such as voice and data over different paths	MPLS TE combined with tunnel priories using DiffServ-aware DS-TE and/or link affinity attributes along with a path/TE selection (QoS DiffServ based) to forward traffic based on QoS marking (You can use MPLS-TE FRR to protect critical TE tunnels only, ideally PE to PE.)

Table 5-3 *MPLS-TE Solution with Regard to TE Tunnel Placement*

Routing Domains

One of the primary and critical MPLS-TE design factors is the Layer 3 routing design for both IGP and external gateway protocol (EGP) domains. As discussed earlier in this section, CSPF typically relies on the information flooded by the link state with the relevant TE extension that carries specific information for the MPLS-TE commutation. In addition, earlier in this book, the term *interdomain routing* described the concept of routing between two independent routing domains via an EGP protocol. However, in this section, interdomain routing *for the purpose of MPLS-TE design* refers to the routing between either two BGP domains (autonomous systems) or different IGP flooding domains.

Typically, in both cases, the MPLS-TE ingress node "headend node" does not have the required information in its TED to build an LSP to the destination (tail-end) node that resides in another routing domain. Therefore, the headend node cannot calculate the path to the tail end by itself with this limited vision. Therefore, the border routers (Area Border Routers [ABRs] or Autonomous System Border Routers [ASBRs]) along the path can complement the computation of the path segment relevant to the "downstream" flooding domain, because only the ABR knows the TE topology of the area to which it belongs. Based on that, the headend node will find the path to the next ABR, then this ABR will perform the RSVP-TE explicit route object (ERO) to specify the path of an MPLS LSP via a sequenced list of label-switching routers (LSRs) and will be expanded by the next ABR or LSR, also known as *ERO expand*. As illustrated in Figure 5-26, with the ERO expanded, when an LSR node receives an RSVP Path message, if the next hop in the ERO is specified as a "loose subobject," it should compute a path segment to that loose hop (typically an ABR).

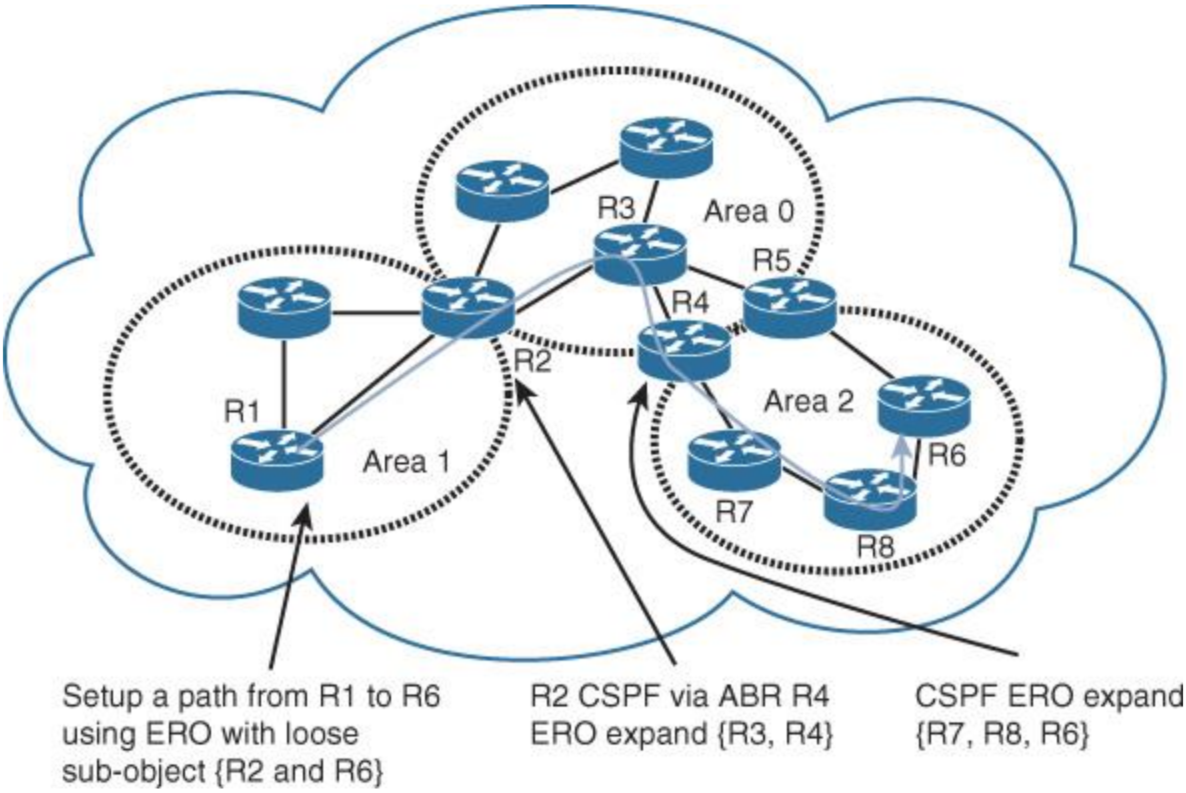

Figure 5-26 *MPLS-TE Interdomain*

One of the main limitations of this approach is it cannot guarantee optimal end-to-end shortest path selection (CSPF based). This is because each node will calculate the path up to the next loose subobject in the ERO path list (headend to the next ABR and ABR to the next ABR), which may not always lead to a desirable outcome.

For instance, in Figure 5-26, the end-to-end shortest path optimally should go via R5 to reach R6 (assuming the path to R5 meets other CSPF requirements such as minimum available bandwidth). However, because the calculation of the path is performed up to the ABR, suboptimal routing in situations like this is expected.

Alternatively, you can use a network component with the ability to calculate routes based on network graph known as*path computation element* (PCE) and described in RFC 4655. In PCE architecture, the MPLS-TE headend is technically considered a path computation client (PCC) and communicates over PCEP (Path Computation Element Protocol) with the PCE. The PCE component can be any node sitting somewhere in the network (in a router or as a software running on a server), ideally providing full visibility of the network topology. This approach can offer a more optimal end-to-end path computation, because the PCE typically has a better view of the network topology; however, network nodes must support this protocol for the PCE architecture to function properly because it requires additional control system integration (PCC to PCE). This might not be achievable in some situations (for example, the business cannot afford any additional cost to upgrade any hardware or software to support PCE), or it might be seen as an added complexity to the design (additional components and additional control protocol).

Furthermore, BGP link state (BGP-LS, described in IETF draft idr-ls-distribution-impl-xx) introduces another mechanism in which BGP can be used to collect link-state (link-state database [LSDB]) information and traffic engineering (TED) information from different routing domains and share it with external components such as external BGP speaker (different IGP domains or multi-AS BGP scenario) using a new BGP network layer reachability information (NLRI) encoding. As a result, multiple paths and links attributes stored in any given routing domain LSDB such as link metric, link bandwidth, preemption, and SRLG can now be shared with other domains over specific BGP NLRI. With this approach, the IGP collected information per IGP domain such as link metrics and TED-related information can be exported to the BGP control plane. BGP, however, will be able to reliably and selectively share their information with other external BGP speakers such as a peering SP, and more efficiently than typical LSDB information sharing (support information aggregation of multiples IGP flooding domains and autonomous systems). However, due to the fact that with this approach BGP will be carrying information from links-state IGP, which may lead to more frequent NLRI updates, it is better to avoid using an existing BGP RR for this purpose (peering directly to BGP RR), to avoid any impact on the existing BGP stability in large networks.

It is obvious that MPLS-TE design and deployment is more complicated over multiple routing domains. In addition, path reoptimization (TE LSP reoptimization) in interdomain types of scenarios can prove challenging. This may introduce confusion when designing a large network as to whether to optimize routing design with more modular (multiple logical flooding domains) or to keep it under one single flooding domain. The solution to this dilemma depends completely on the design requirements, goals, and priorities (more scalable module routing design versus simplified MPLS-TE design, with flatter routing design under a single flooding domain).

Forwarding Traffic via the TE Tunnel

The other vital element of MPLS-TE design is how to place or forward traffic through the TE tunnel. Although this element might seem like an implementation decision; in fact, the method used to steer traffic through the TE tunnel can impact not only the MPLS-TE design, but it can also impact other routing and forwarding behavior if it is not aligned well (considering the holistic principle discussed in Chapter 1). Table 5-4 summarizes and compares the common and primary mechanisms that you can use for this purpose.

Mechanism	Characteristic	Design and Operational Considerations	Possible Use Scenario
Static route	Manually and selectively point traffic flows through TE tunnel	Configuration and design complexity Operational complexity with large-scale networks	Small number of flows or a temporary redirection of traffic over a TE tunnel such as P2P L2VPN PW
Policy-based routing (PBR)	Manually and selectively select flows (sourced based)	Configuration and design complexity Operational complexity with large-scale networks May introduce some routing loops Can significantly increase hardware utilization in large deployments	Small number of flows or to temporarily redirect traffic over a TE tunnel
Autoroute	Automatic injection of the TE-tunnel tail end and the downstream destinations behind the tail end	Simple and dynamic Easier to deploy Lower operational complexity Less control of what traffic to use which TE when not combined with other capabilities like CBTS	Can support various scenario that need automatic injection of next hop reachability information via TE tunnel

Class-based tunnel selection (CBTS)	TE selection QoS DiffServ based (typically based on EXP marking value)	Simple and dynamic Provides high degree of service Differentiation by routing traffic based EXP marking Offers more granule bandwidth control when combined with MPLS DS-TE	Ideal for QoS-enabled solutions that need service differentiation such as voice, video, and data
Forwarding adjacency (FA)	Replicate the typical IGP behavior in term of reachability to remote prefixes (The TE tunnel will be seen as a direct link that will be considered in the IGP's path calculation.)	Can impact the entries' IGP routing domain of a network, which may lead to suboptimal routing across the network Introduces scalability limitation to the IGP because it significantly increases the IGP (typically LSDB) size Facilitates load sharing across physical and TE tunnel paths in some scenarios	Limited scenarios such as load sharing over physical and TE tunnel paths

Table 5-4 *Primary Traffic Forwarding Mechanisms into MPLS-TE Tunnels*

Based on the information highlighted in Table 5-4, each mechanism of routing or placing traffic into the TE tunnel has different characteristics and will logically impose different design considerations that network designers must be aware of, because it may impact other protocols such as IGP stability and lead to a more complex network to operate.

Summary

SP network architectures may look more flat than enterprise network architectures in terms of the depth of tiers. However, the number of control planes, virtualization, and overlays integrated across SP networks can result in a complicated "layer in layer" multitier architecture. In addition, the large scale of SP networks, from both the infrastructure and number of prefixes perspectives, makes the operational complexity a critical aspect to be considered by the network designer when adding and integrating any new feature or protocol.

This chapter also covered the common physical architectural components of SP networks, along with the primary control plane protocols that all can construct a modern converged SP-grade network. In addition, this chapter highlighted how a technical design decision can directly impact the SP business if it is driven only from a technical point of view. Last but not least, this chapter explained how MPLS-based SP networks can take advantage of MPLS-TE and add several benefits and optimization to the overall network design and performance if it is planned well using the top-down approach, taking into account the different design considerations of MPLS-TE covered in this chapter.

Further Reading

RFC 4105, *Requirements for Inter-Area MPLS Traffic Engineering*, http://www.ietf.org

"PCE-Initiated Traffic Engineering Path Setup in Segment Routed Networks," http://tools.ietf.org/html/draft-sivabalan-pce-segment-routing-00

RFC 3031, *Multiprotocol Label Switching Architecture*, http://www.ietf.org

"PCEP Extensions for MPLS-TE LSP Automatic Bandwidth Adjustment with Stateful PCE,"https://tools.ietf.org/html/draft-dhody-pce-stateful-pce-auto-bandwidth-00

RFC 1771, *A Border Gateway Protocol 4 (BGP-4)*, http://www.ietf.org

RFC 7149, *Software-Defined Networking: A Perspective from Within a Service Provider Environment*,http://www.ietf.org

Chapter 6. Service Provider MPLS VPN Services Design

Chapter 4, "Enterprise Edge Architecture Design," discussed the various design considerations, advantages, and limitations of the modern Multiprotocol Label Switching (MPLS) virtual private networking (VPN) solutions as a wide-area network (WAN) transport from the enterprise network design perspective. This chapter covers the different design options and considerations of MPLS VPN (both Layer 2 and Layer 3) from the service provider perspective.

MPLS VPN (L3VPN)

MPLS L3VPN is the most popular and proven service offered by the majority of today's service providers. This is because of several benefits (covered in Chapter 4) that enterprises can leverage to meet their business and technology requirements, specifically for the modern converged enterprise customer networks. Typically, the aim of any service provider is to meet the varied requirements of their customers, which will ultimately lead to maximizing their profit. Generally, service provider businesses aim to offer services that can

- Support a large number of customers and large number of sites per customer (scalable).
- Provide customers with value-added services that can create new revenue-generation sources such as service differentiation capable transport with end-to-end quality of service (QoS) (flexible and reliable).
- Support various services for all of their customers over one unified infrastructure. Having a single consolidated network reduces capital expenditure (capex) and can offer a significant return on investment (ROI) to the service provider on their hardware equipment (with the virtualized or overlaid architectures).
- Provide flexible service provisioning over various media access methods.

In addition, the MPLS L3VPN peer model has proven its flexibility and reliability in fulfilling these goals for service providers and many enterprise customers by offering

- Single infrastructure that can serve all VPN customers (as shown in Figure 6-1)

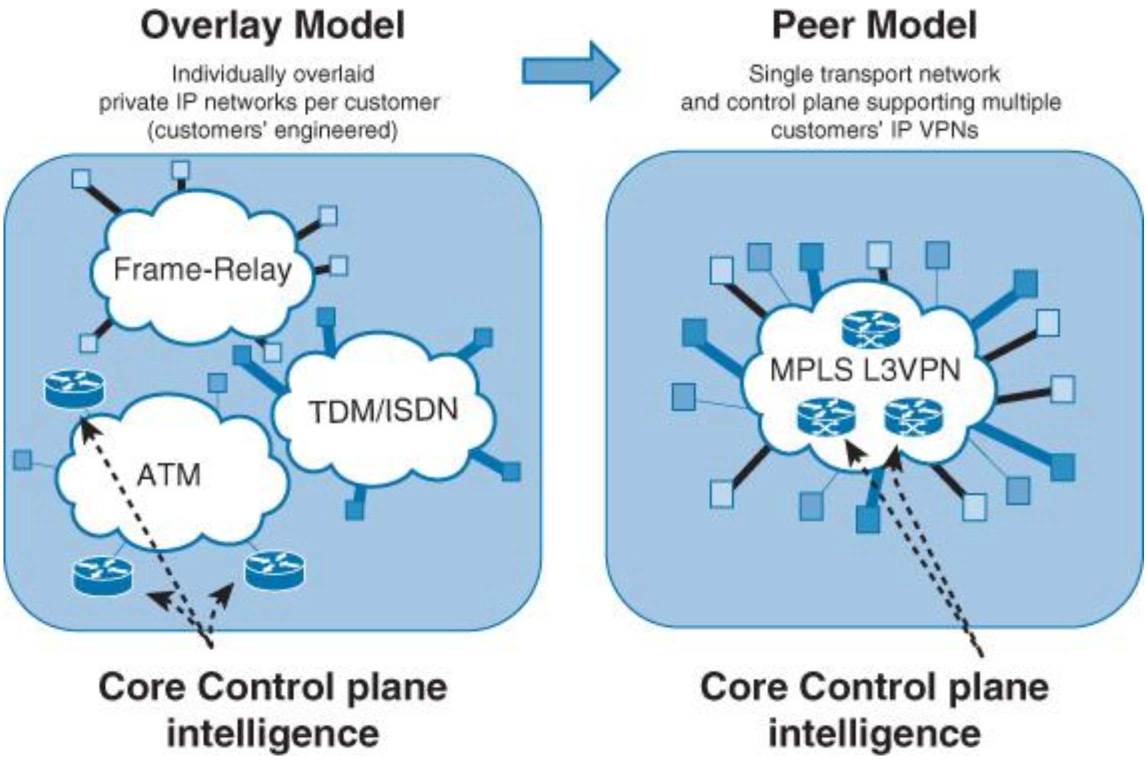

Figure 6-1 *MPLS L3VPN Peer Model*

- The optimization of OPEX. For instance, adding a new customer or a new site for an existing customer will require simple changes to the relevant edge nodes (provider edge [PE] nodes) only as the core control plane intelligence is pushed to the provider rather cloud, as shown in Figure 6-1.
- The opening of new revenue-generation sources to the business by offering differentiated services for their customers such as prioritization and expedited forwarding for voice.
- The optimization of time to market to introduce new services to their L3VPN customers, such as IPv6 and multicast support.
- A high degree of flexibility to service providers in offering various media access methods for their customers, such as legacy equipment, Ethernet over copper or fiber, and Long Term Evolution (LTE) or 3G.

MPLS L3VPN Architecture Components

In a typical MPLS L3VPN environment, the architecture is constructed of the following components:

- Customer edge (CE)
- Provider edge (PE)
- Provider nodes (P)

In the typical MPLS L3VPN architecture, the provider edge nodes (PEs) carry customer routing information to inject customer routes from the directly connected customer edge nodes (CEs), each to the relevant Multiprotocol Border Gateway Protocol (MP-BGP) VPNv4/v6, along with the relevant VPN and transport MPLS labels (label edge router [LER]). This achieves the optimal routing of traffic that pertains to each customer within each VPN routing domain. However, provider routers (Ps) at the core of the network are mainly responsible for switching MPLS labeled packets. Therefore, they are also known as *label switching routers* (LSRs). Figure 6-2 illustrates the primary component of an MPLS L3VPN architecture:

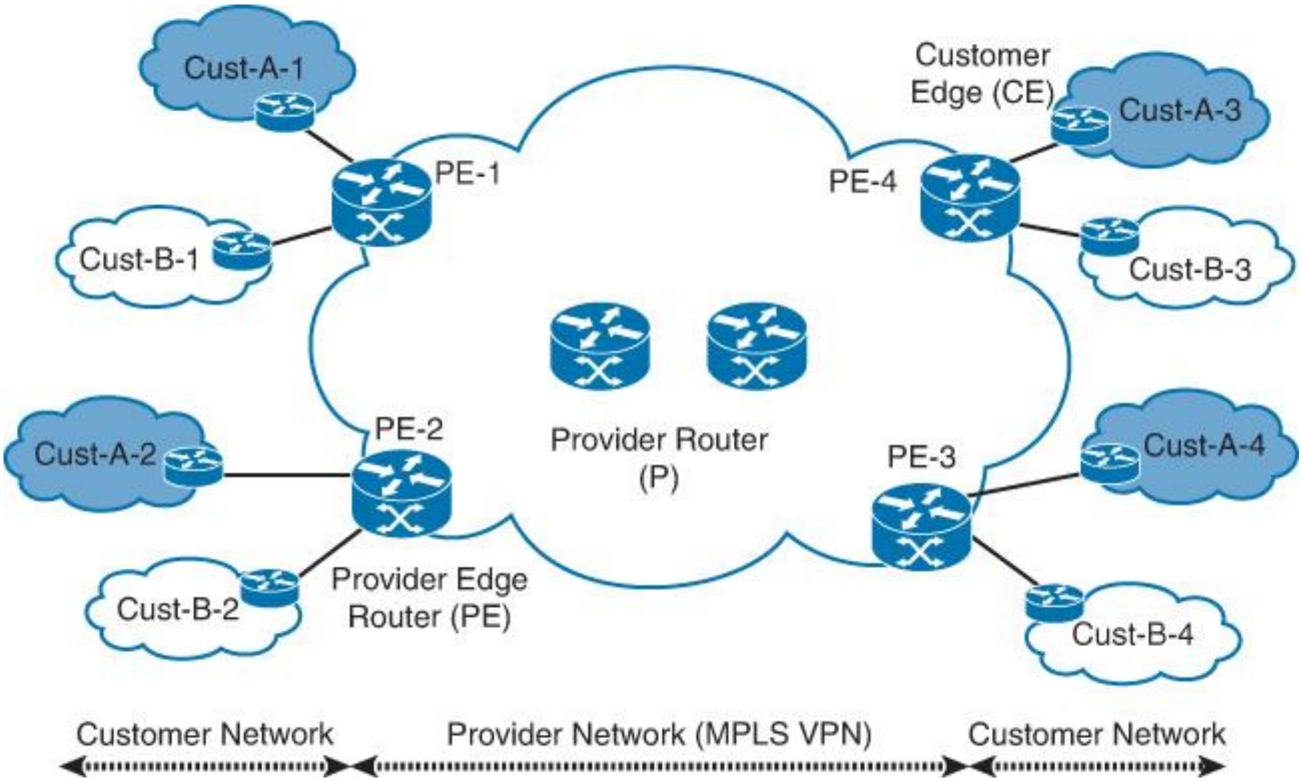

Figure 6-2 *Primary Elements of an MPLS L3VPN Architecture*

On top of the architectural components shown in Figure 6-2, the different control plane protocols discussed in Chapter 5, "Service Provider Network Architecture Design," are overlaid on top of this architecture to construct the control and forwarding planes for each VPN network (per customer). Figure 6-3 shows the relationship between the different control and forwarding plane components in an MPLS L3VPN architecture.

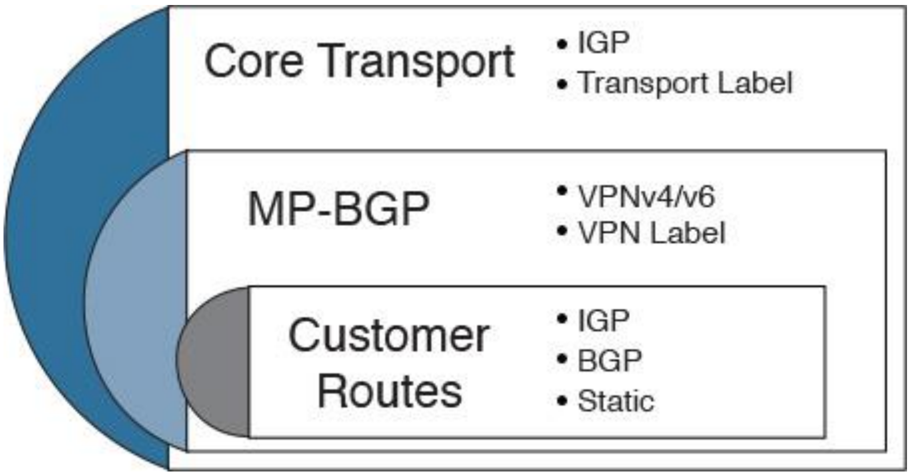

Figure 6-3 *MPLS L3VPN Control Plane Components*

The actual communication in a typical MPLS L3VPN environment is driven by following three primary elements:

- Routing information isolation between different VPNs (for example, virtual routing forwarders [VRFs])
- Controlled sharing of routing information to sites within a VPN or between different VPNs (for example, Route distinguisher [RD] + Route target [RT] + MP-BGP)
- MPLS L3VPN traffic forwarding of packets across the service provider core (for example, VPN and transport labels)

L3VPN Control Plane Components

This section covers the primary control plane elements of an MPLS L3VPN environment.

Virtual Routing and Forwarding

As discussed in Chapter 3, "Enterprise Campus Architecture Design," VRFs are one of the primary mechanisms used in today's modern networks to maintain routing isolation on a Layer 3 device level. In MPLS L3VPN architecture, each PE holds a separate routing and forwarding instance per VRF per customer, as shown in Figure 6-4. Typically, each customer's VPN is associated with at least one VRF. Maintaining multiple VRFs on the same PE is similar to maintaining multiple dedicated routers for customers connecting into the provider network. In addition, maintaining multiple forwarding tables at the PE is essential to support overlapping address spaces. Normally, the routing information of each customer (VPN) is installed at the relevant VRF routing tables of a PE, either from directly connected CEs (using a VRF-aware interior gateway protocol [IGP], BGP or static route) or routes of other CEs learned via remote PEs over MP-BGP VPNv4/v6.

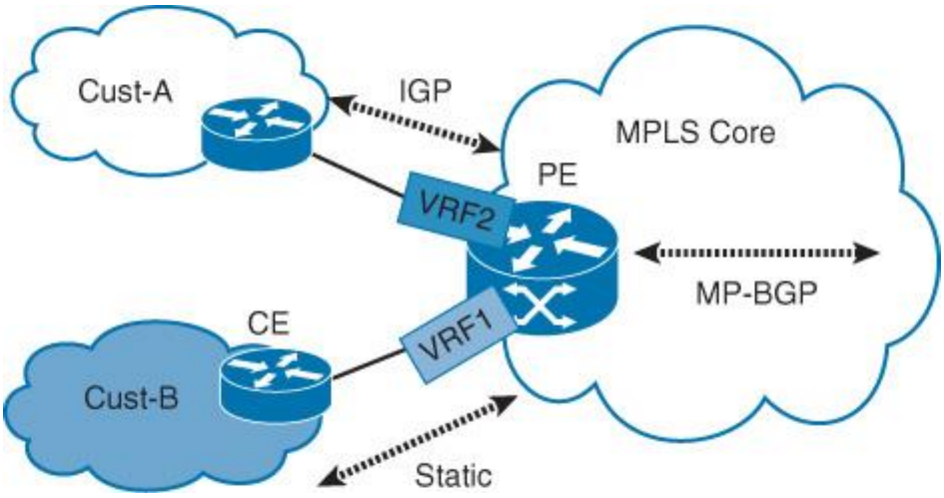

Figure 6-4 *MPLS L3VPN: Virtual Routing and Forwarding*

Route Distinguisher

For an MPLS L3VPN to support having multiple customer VPNs with overlapping addresses and to maintain the control plane separation, the PE router must be capable of using processes that enable overlapping address spaces of multiple customer VPNs. In addition, the PE router must also learn these routes from directly connected customer networks and propagate this information using the shared provider backbone. This is accomplished by using a route distinguisher (RD) per VPN or per VRF. As a result, service providers can seamlessly transport customers route (overlapped and nonoverlapped) over one common infrastructure and control plane protocol to take advantage of the RD prepended per MP-BGP VPNv4/v6 prefix. Normally, the RD value can be allocated using different approaches. Each approach has its strengths and weaknesses, as covered Table 6-1.

RD Model	Strength	Weakness	Suitable Scenario
Unique RD per VPN	Simple to design and manage Lower hardware resource consumption compared to other models	Lacks the ability to support load-balancing capability when VPN route reflection (RR) is used in the MPLS VPN network and there are customers that have multihomed CE routes	Very large-scale MPLS VPN without load-balancing or load-sharing requirements toward multihomed sites
Unique RD per VPN per PE*	Offers the ability to load balance traffic toward multihomed sites that are part of the same VPN but connected to different PEs	Requires more hardware resources such as memory to store the additional VPN routes Higher design and operational complexity than the unique RD per VPN model	Large-scale MPLS VPN with load-balancing or load-sharing requirements toward multihomed sites**
Unique RD per VPN per interface of each PE	Simplifies identifying sites within a VPN	Highest hardware resources utilization (a separate VRF per CE) High design and operational complexity	MPLS VPN of an enterprise with small number of VPNs with requirements to have a simplified mechanism for the operation team to identify the origin of any route (from which site per VPN) through its RD value***

* Load balancing or load sharing for multihomed sites using unique RD per VPN per PE is covered in more detail later in this chapter.

** In large-scale networks with a large number of PEs and VPNs, unique RD per VPN RD allocation should be used. The unique RD per VPN per PE RD allocation model can be used only for multihomed sites if the customer needs to load balance/share traffic toward these sites.

*** BGP site of origin (SoO) can be used as an alternative to serve the same purpose without the need of a unique RD per interface/VRF.

Table 6-1 *MPLS L3VPN RD Allocation Models*

Figure 6-5 illustrates these different RD allocation models discussed in Table 6-1.

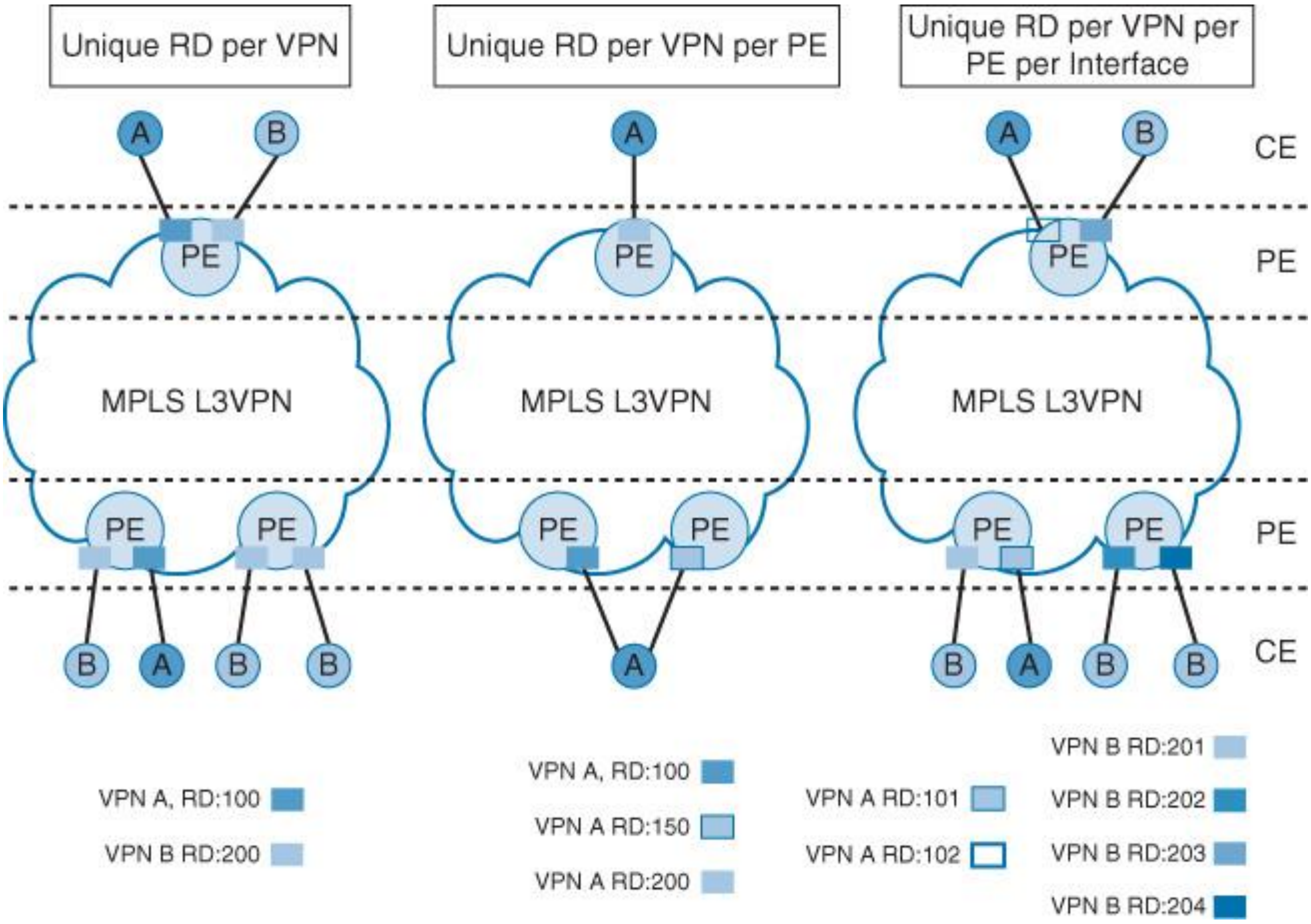

Figure 6-5 *MPLS L3VPN RD Allocation Models*

Note

Based on the RD allocation models covered in Table 6-1, a single VPN may include multiple RDs with different VRFs. However, the attributes of the VPN (per customer) will not change and is still considered an intra-VPN because technically the route propagation is controlled based on the import/export of the RT values.

Route Targets

Route targets (RTs) are an additional identifier and considered part of the primary control plane elements of a typical MPLS L3VPN architecture because they facilitate the identification of which VRF can install which VPN routes. In fact, RTs represent the policies that govern the connectivity between customer sites. This is achieved via controlling the import and export RTs. Technically, in an MPLS VPN environment, the export RT is to identify a VPN membership with regard to the existing VRFs on other PEs, whereas the import RT is associated with each PE local VRF. The import RT recognizes and maps the VPN routes (received from remote PEs or leaked on the local PE from other VRFs) to be imported into the relevant VRF of any given customer. In other words, RTs can offer networks designers a powerful full capability to control what MP-BGP VPN route is to be installed, in any given VRF/customer routing instance. In addition, it provides flexibility to create various logical L3VPN (WAN) topologies for the enterprise customer, such as any to any, hub and spoke, and partially meshed, to meet different connectivity requirements.

L3VPN Forwarding Plane

In addition to the control plane discussed earlier, the data or forwarding plane forms the other major component of typical MPLS L3VPN building blocks. In MPLS VPN environments, the data plane is based on forwarding packets based on labels (transport labels, VPN labels, MPLS Traffic Engineering [MPLS-TE] labels, and so on). This section focuses on the VPN label.

VPN Label

Typically, VPN traffic is assigned to a VPN label at the egress PE (LER) that can be used by the remote ingress PEs (LER), where the egress PE demultiplexes the traffic to the correct VPN customer egress interface based on the assigned VPN label. In other words, the VPN label is generated and assigned to every VPN route by the egress PE router, then advertised to the ingress PE routers over an MP-BGP update. Therefore, it is only understood by the egress PE node that performs demultiplexing to forward traffic to the respective VPN customer egress interface/CE based on its VPN label.

The VPN labels in MPLS VPN architectures can be allocated by the PE nodes using different models for MP-BGP L3VPN routes, based on the scenario and the design requirements. It also offers network designers the flexibility to achieve a level of trade-offs between network performance and scalability when possible. The following are the common MPLS VPN label-allocation models.

- **Per prefix:** In this model, a VPN label is assigned for each VPN prefix. Although this model can generate a large number of labels, it is required in scenarios when the VPN packets sent between the PE and CE are label switched.
- **Per VRF:** In this model, a single label is allocated to all local VPN routes of any given PE in a given VRF. This model offers an efficient label space and BGP advertisements, and the lookup does not result in any performance degradation. In addition, some vendor platforms support the same per-VRF label for both IPv4 and IPv6 prefixes.
- **Per CE:** The PE router allocates one label for every immediate next hop; in most cases, this would be a CE router. This label is directly mapped to the next hop, so there is no VRF route lookup performed during data forwarding. However, the number of labels allocated is one for each CE rather than one for each VRF. Because BGP knows all the next hops, it assigns a label for each next hop (not for each PE-CE interface). When the outgoing interface is a multiaccess interface and the media access control (MAC) address of the neighbor is not known, Address Resolution Protocol (ARP) is triggered during packet forwarding [37].

Network designers must be careful if they plan to change the default label allocation, behavior because any inconsistency or simple error can lead to a broken forwarding plane that can easily bring down the entire network or a portion of the network. In a service provider, a PE that goes down this way may result in several customer sites (usually single homed ones) being out of service, which can impact the business significantly, especially if there is a strict service level agreement (SLA) with its customers.

Figure 6-6 shows a summary of end-to-end forwarding and control planes of an MPLS L3VPN architecture.

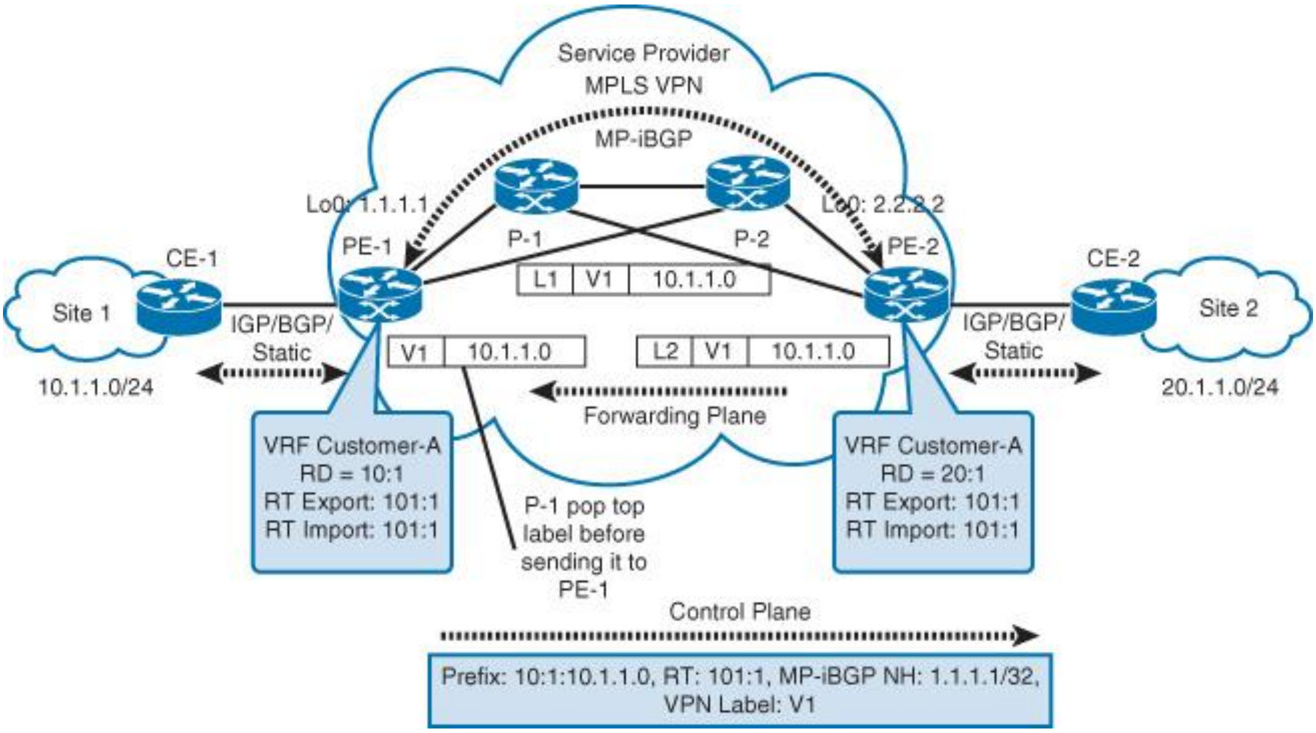

Figure 6-6 *Forwarding and Control Planes of MPLS L3VPN Architecture*